

# Common Package Dimensions for CH Series ICs

Version: 4.0

<https://wch-ic.com>

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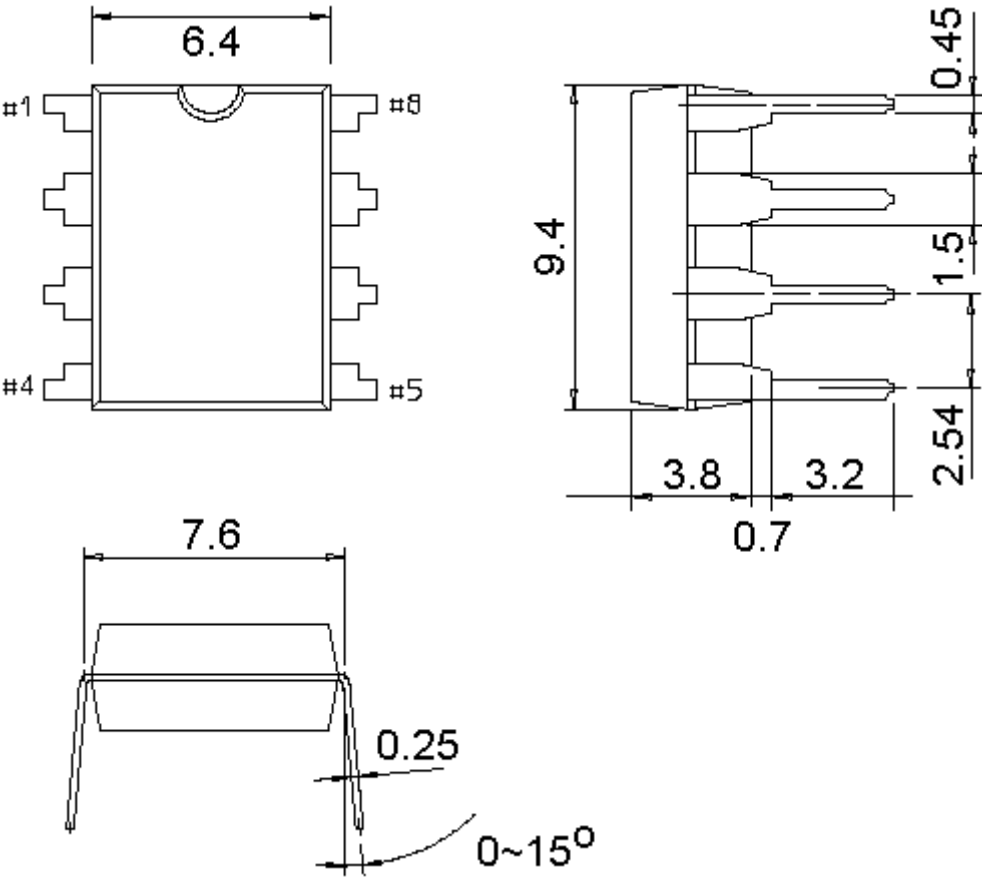
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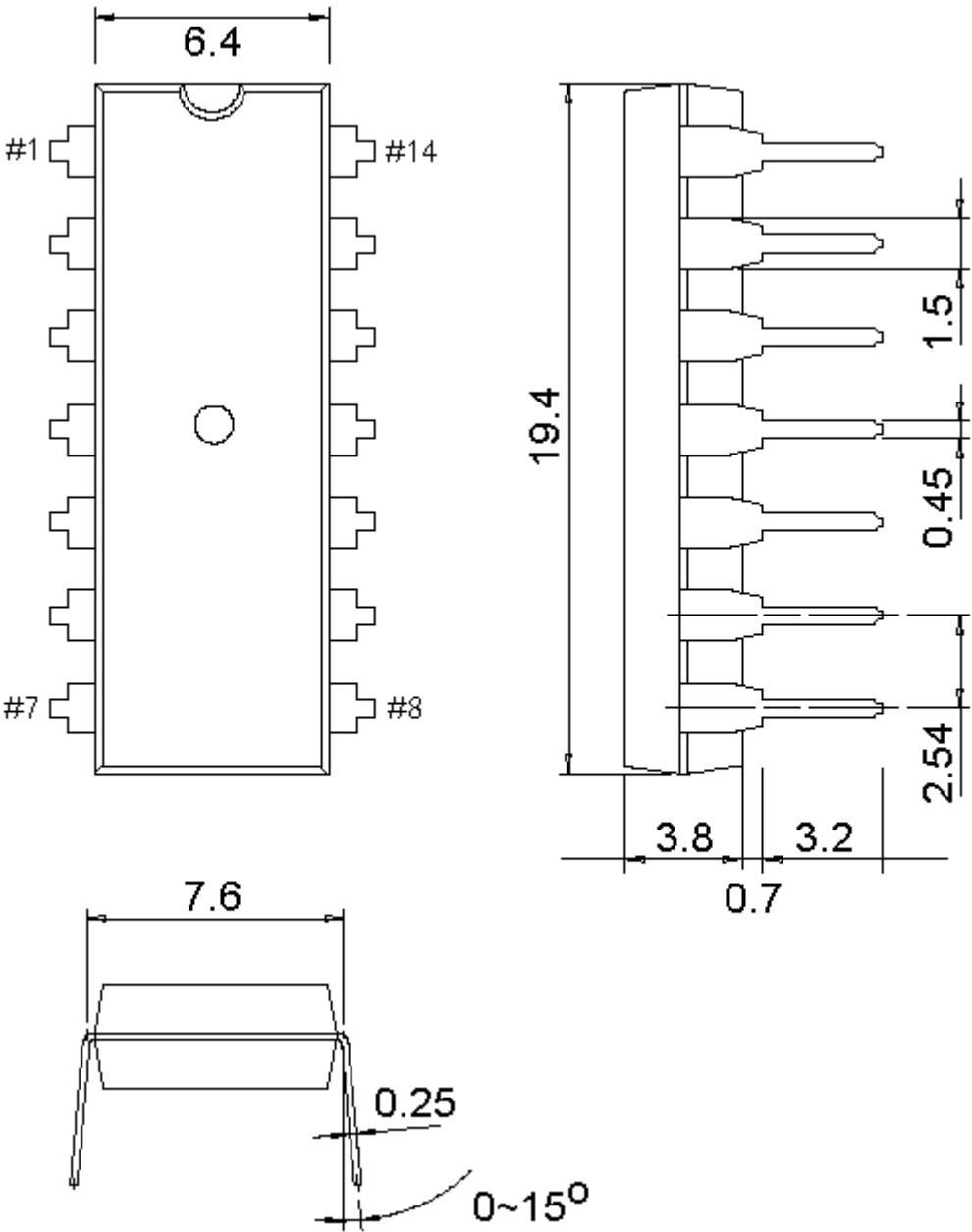
*Note: All dimensions are in mm (millimeters),  
and the lead pitch is always a nominal value, with no error value;  
other than that, the dimensional error is not greater than  $\pm 0.2\text{mm}$  or  $\pm 10\%$ , whichever is greater.*

DIP Type (PDIP)

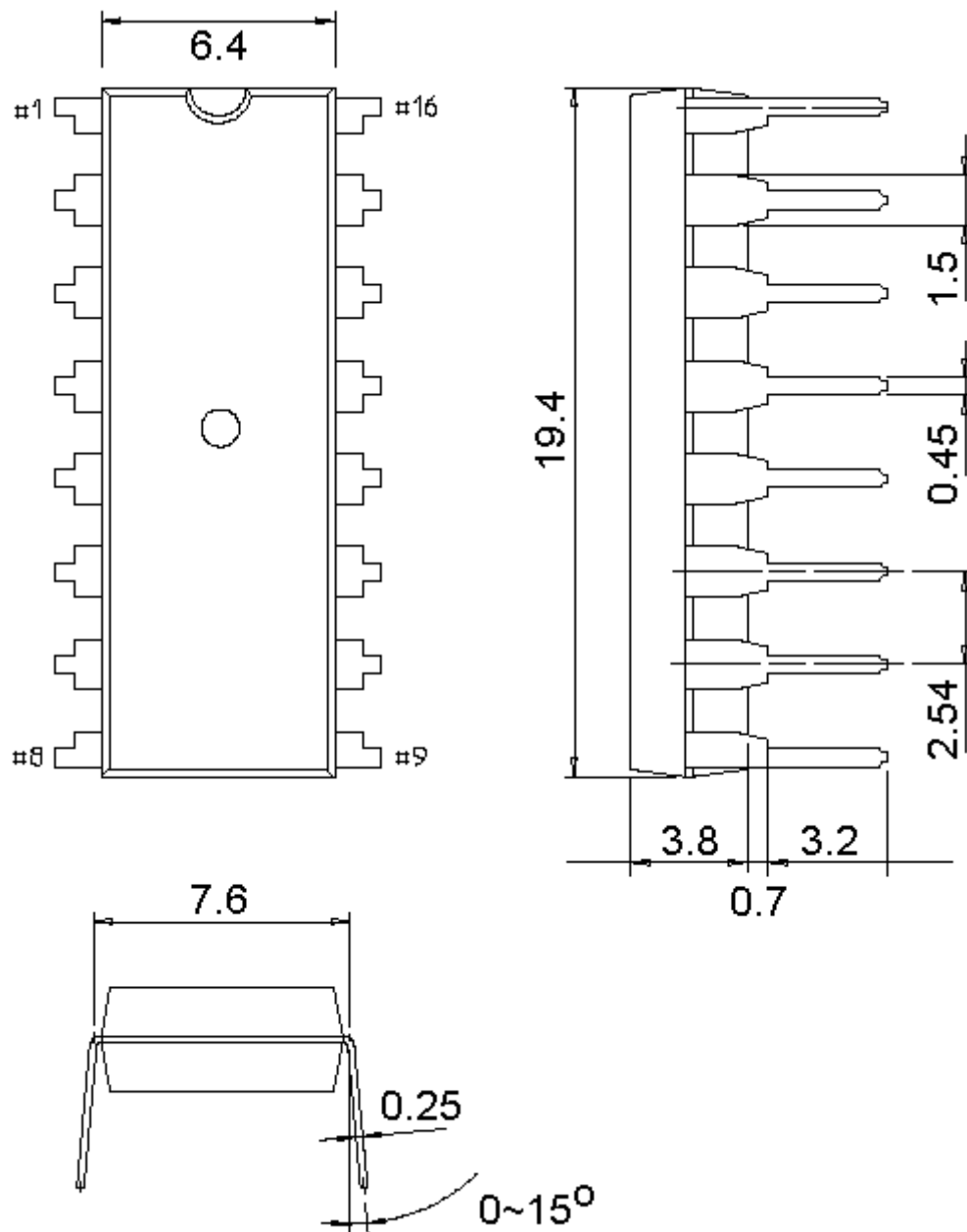
. DIP8



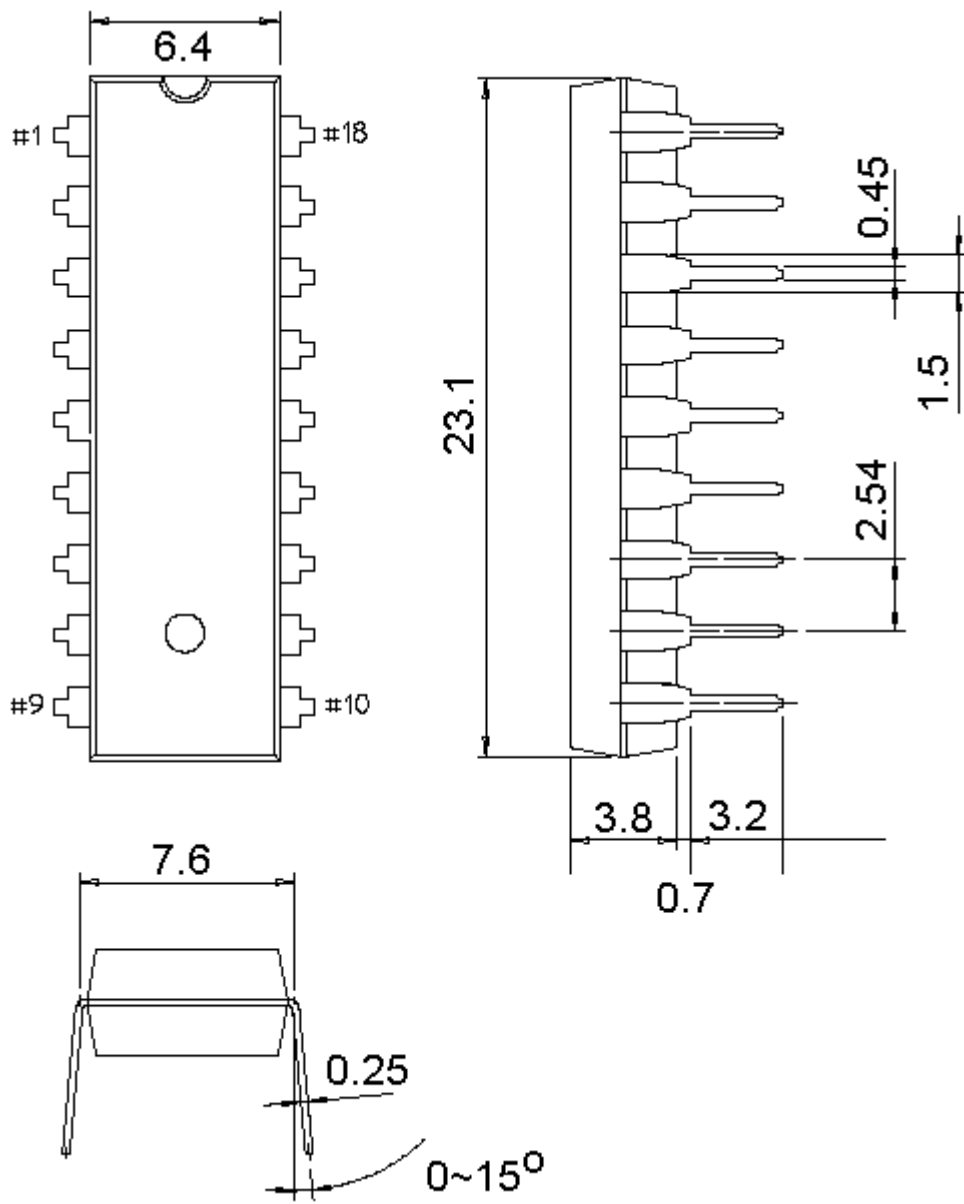
. DIP14



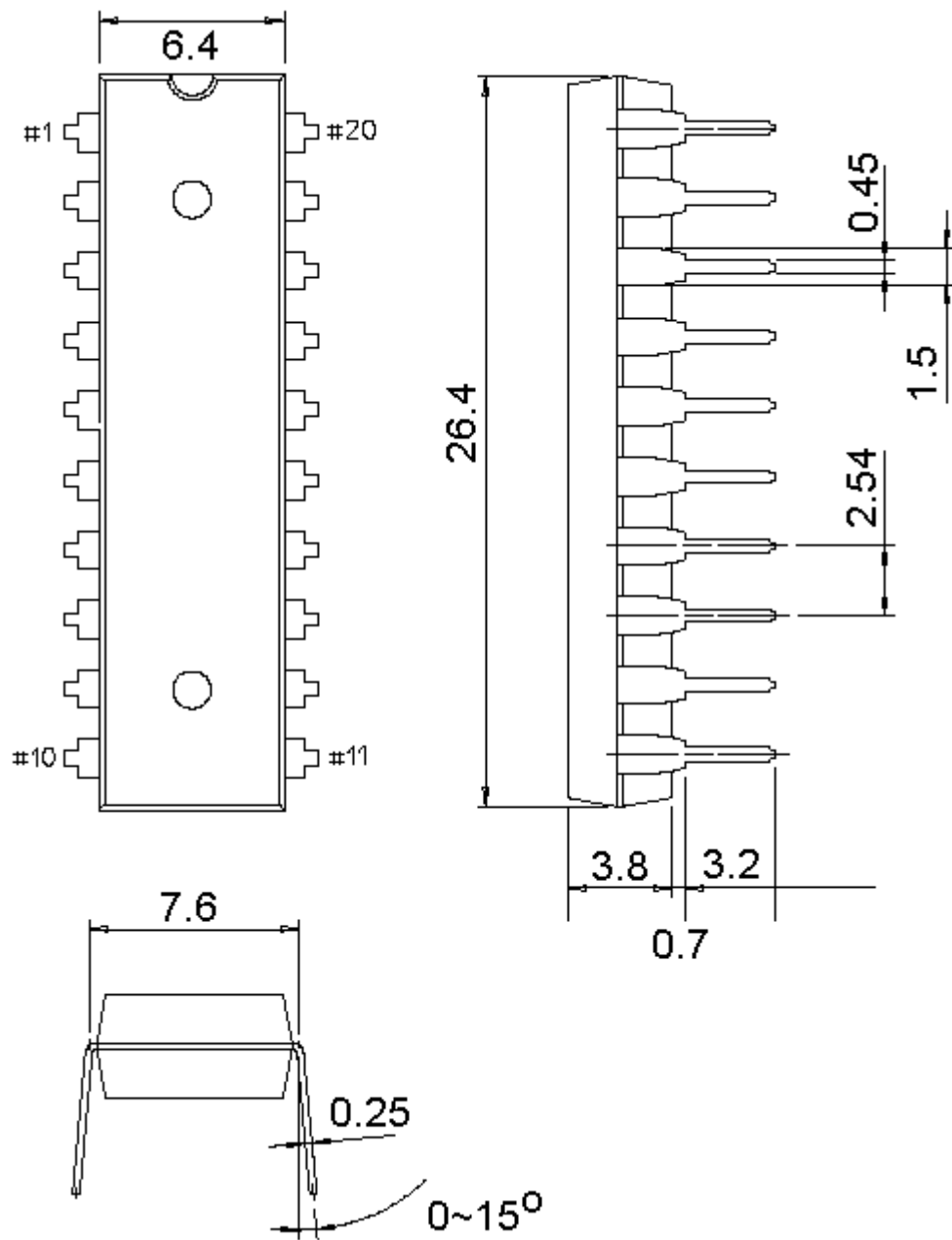
## . DIP16



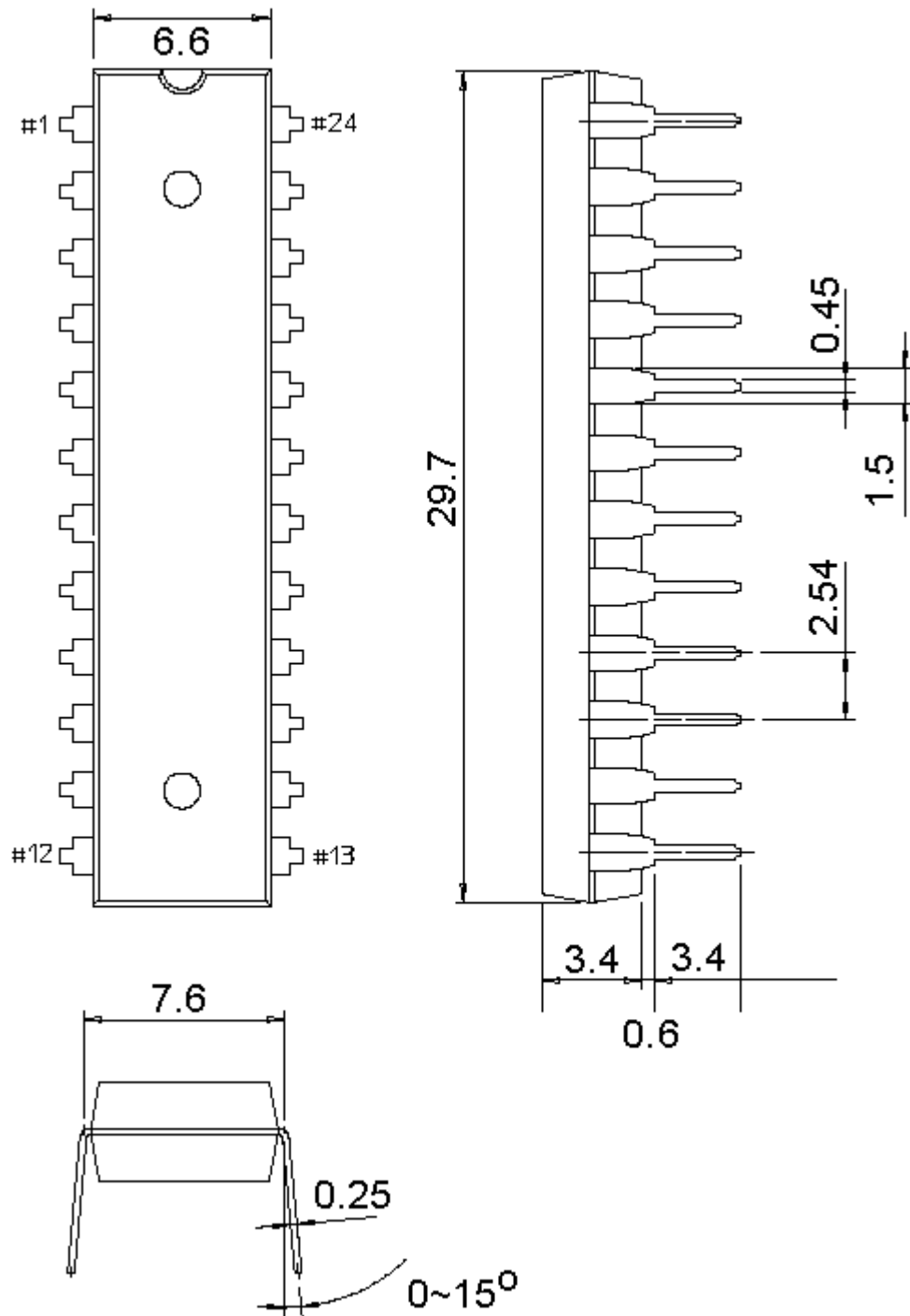
## . DIP18



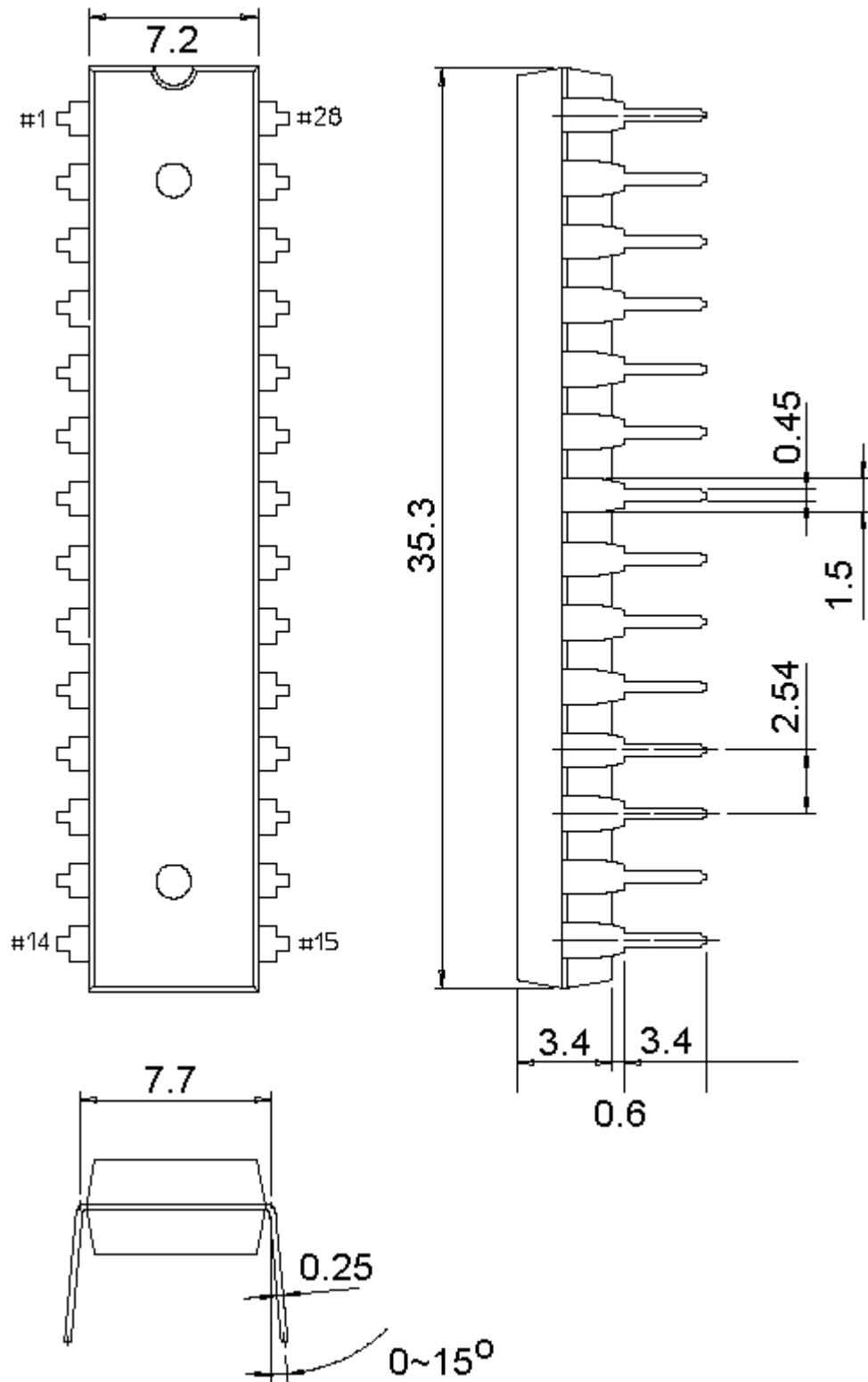
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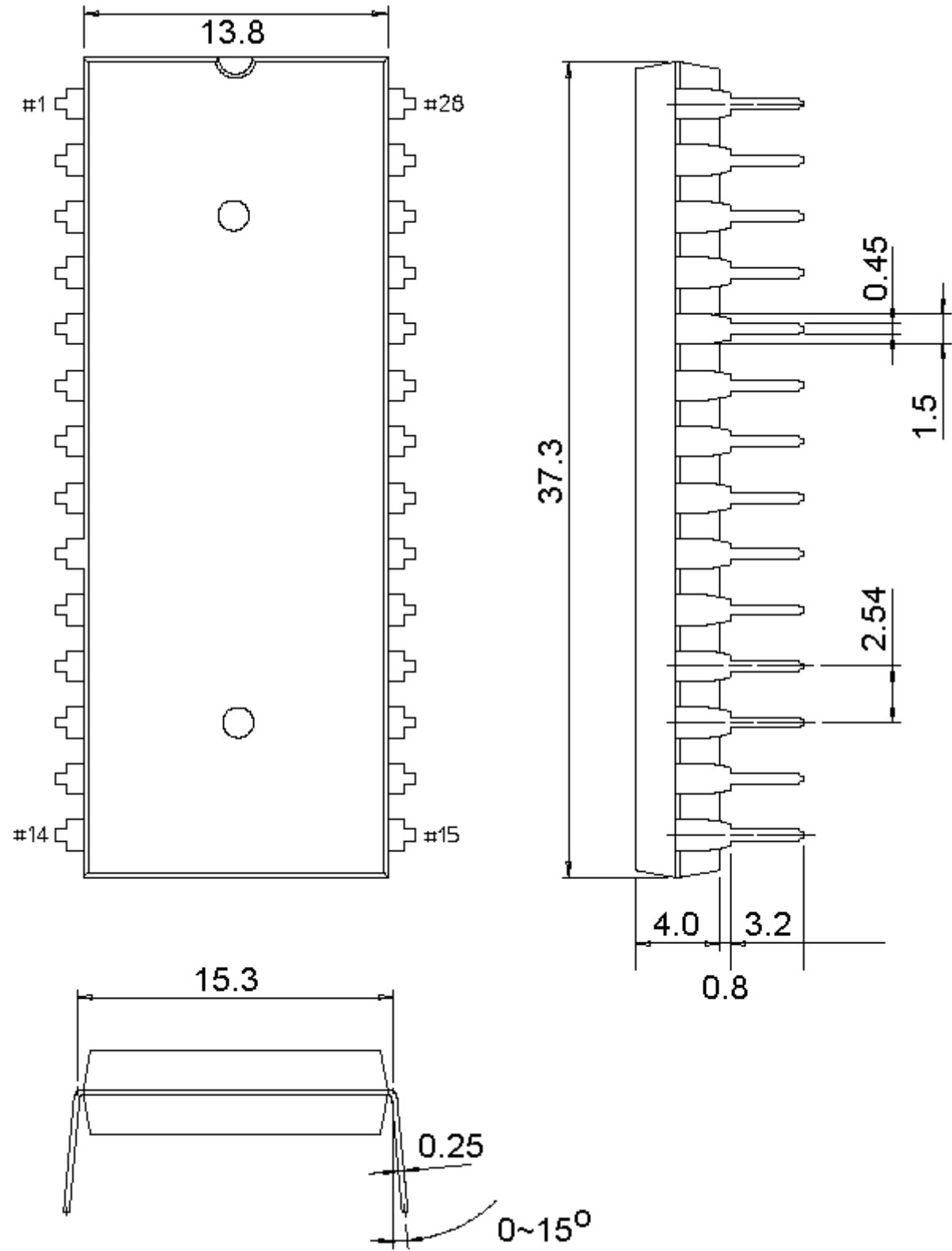
## . DIP24S (SKDIP24, DIP24-300mil)



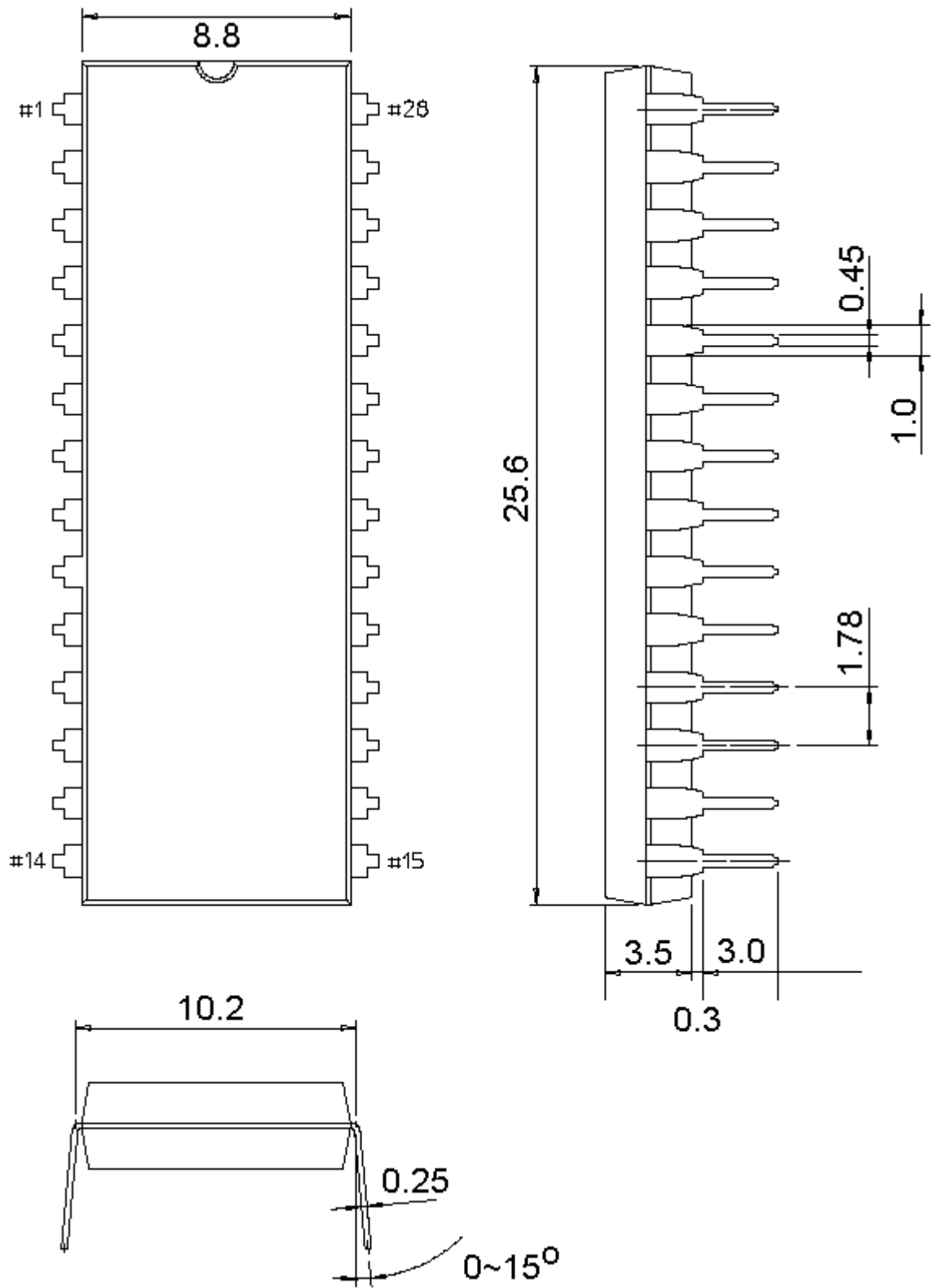
## . DIP28S (SKDIP28, DIP28-300mil)



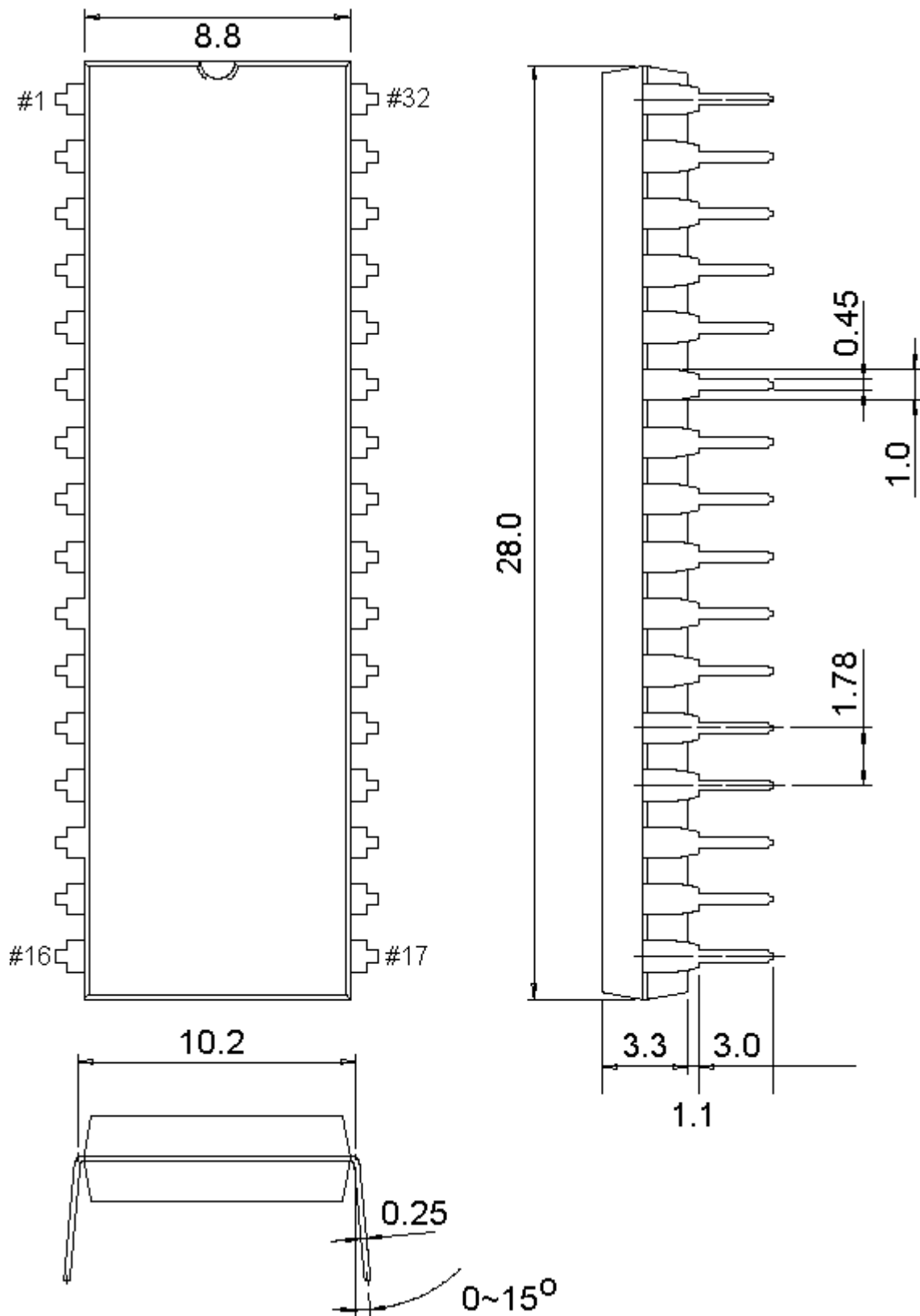
• DIP28 (DIP28W, DIP28-600mil)



• SDIP28 (SDIP28-400mil)



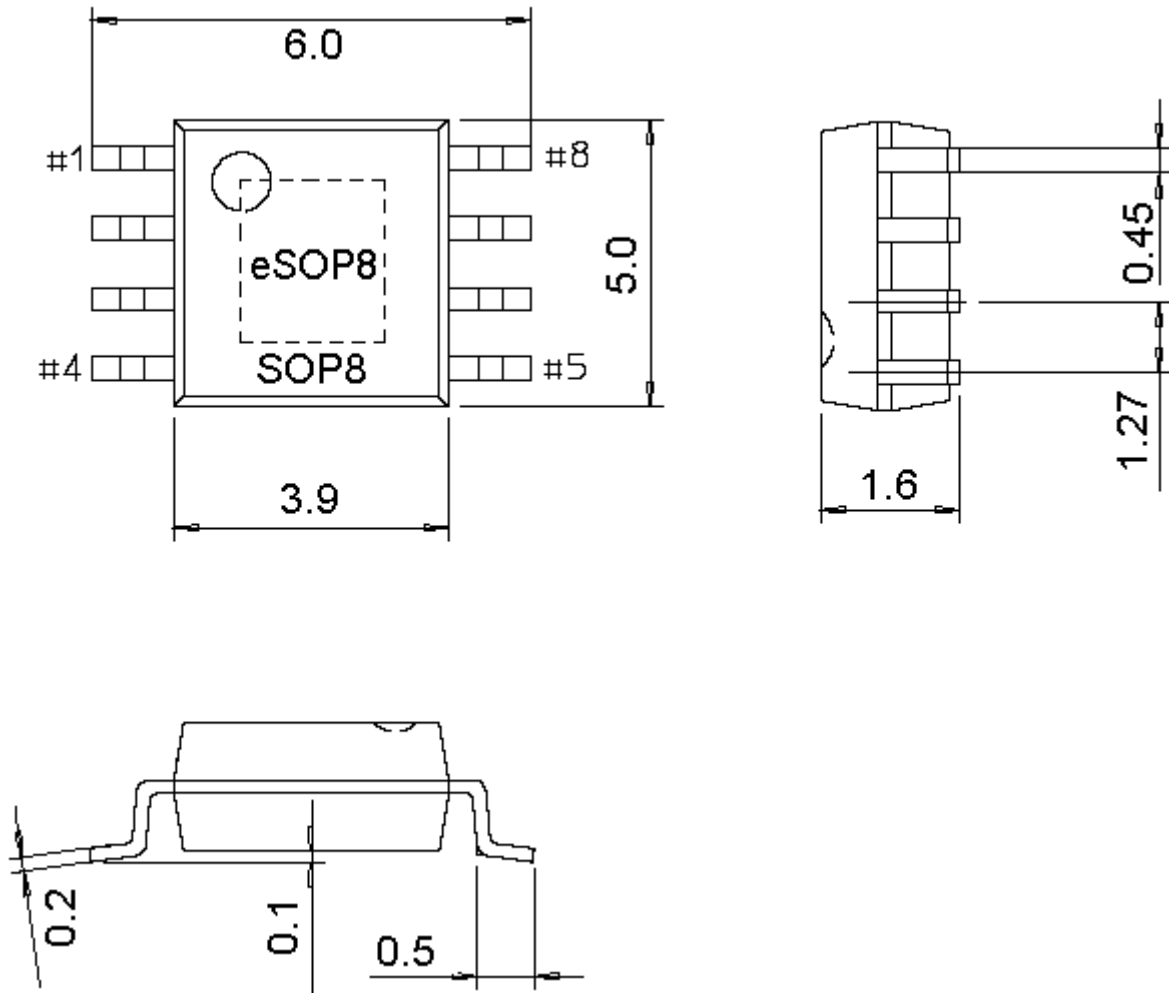
## . SDIP32 (SDIP32-400mil)



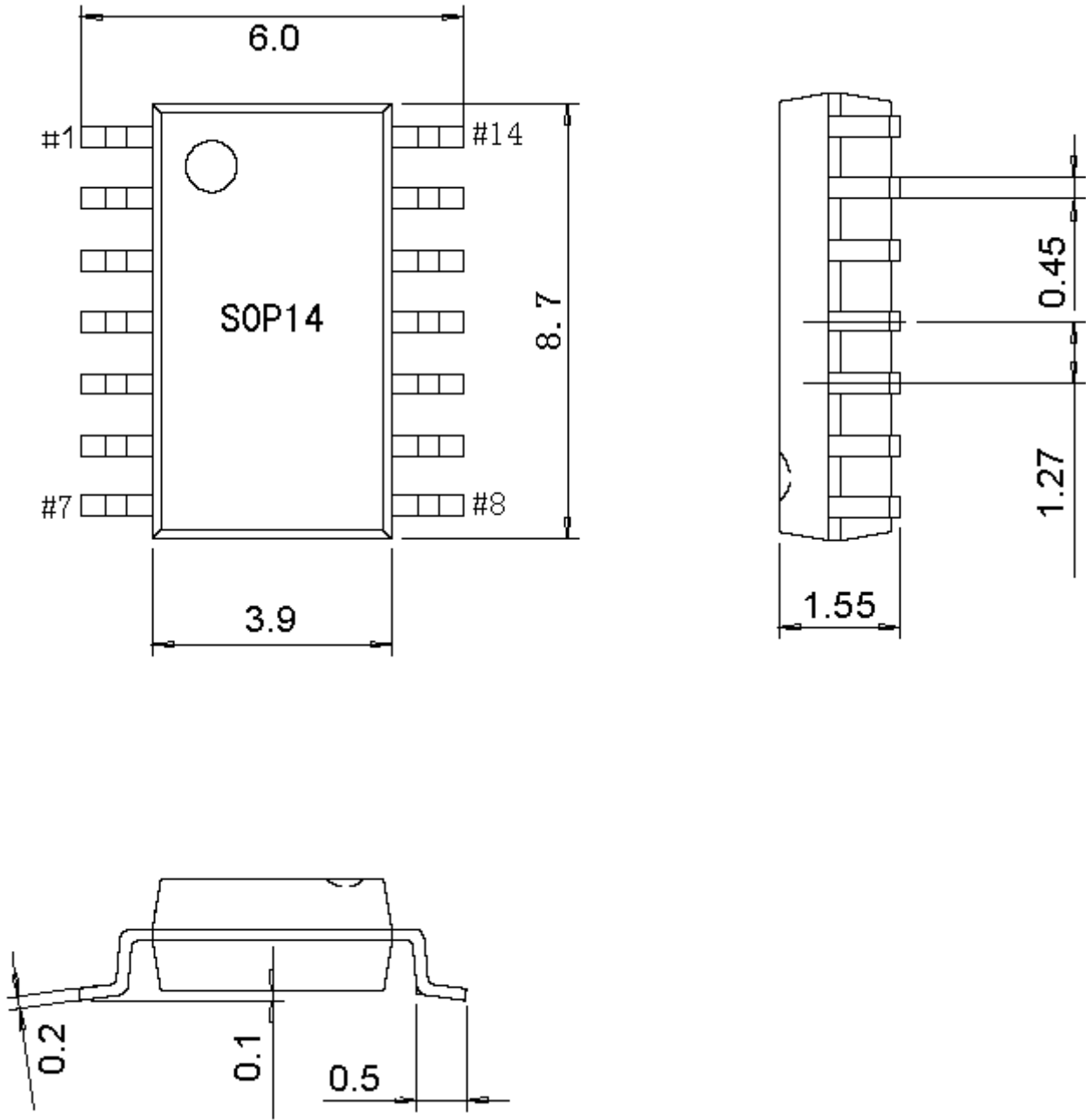
## SOP Type (SOIC)

### • SOP8, eSOP8

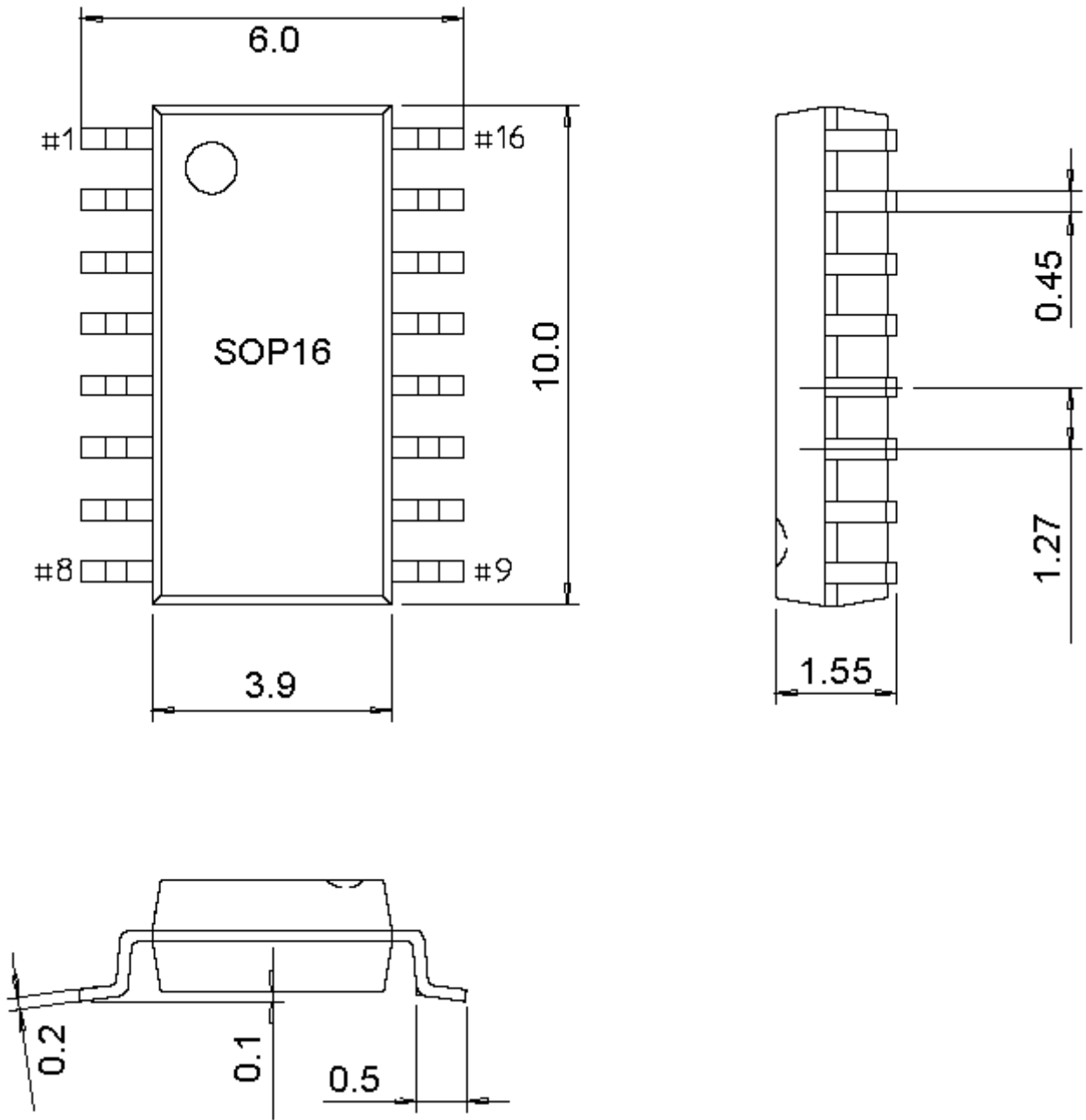
eSOP8 Exposed Pad: 2.1\*3.3, 2.1\*2.1



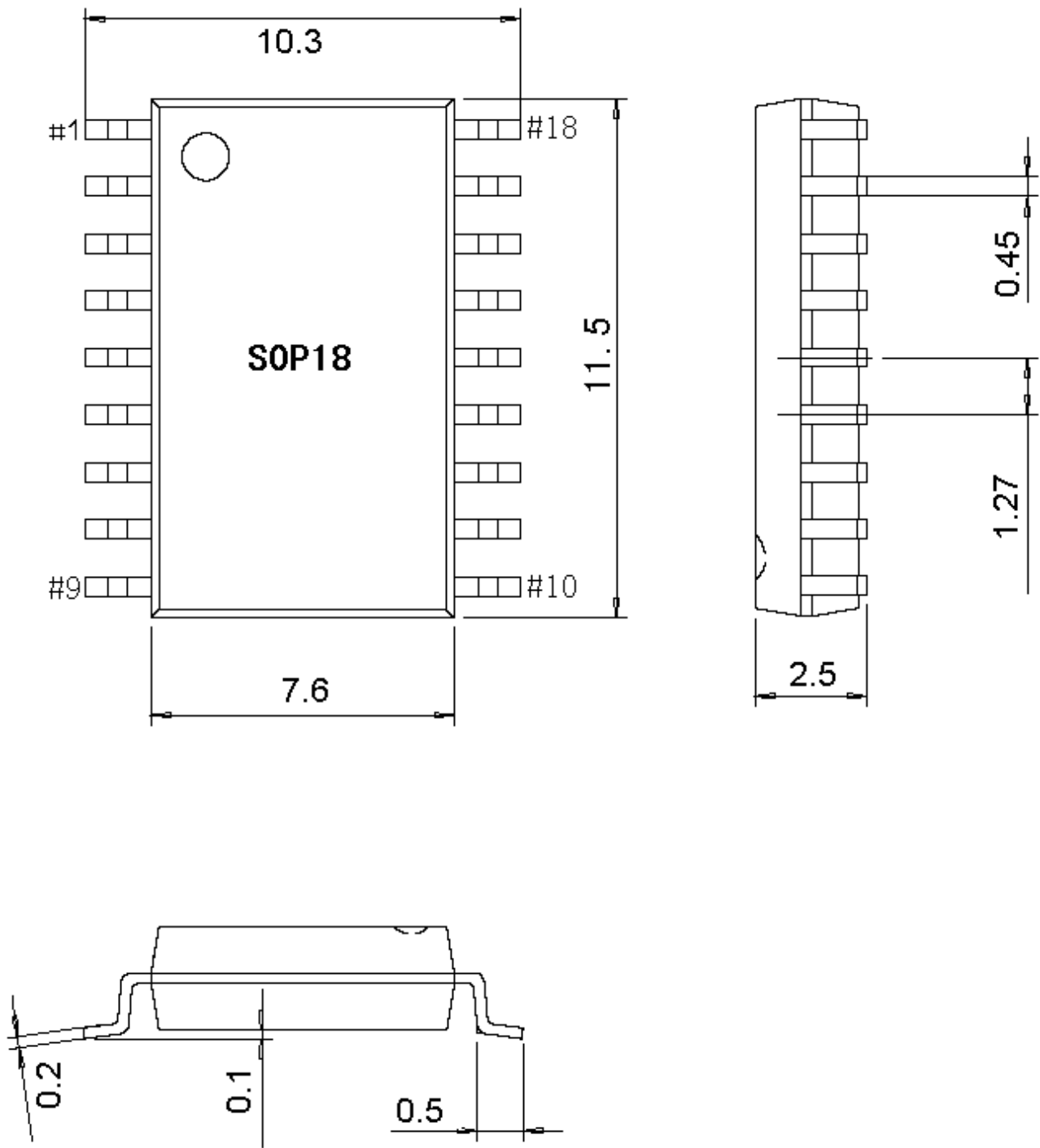
• SOP14



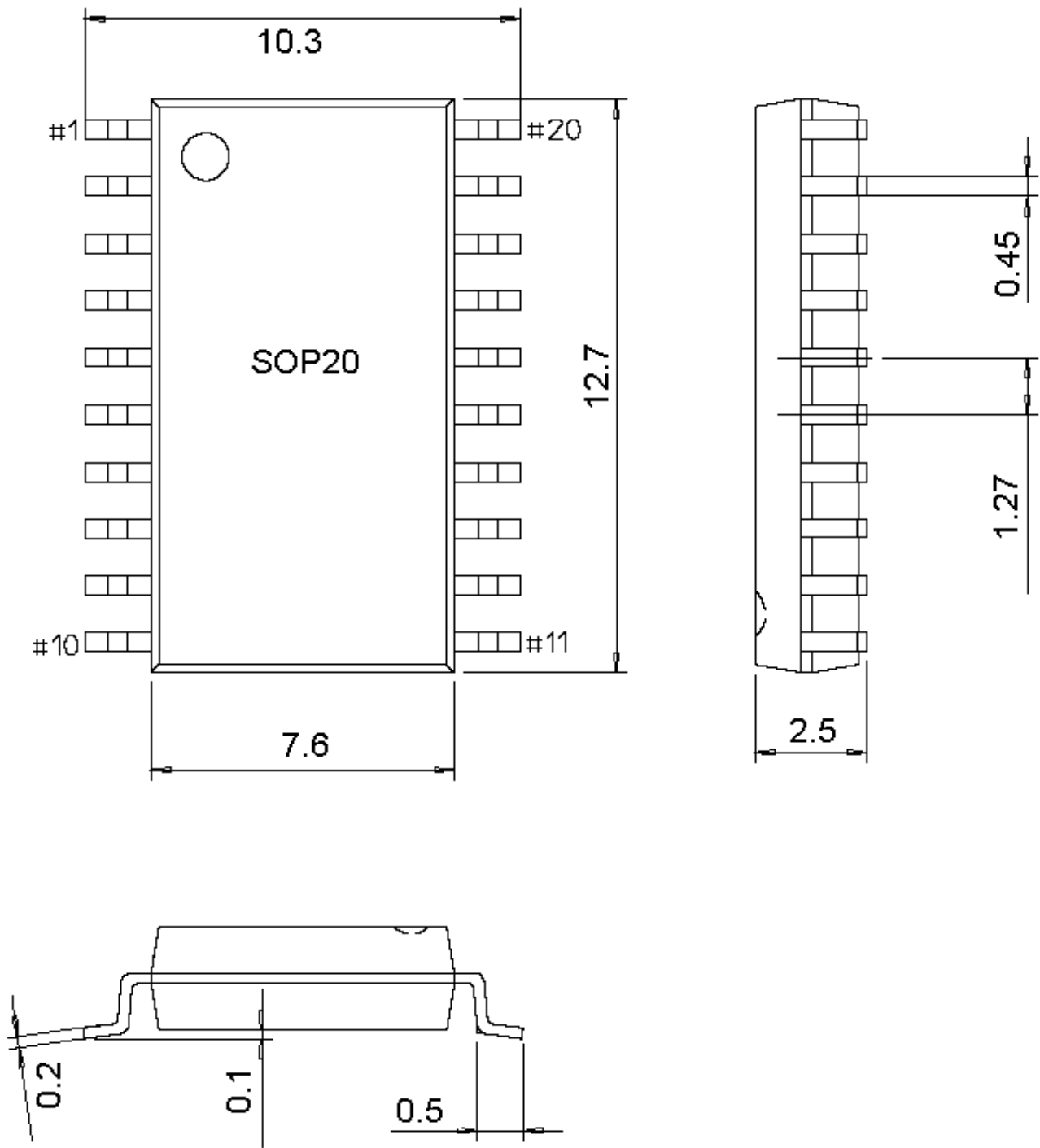
• SOP16 (SOP16-150mil)



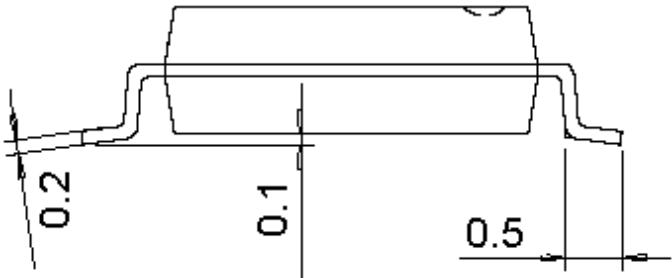
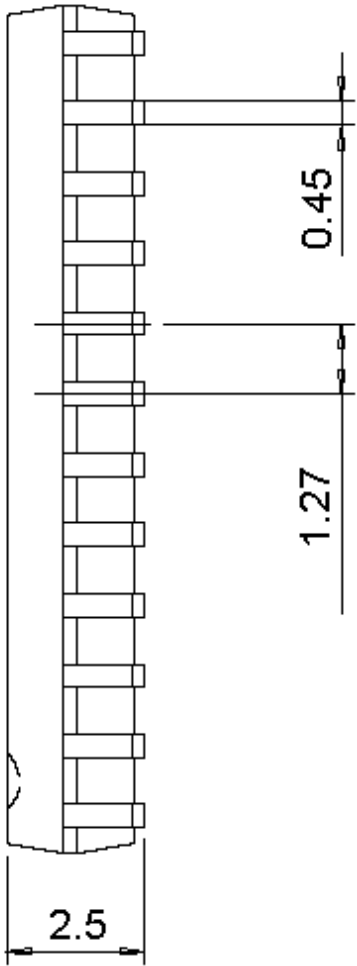
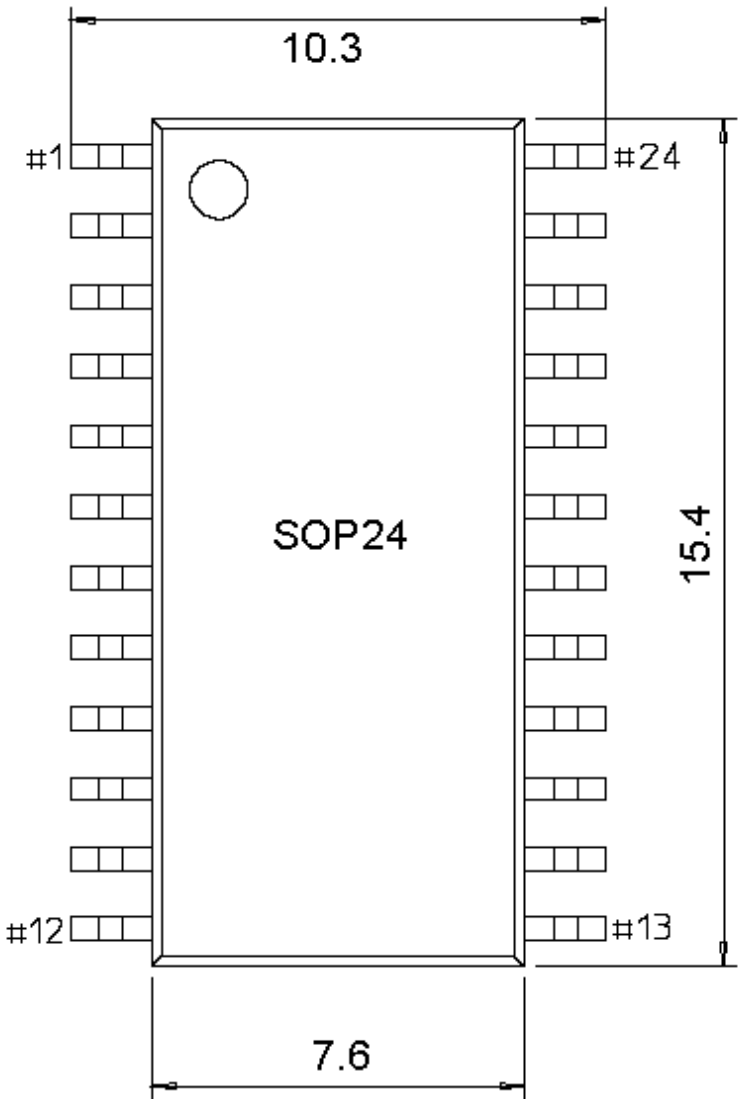
• SOP18



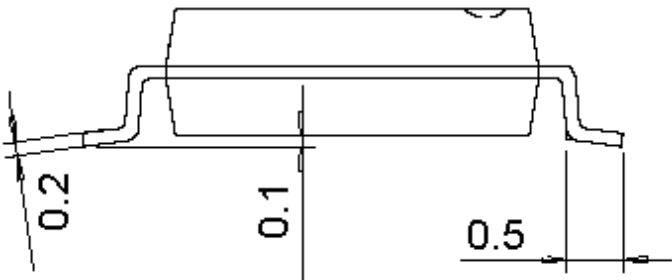
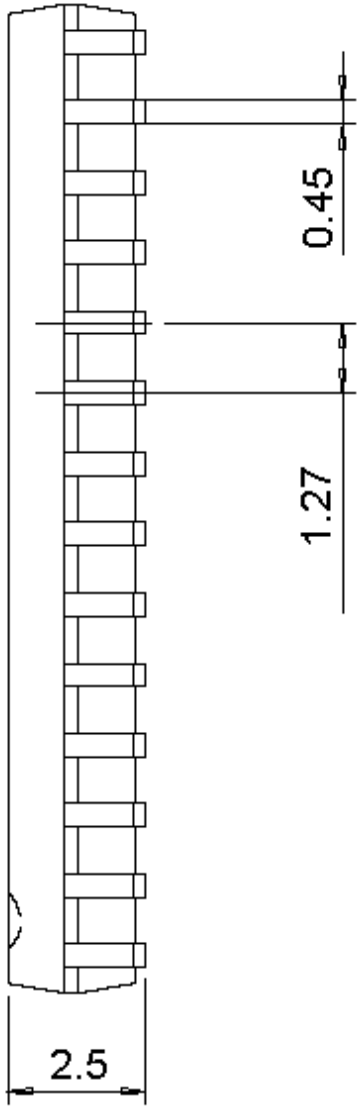
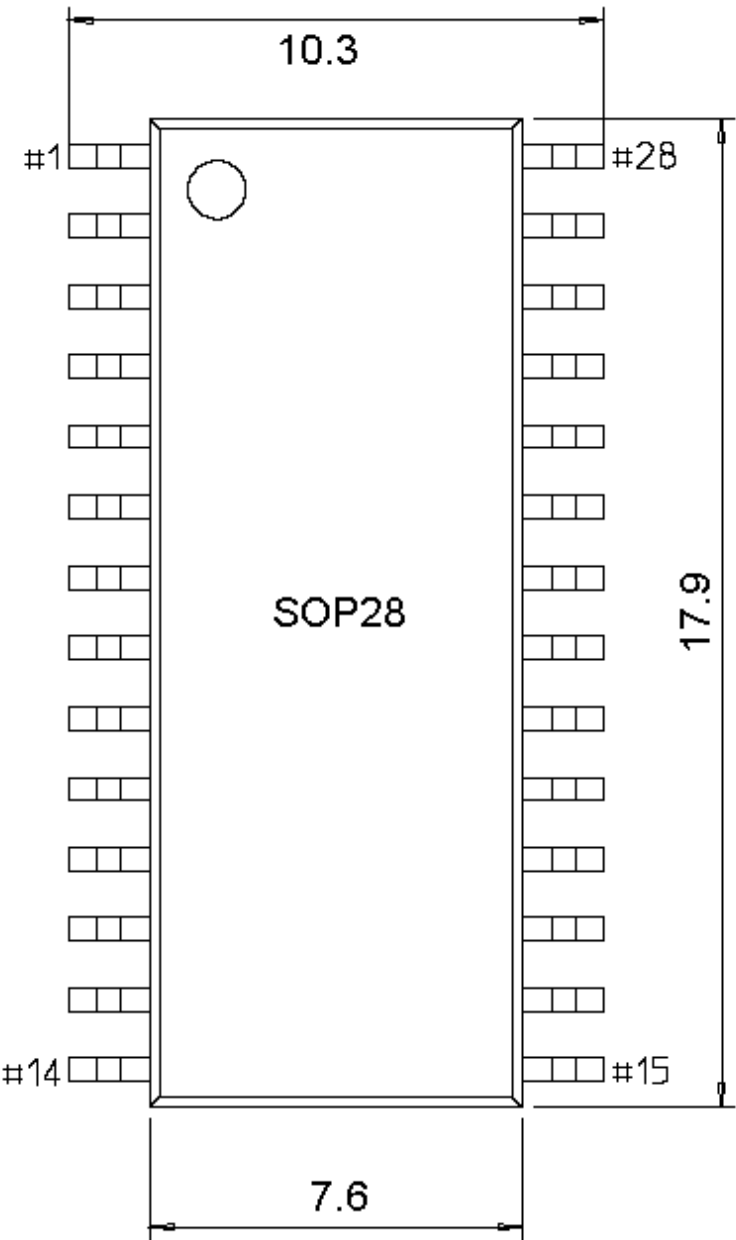
• SOP20



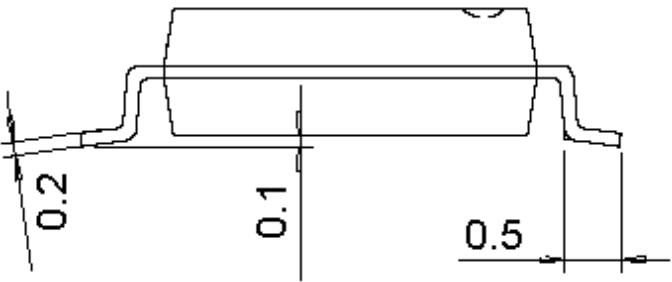
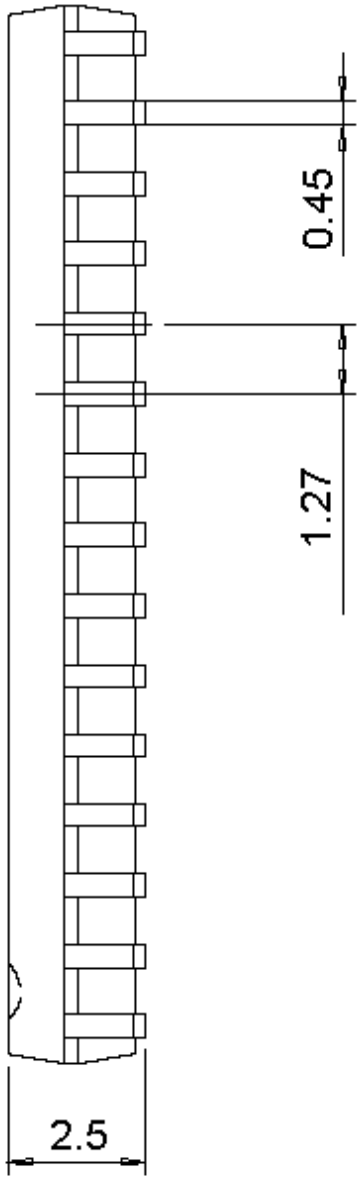
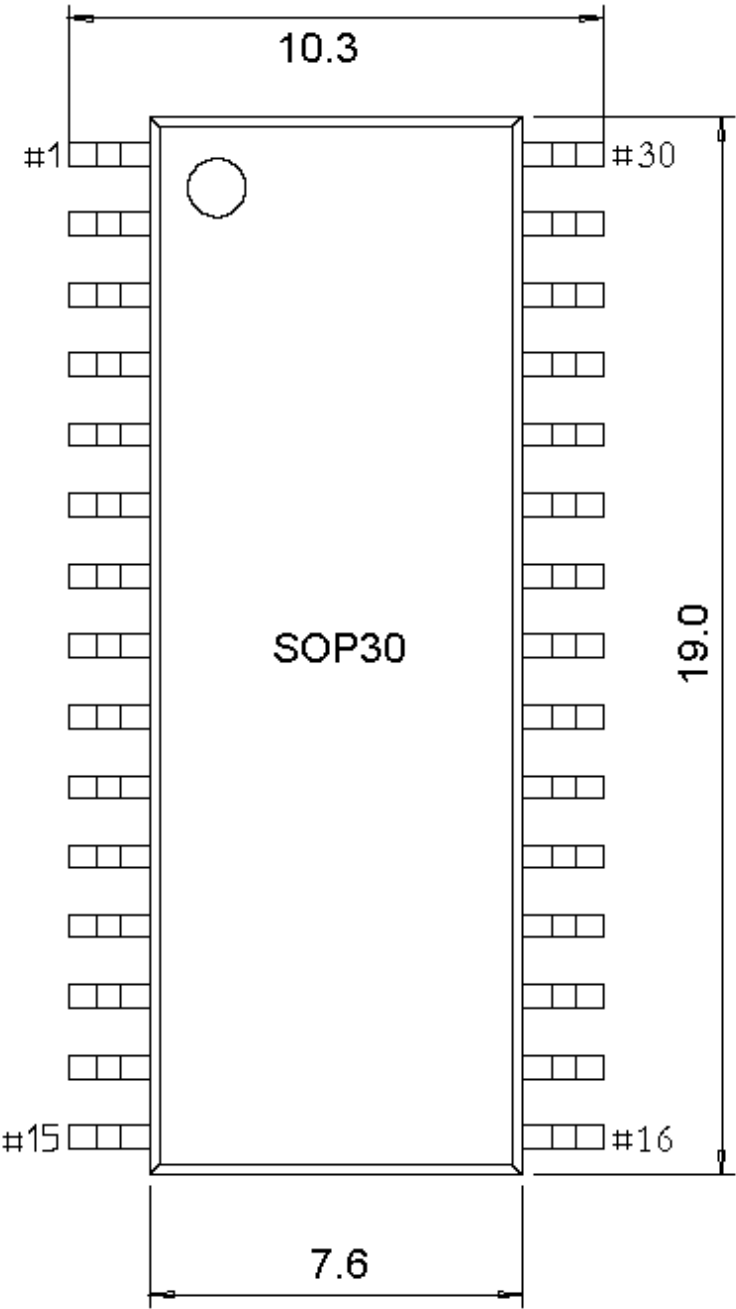
• SOP24



• SOP28



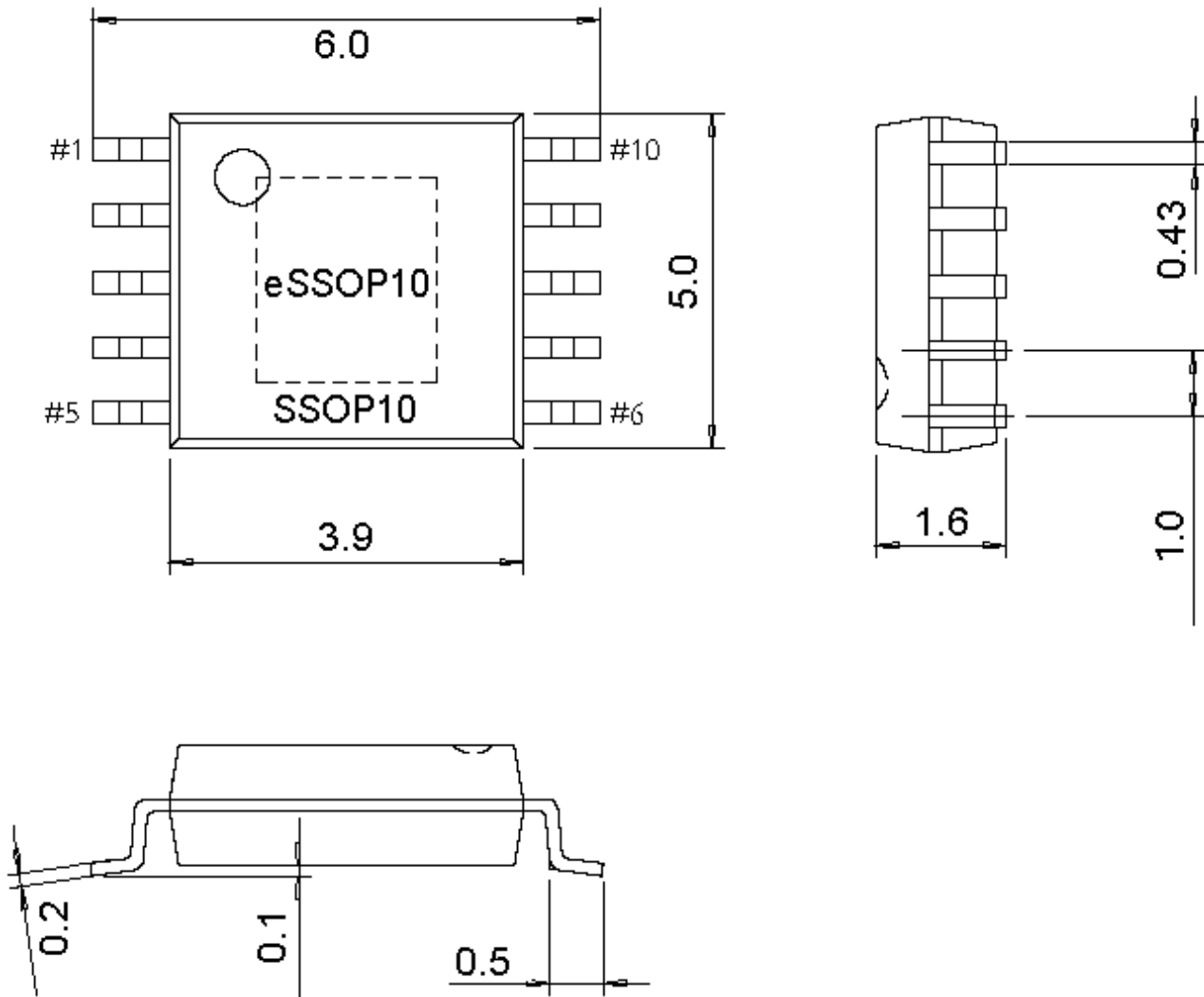
• SOP30



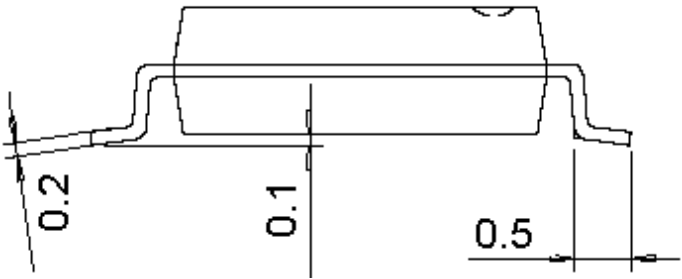
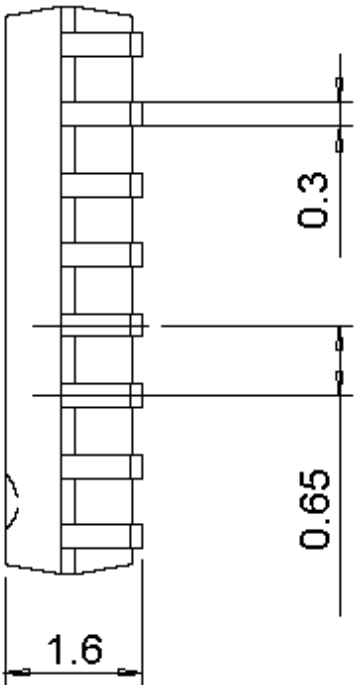
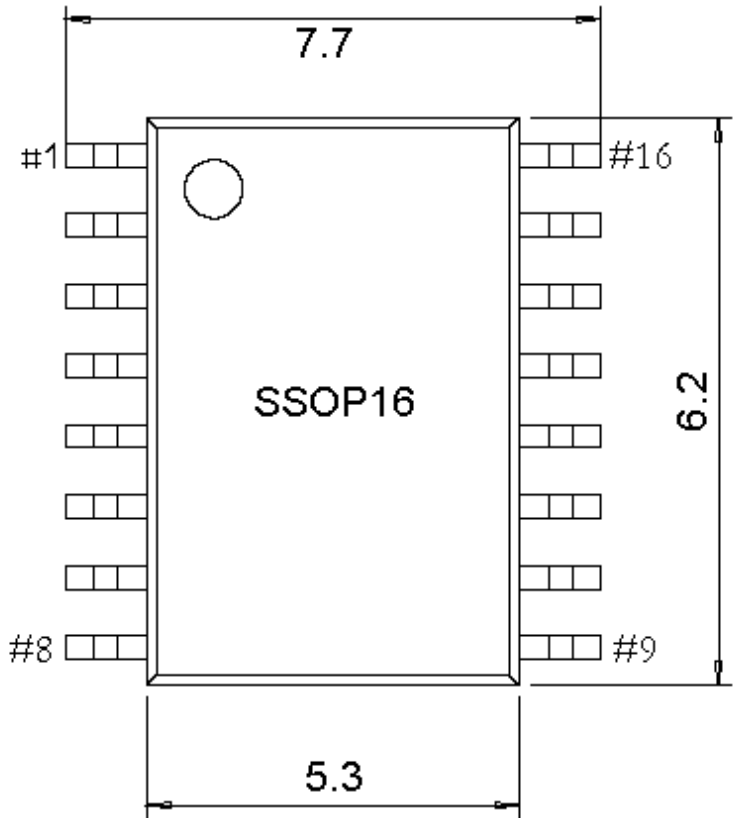
## SSOP Type

### • SSOP10 (SSOP10-150mil-1.00), eSSOP10

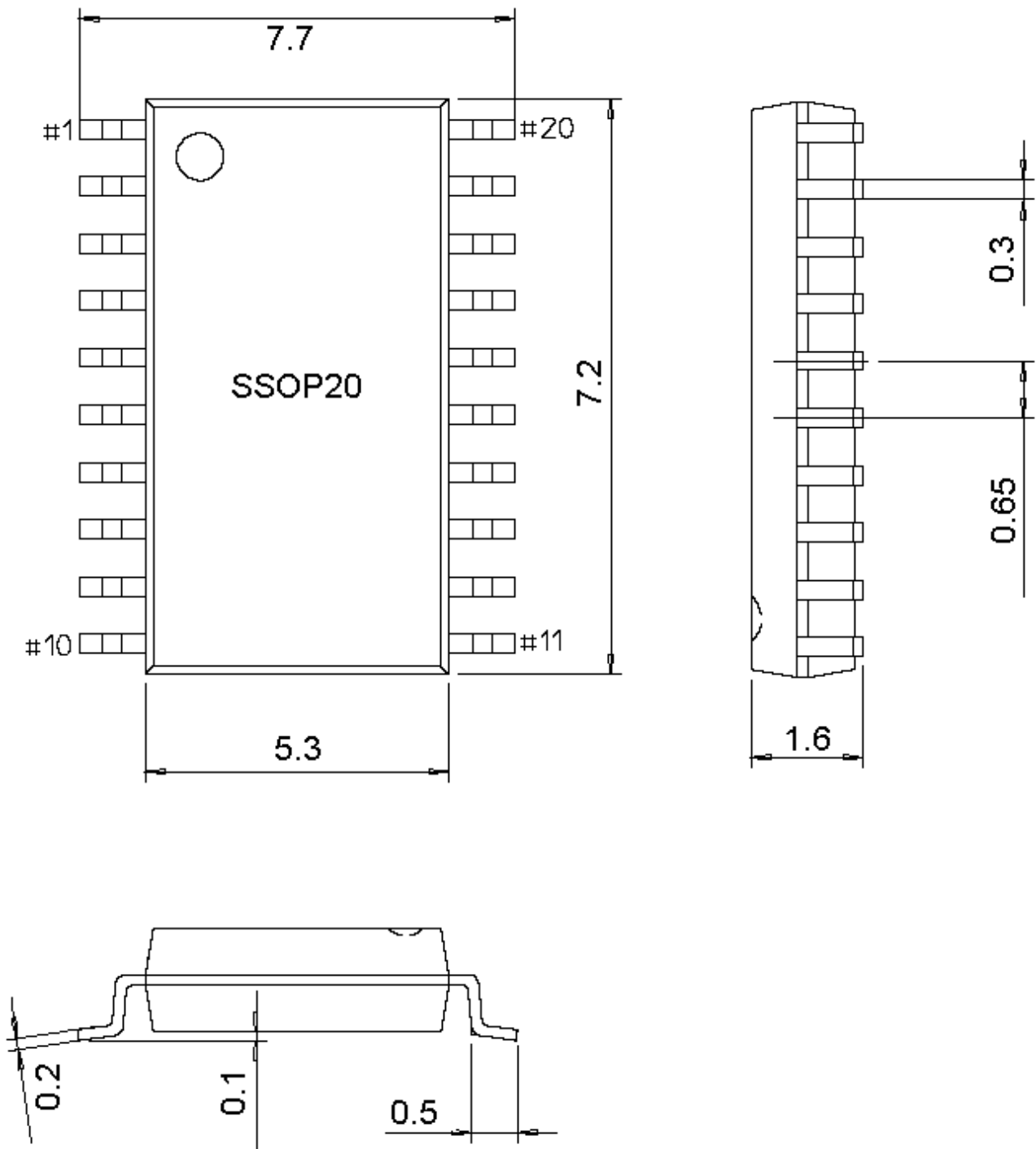
eSSOP10 Exposed Pad: 2.1\*3.3



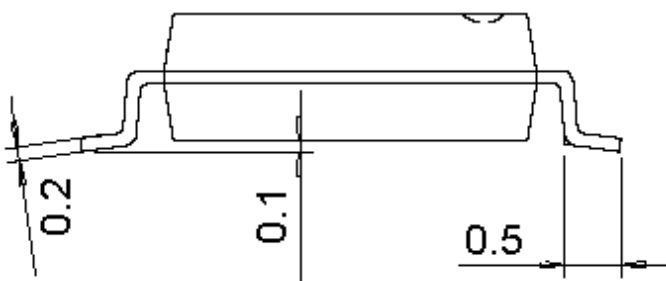
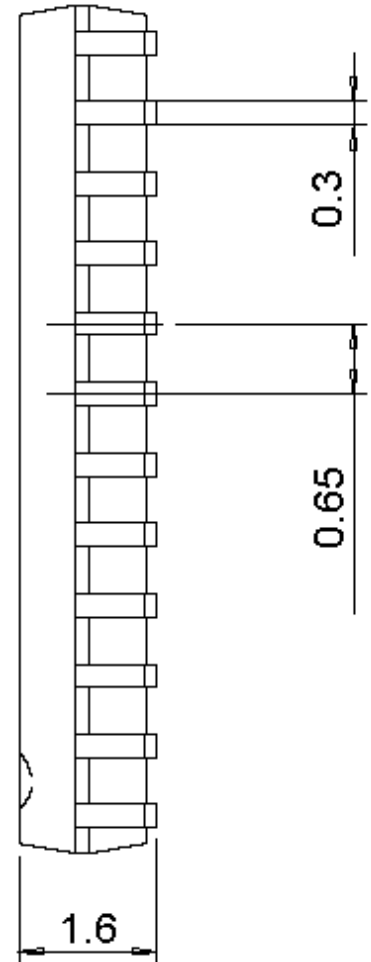
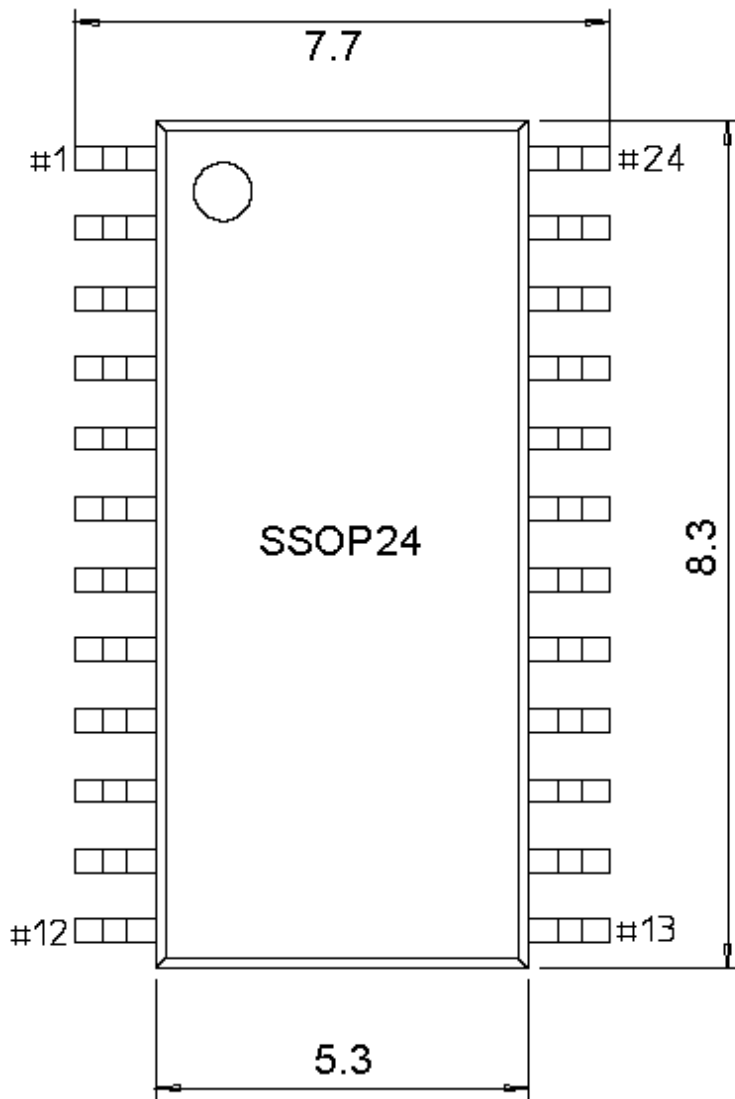
• SSOP16 (SSOP16-209mil-0.65)



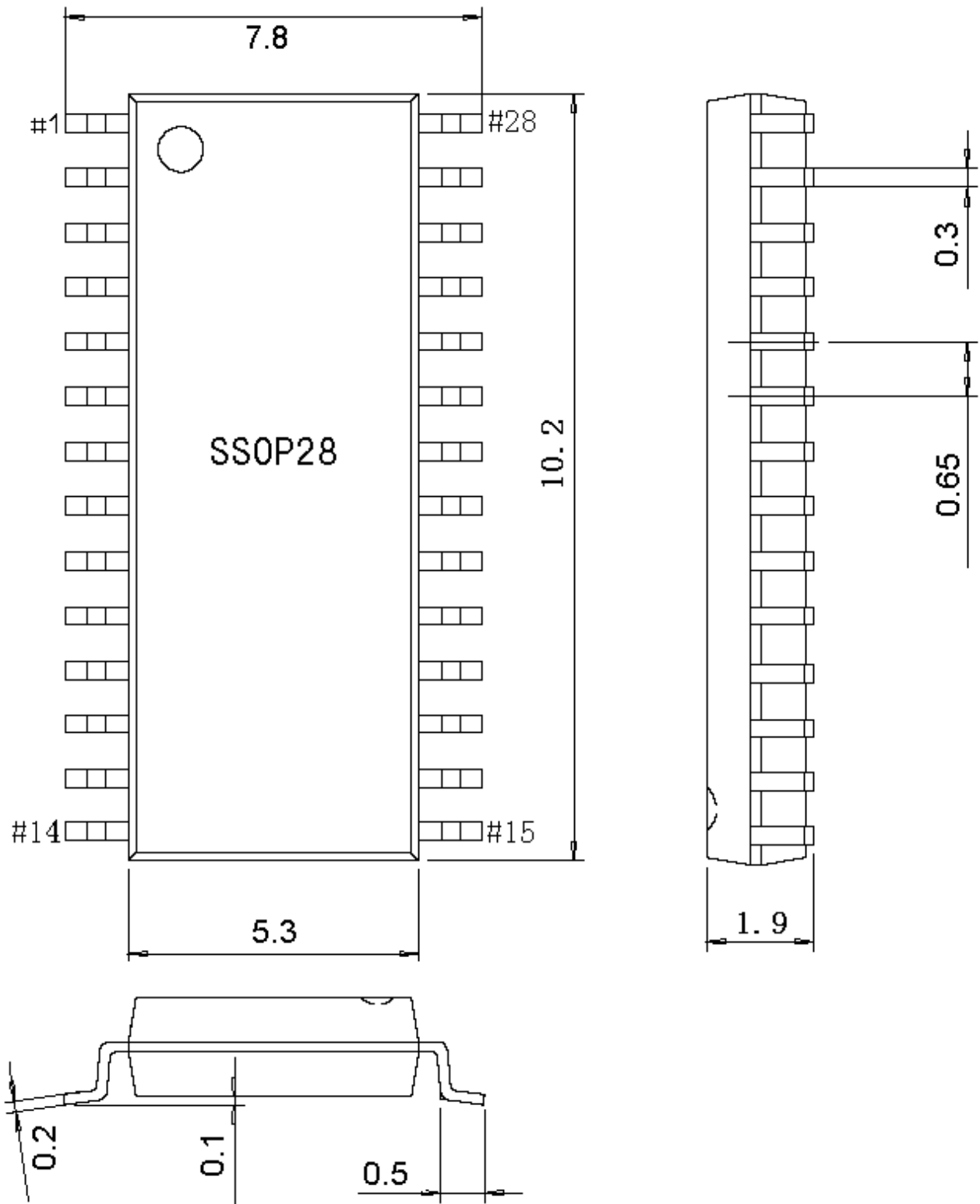
• SSOP20 (SSOP20-209mil-0.65)



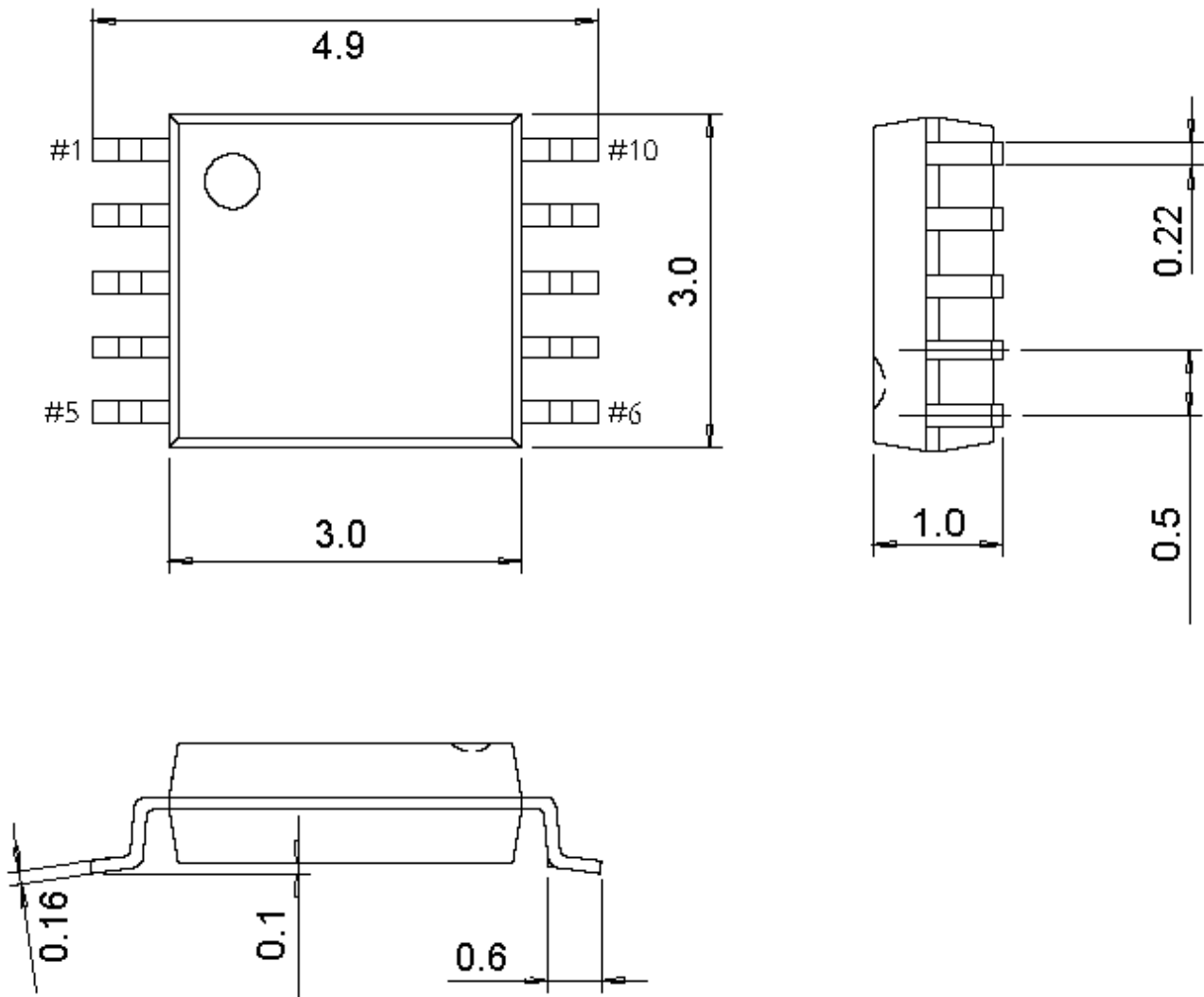
## . SSOP24 (SSOP24-209mil-0.65)



## . SSOP28 (SSOP28-209mil-0.65)

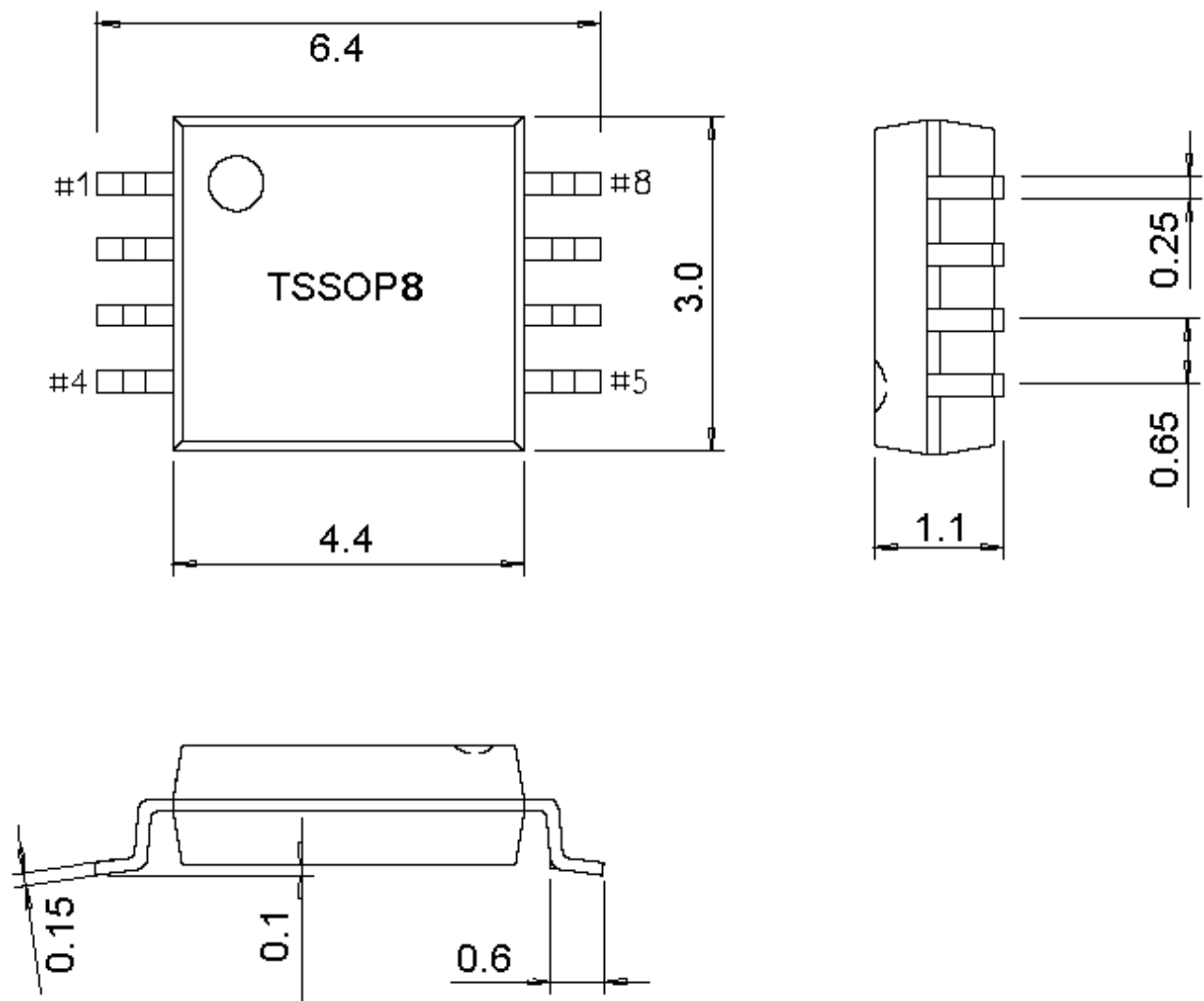


**MSOP Type**  
  **MSOP10**



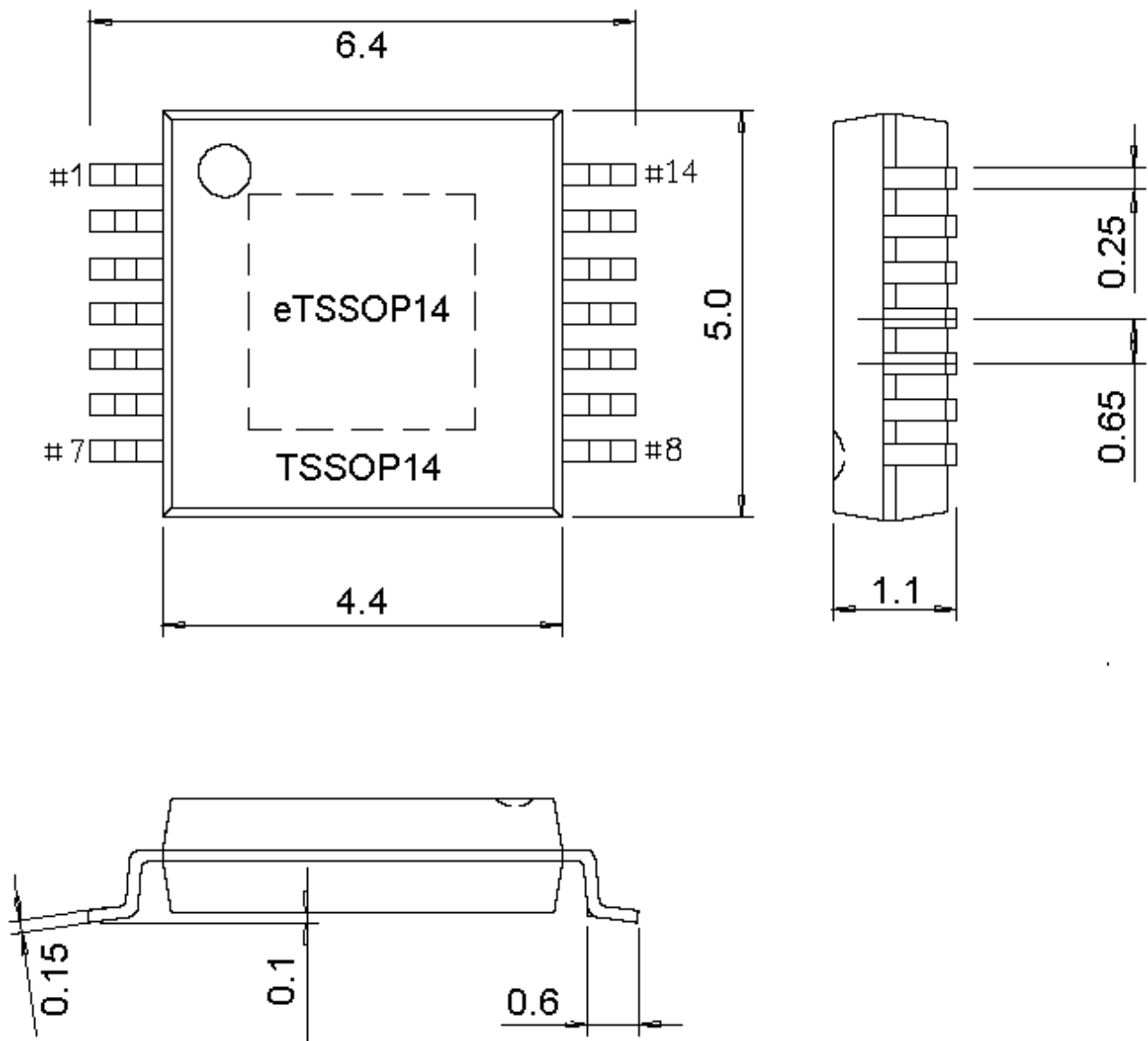
**TSSOP Type(4.4mm, 173mil)**

**· TSSOP8**



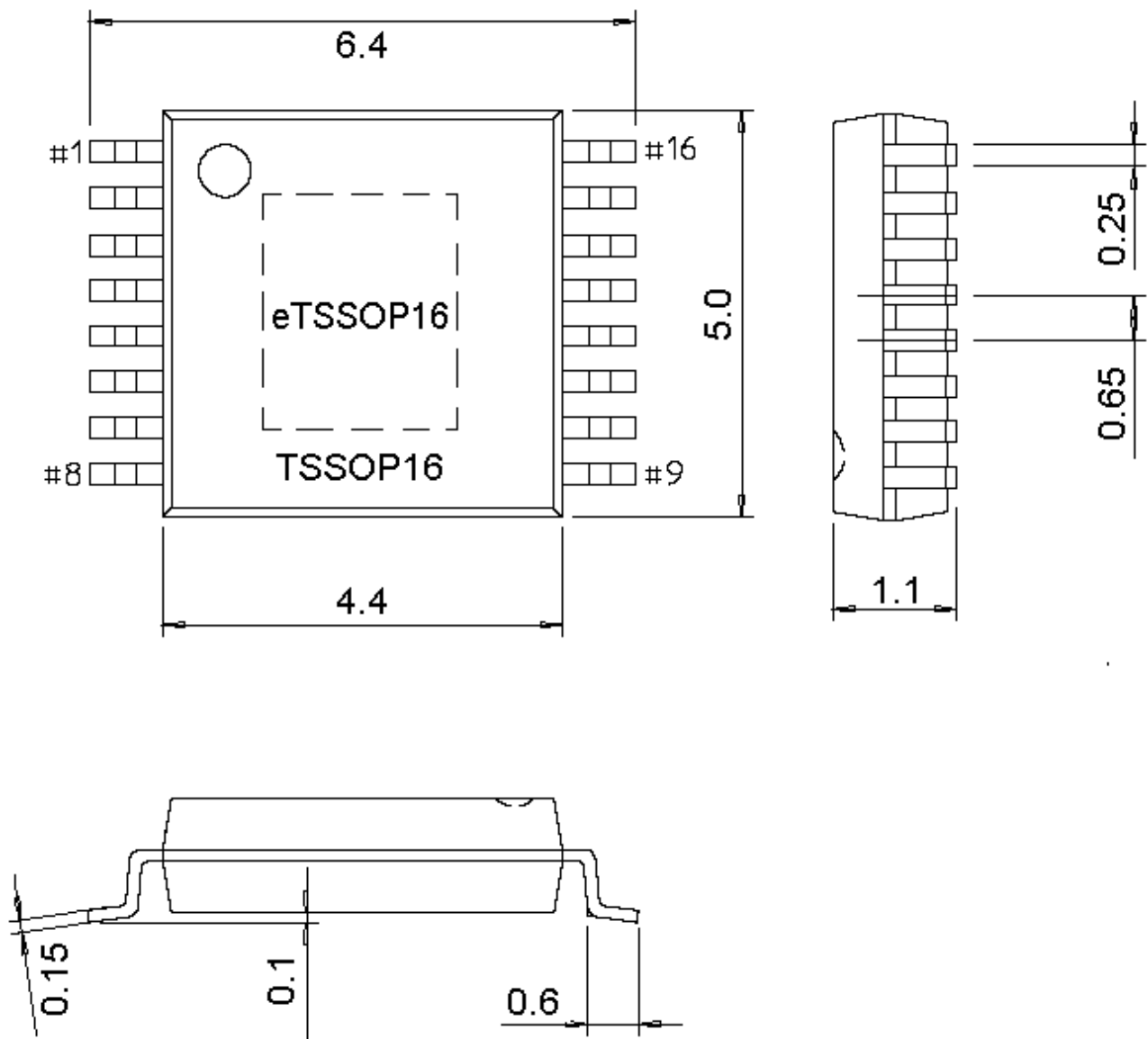
**. TSSOP14, eTSSOP14**

eTSSOP14 Exposed Pad: 3.0\*3.0



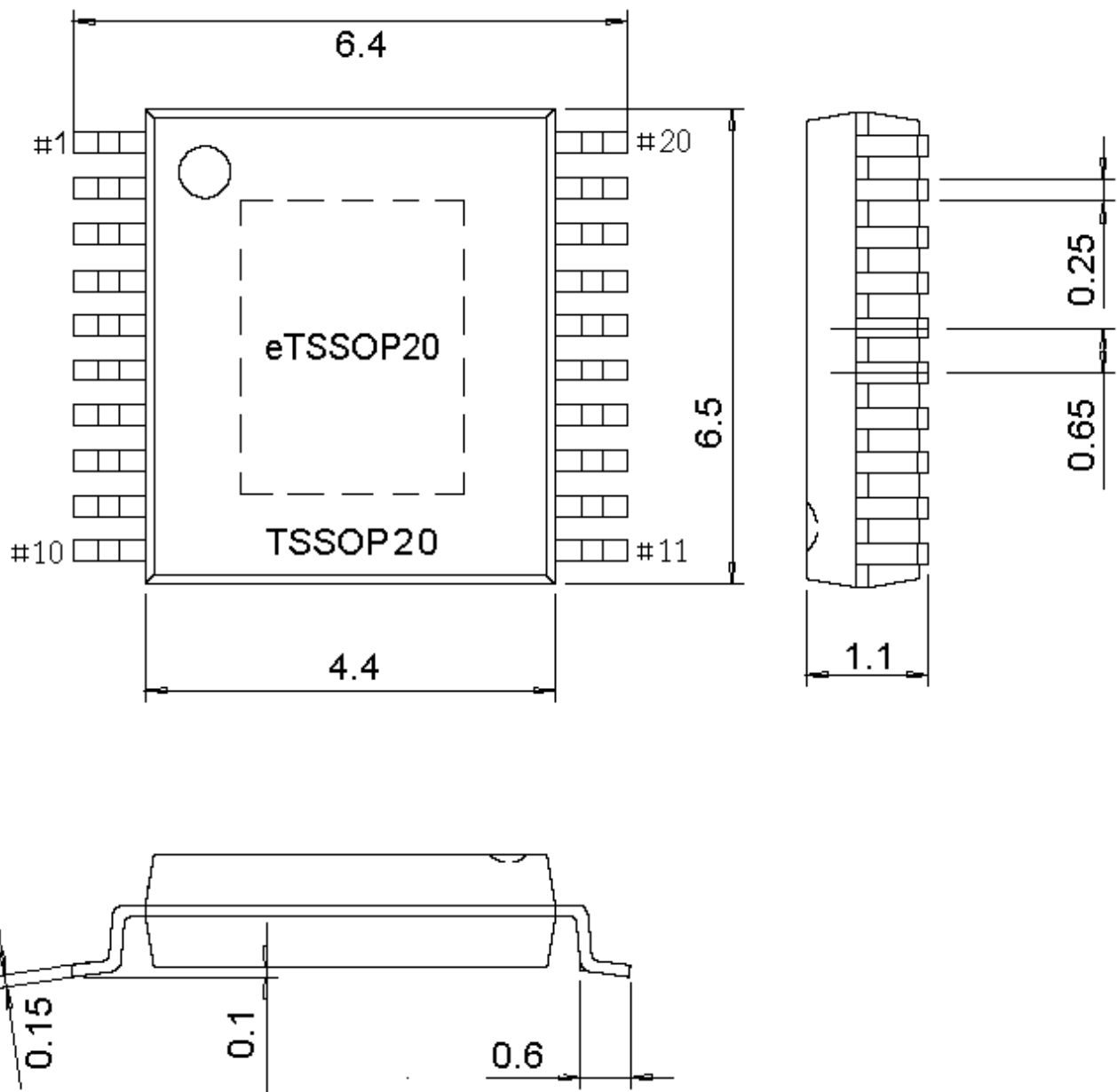
**. TSSOP16, eTSSOP16**

eTSSOP16 Exposed Pad: 2.8\*3.0

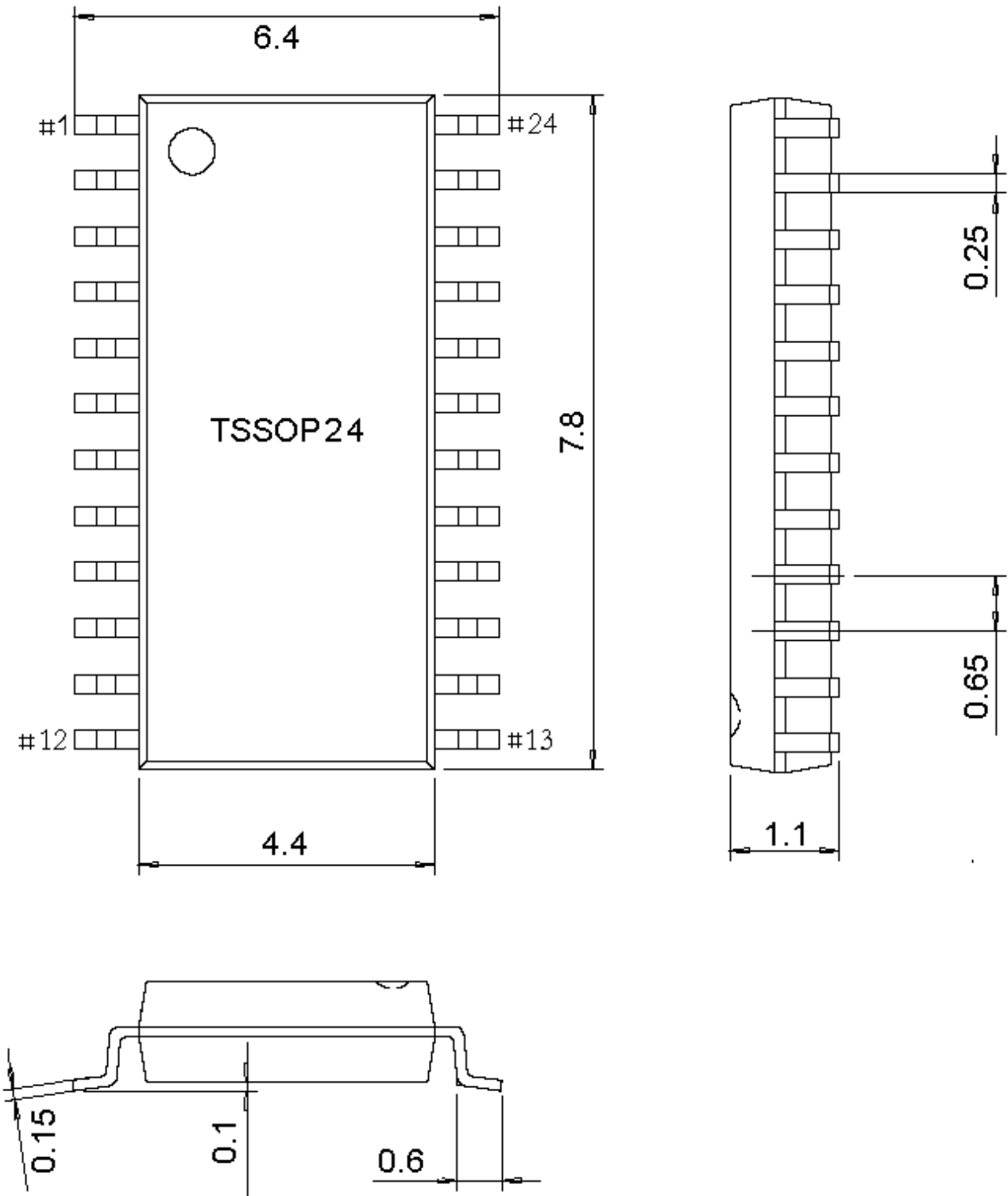


**. TSSOP20, eTSSOP20**

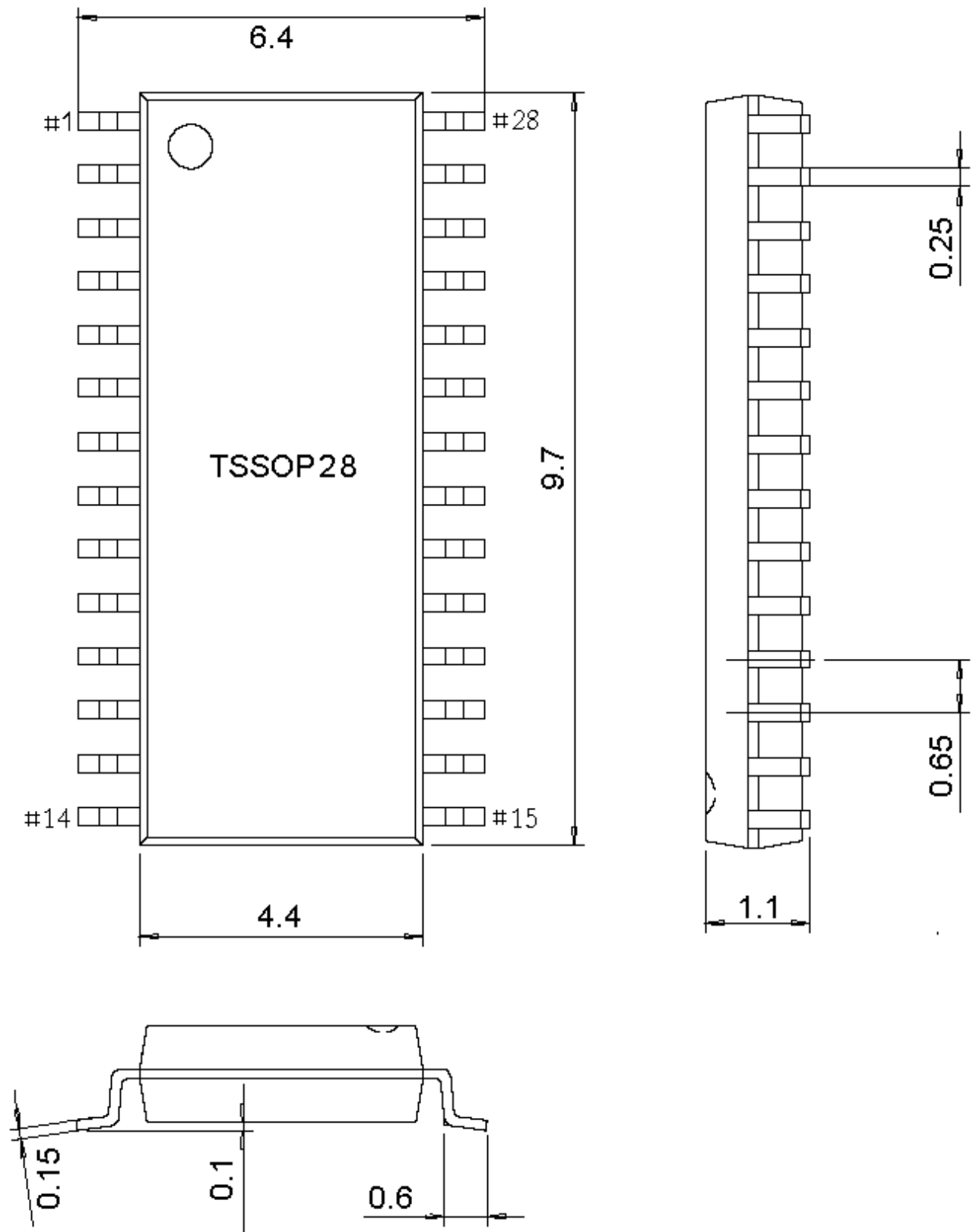
eTSSOP20 Exposed Pad: 3.0\*4.2



· TSSOP24

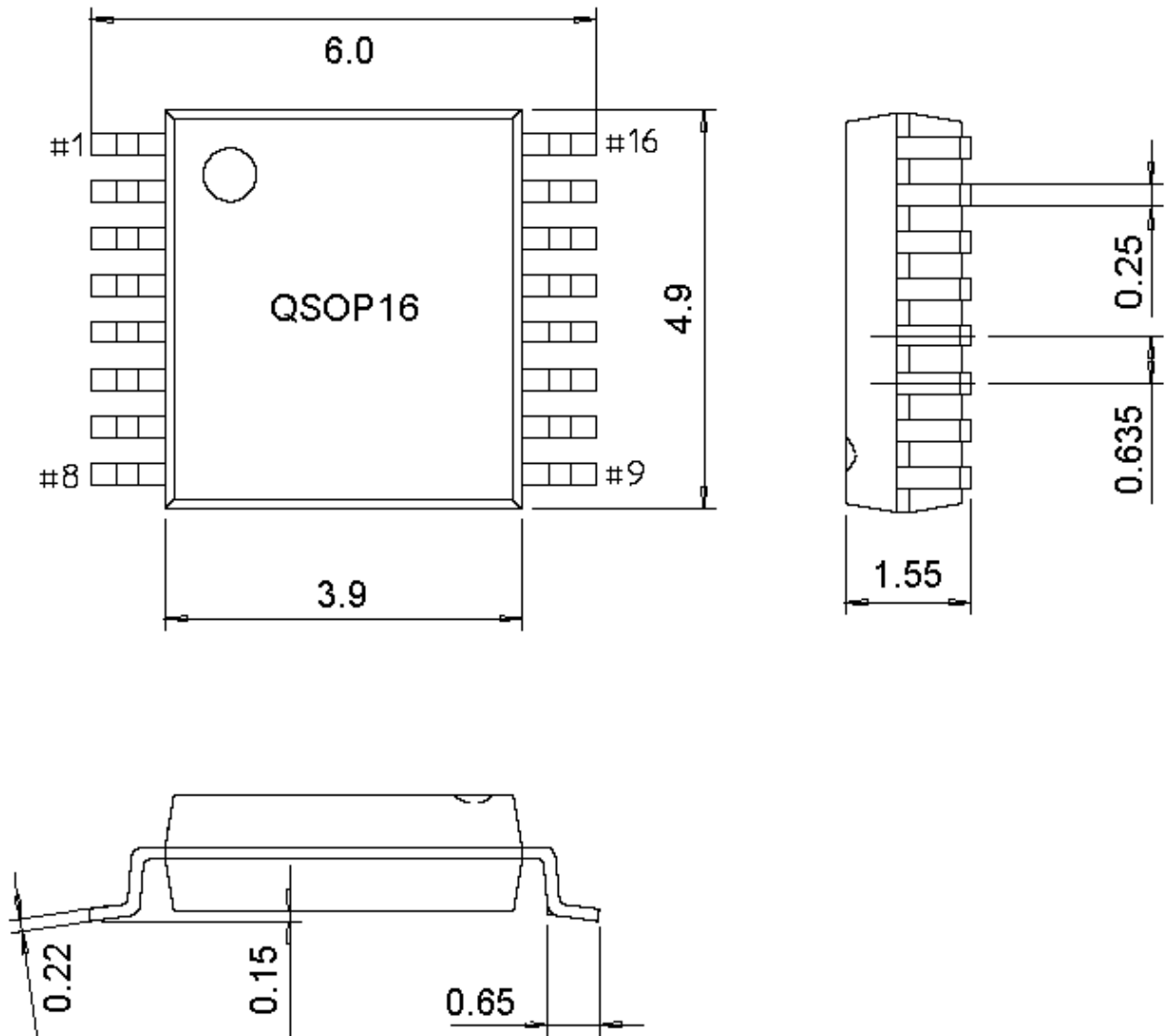


· TSSOP28

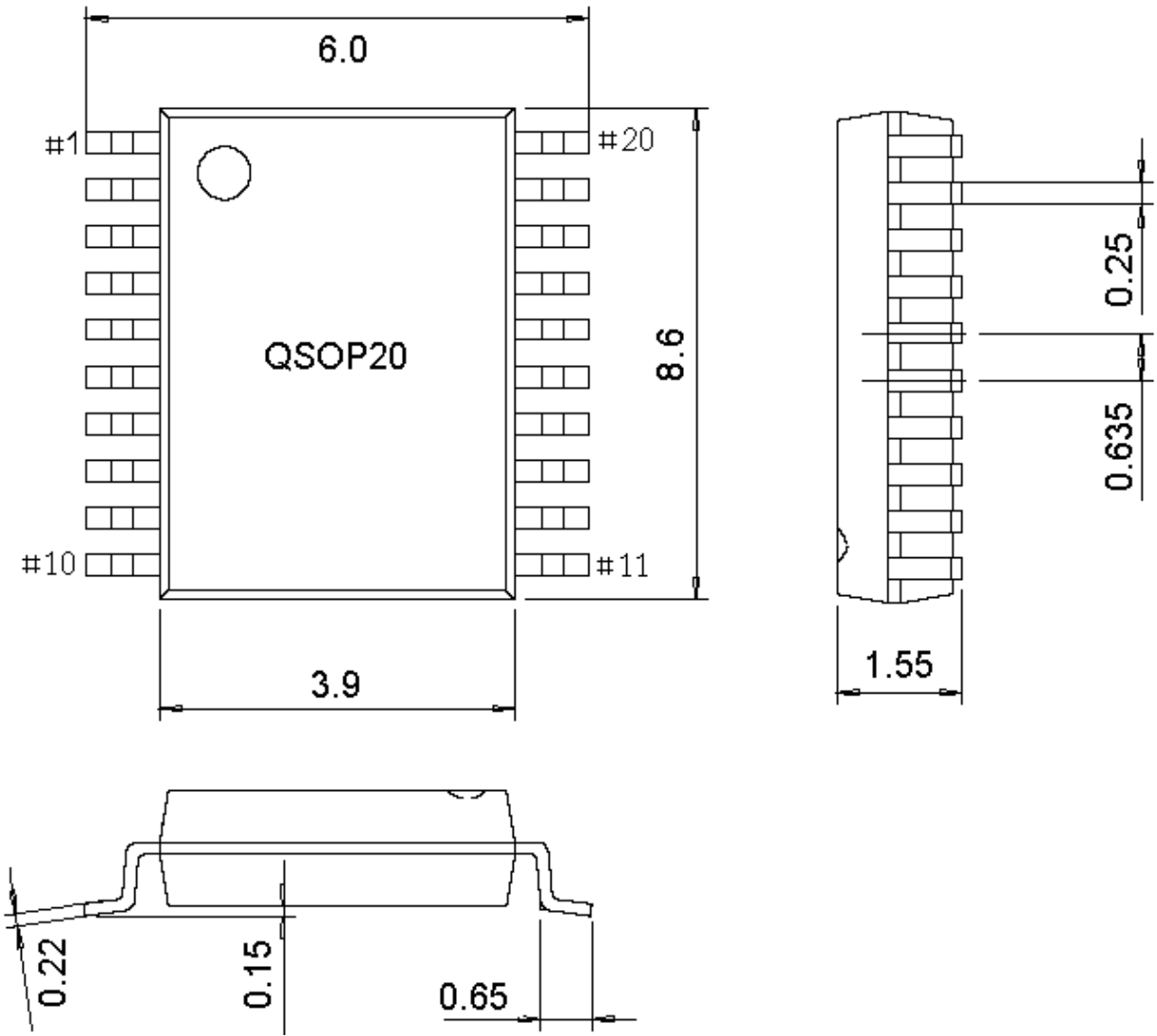


**QSOP Type (3.9mm, 150mil)(SSOP-150)**

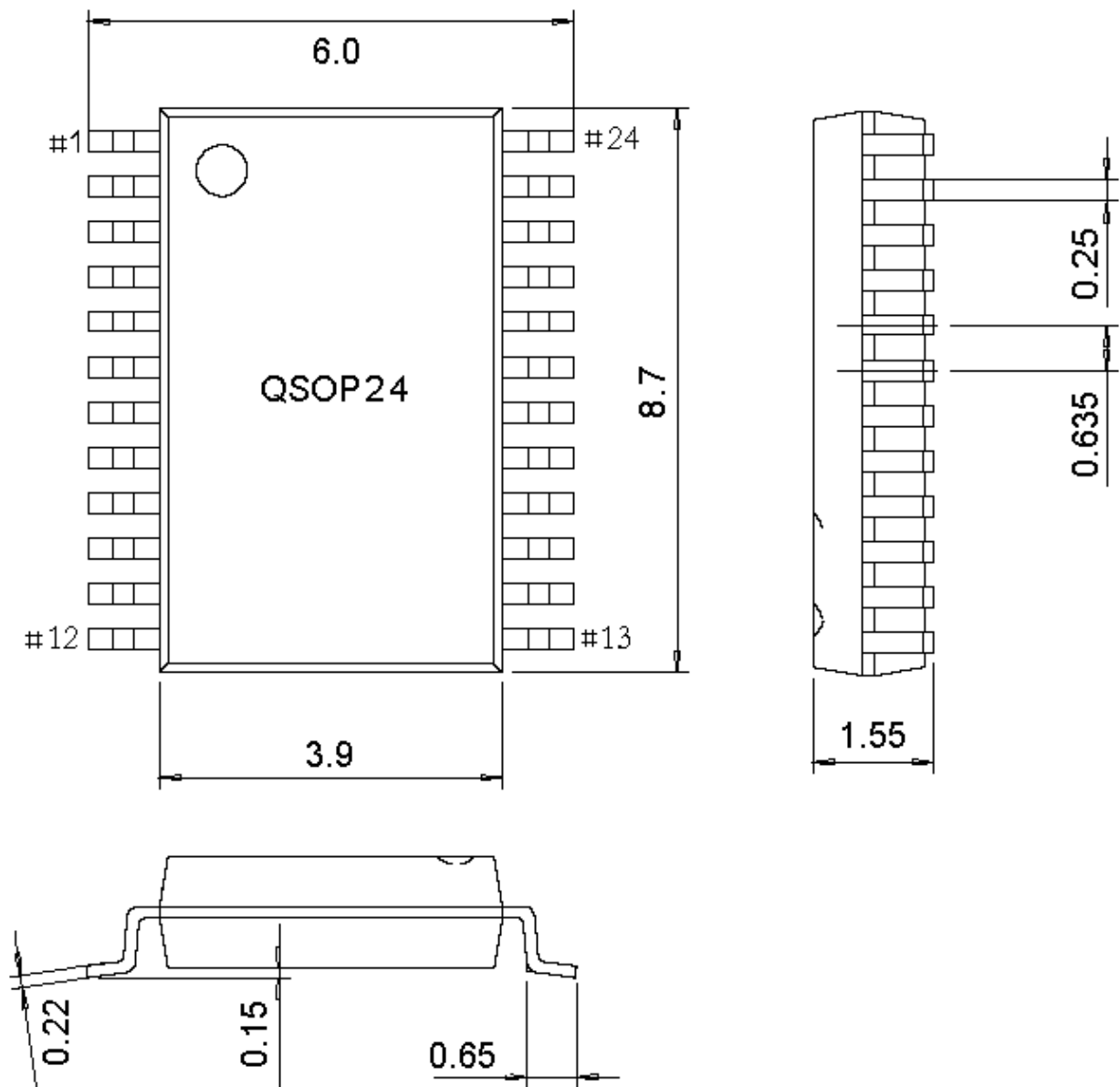
- QSOP16 (SSOP16-150mil-0.635)



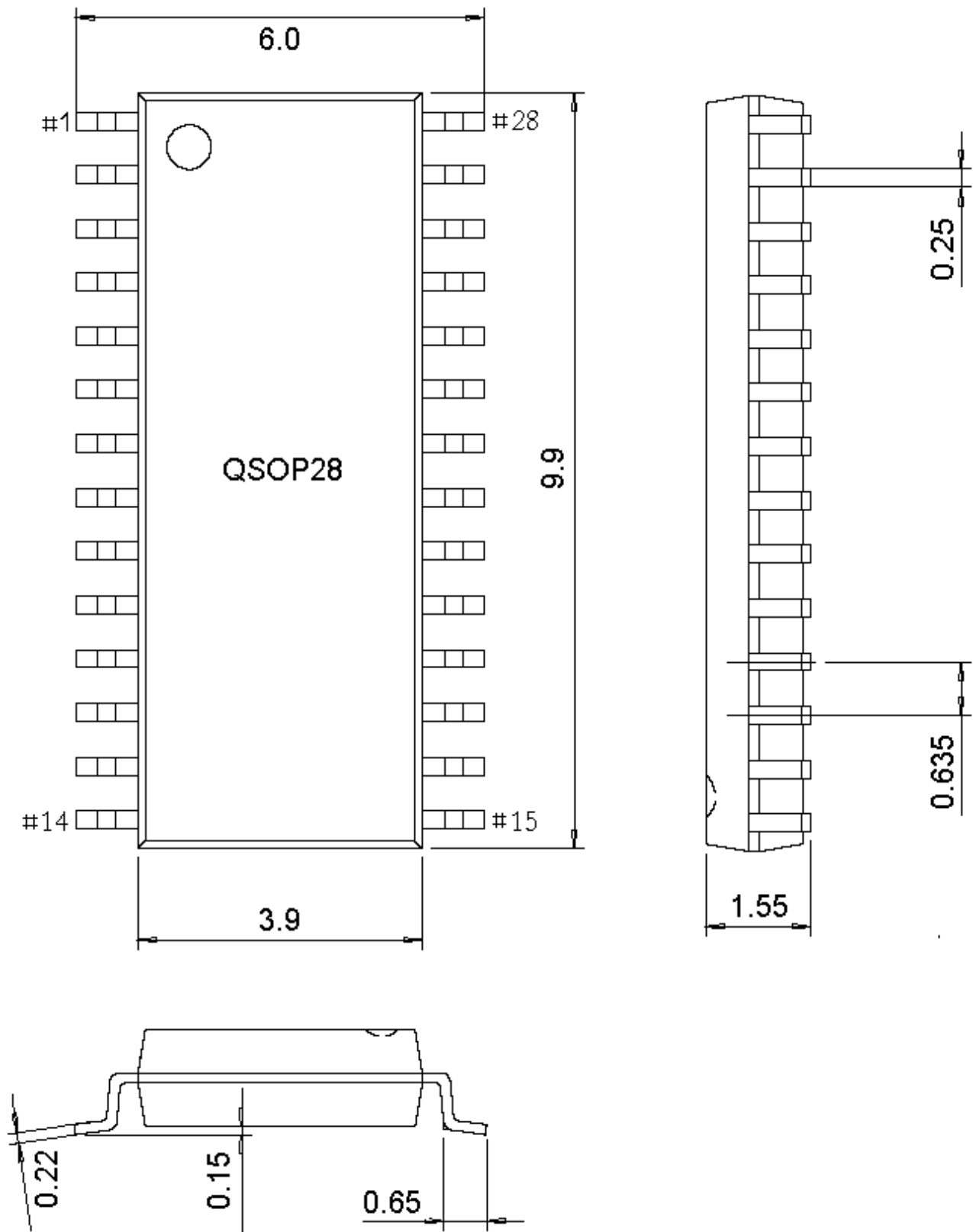
• QSOP20 (SSOP20-150mil-0.635)



## . QSOP24 (SSOP24-150mil-0.635)

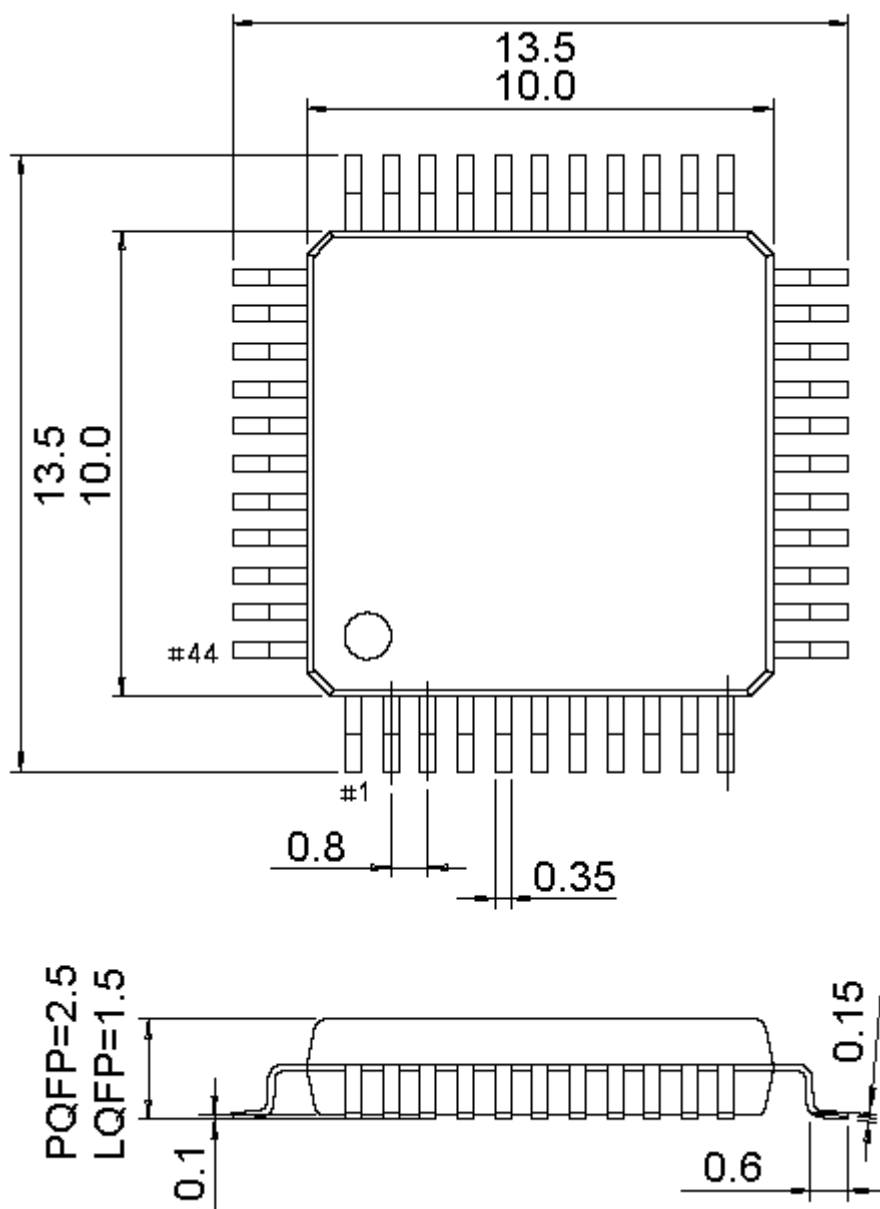


## . QSOP28 (SSOP28-150mil-0.635)

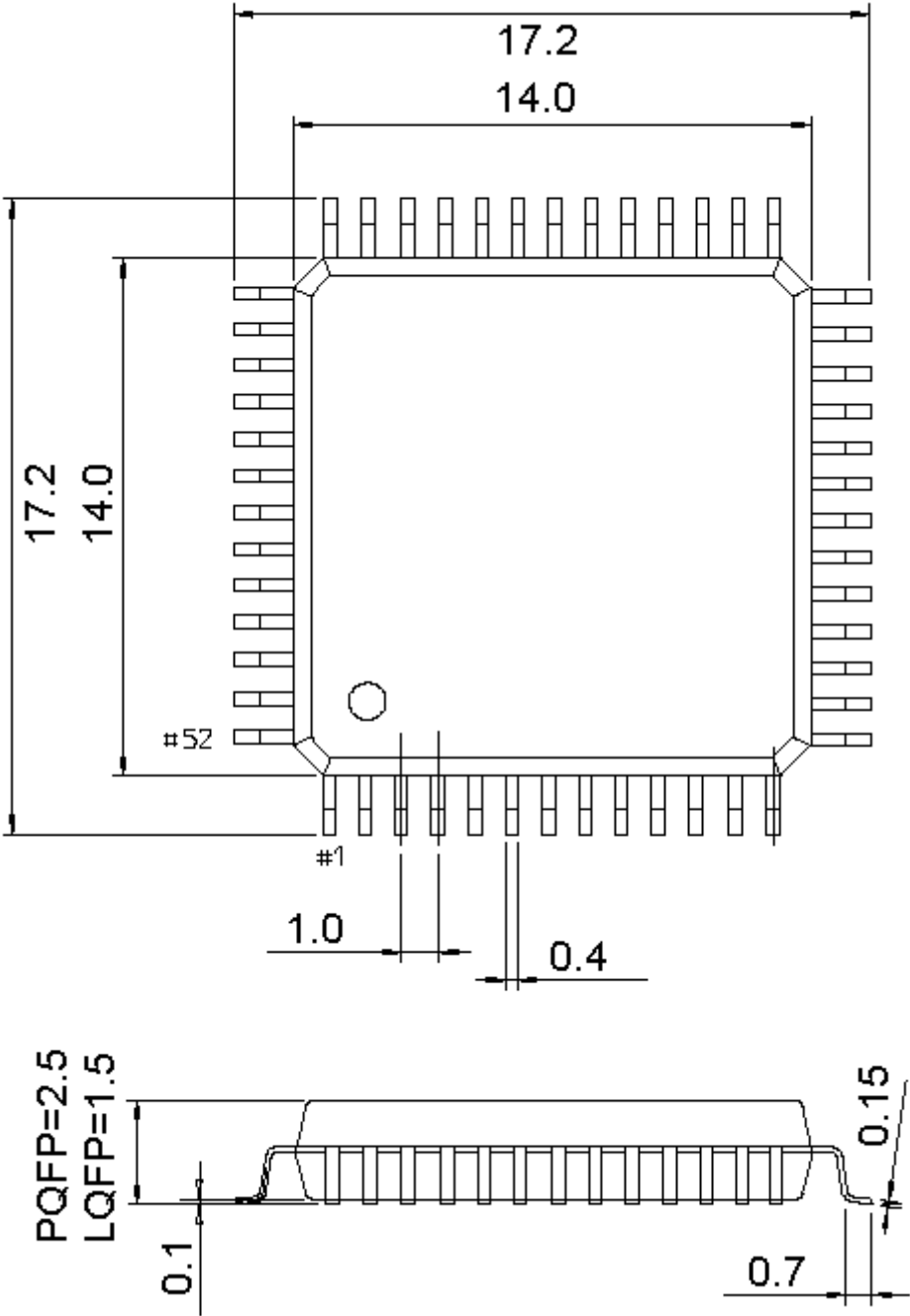


### QFP Type (PQFP, LQFP, TQFP)

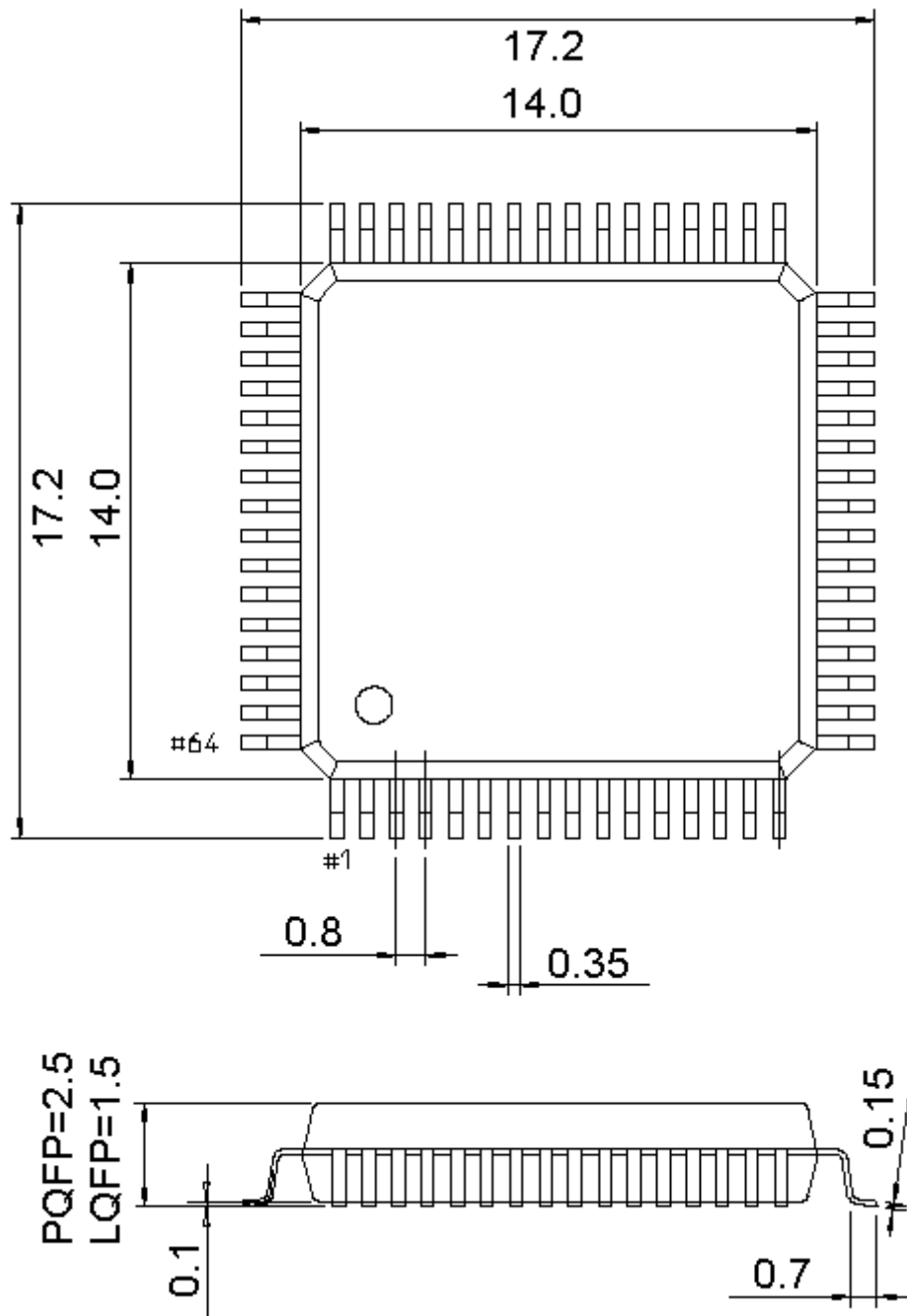
• **PQFP44 (QFP44)**



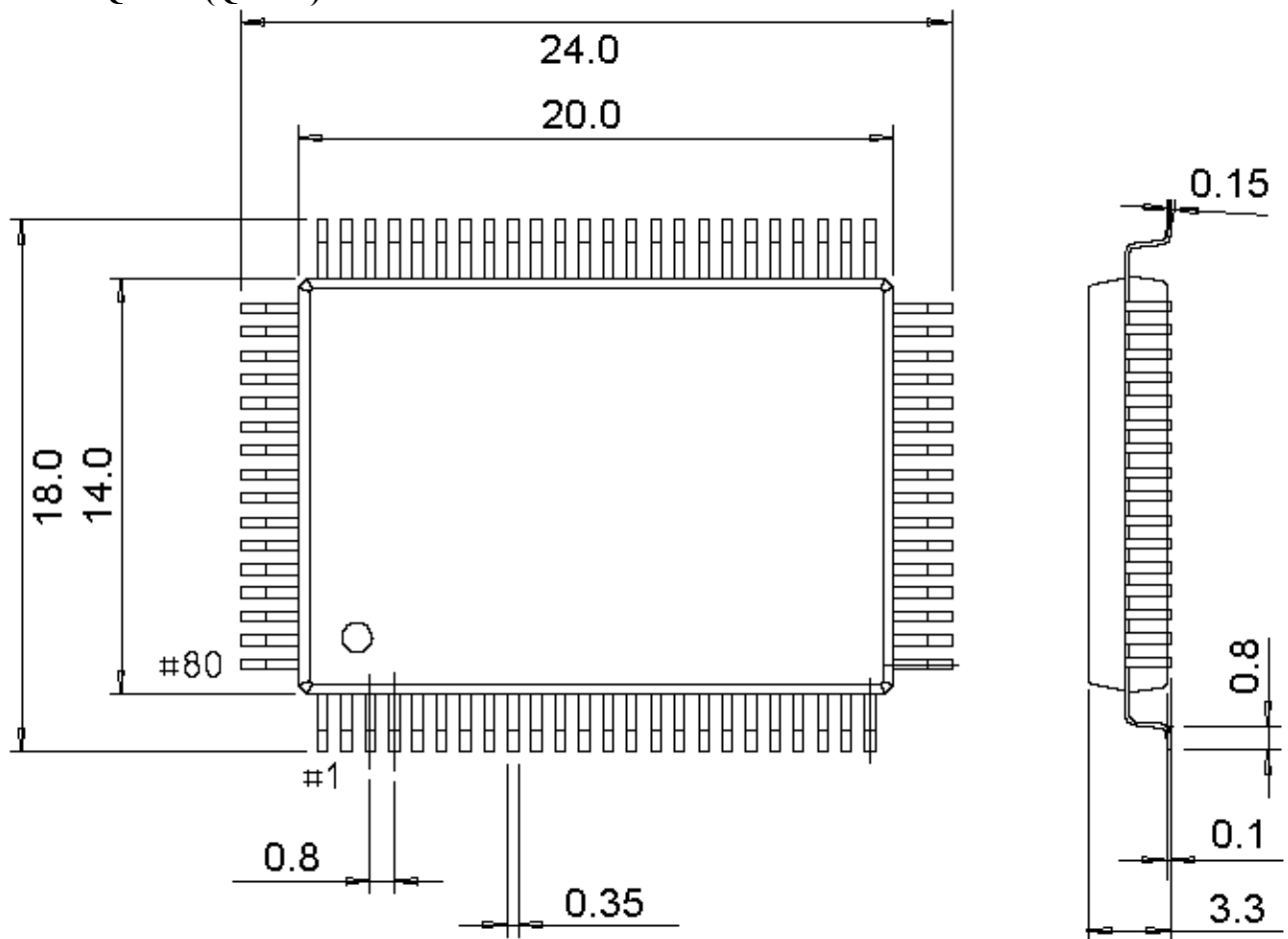
• PQFP52 (QFP52)



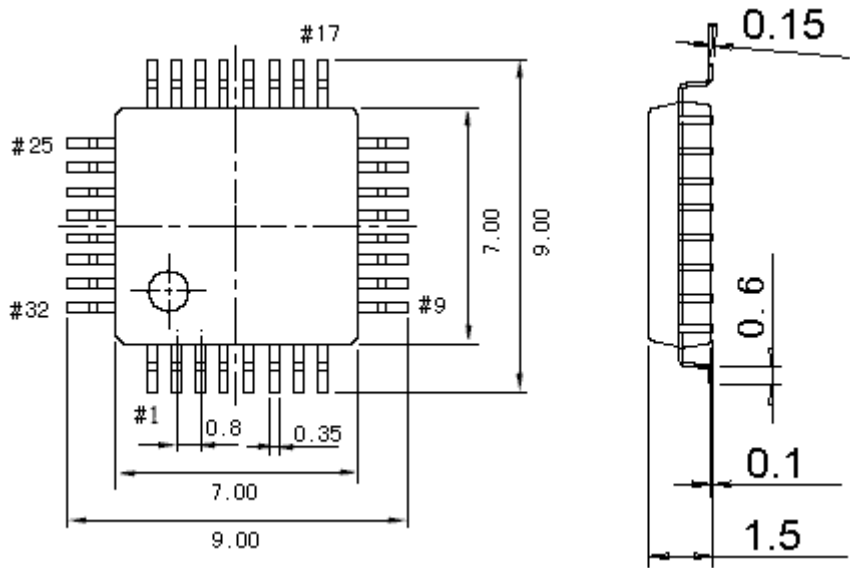
## . PQFP64 (QFP64), LQFP64-14\*14



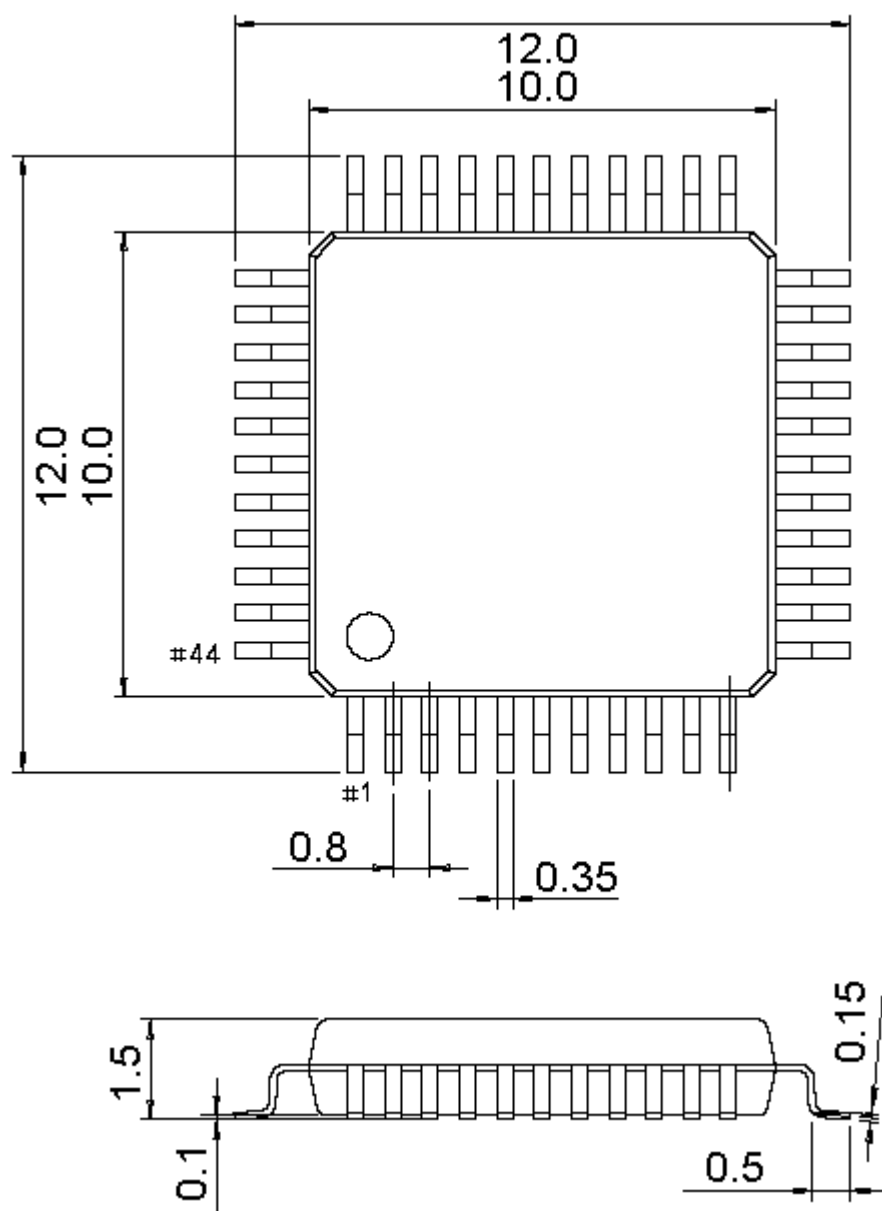
## . PQFP80 (QFP80)



• LQFP32 (LQFP32-7\*7)

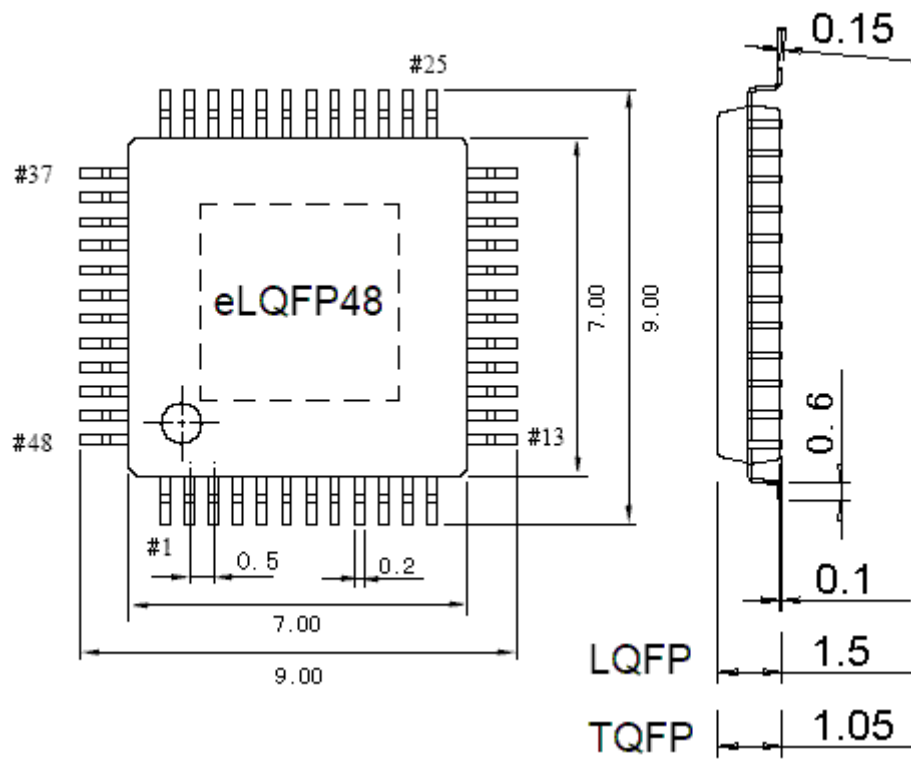


**. LQFP44 (LQFP44-10\*10)**



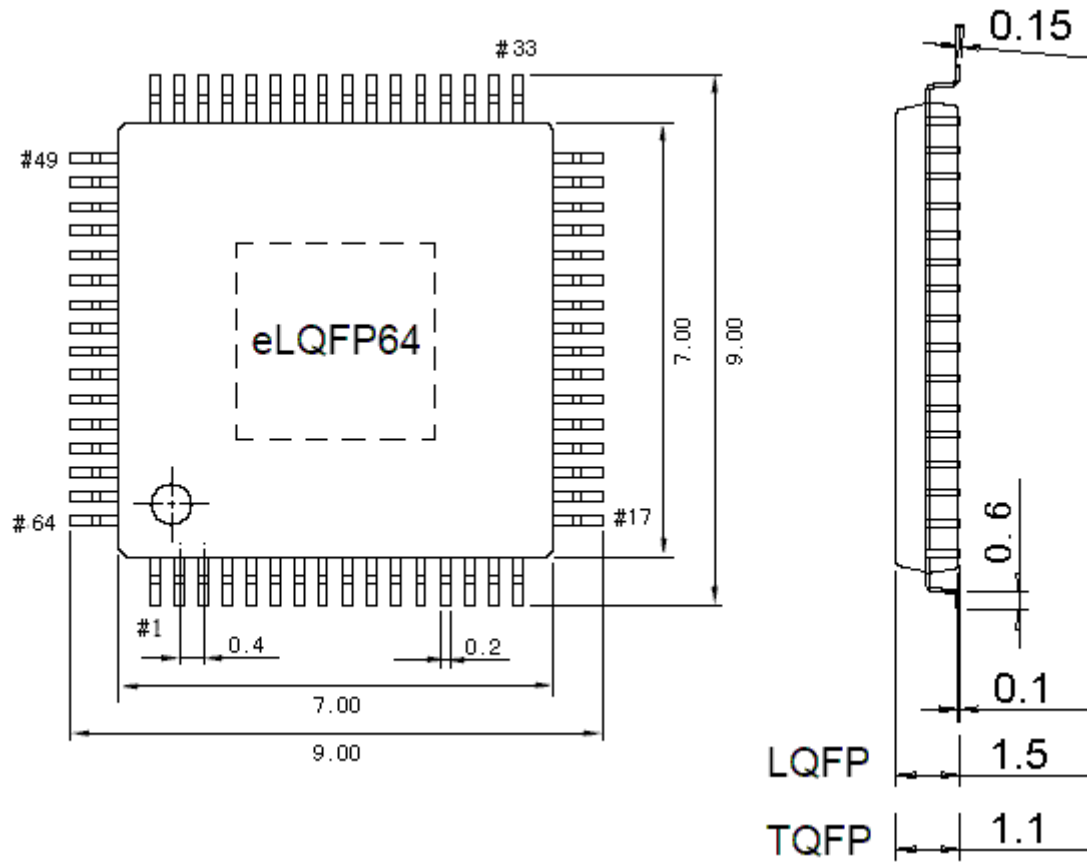
• **LQFP48 (LQFP48-7\*7), TQFP48, eLQFP48**

eLQFP48 Exposed Pad: 3.2\*3.2

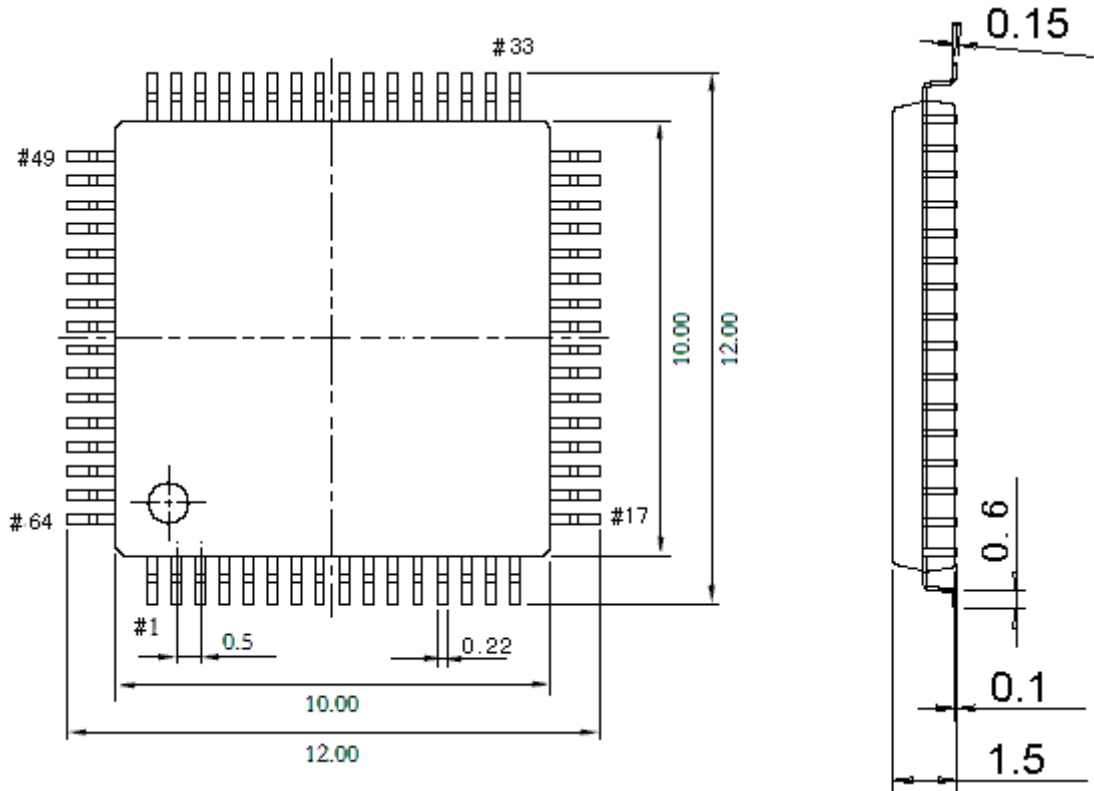


• **LQFP64 (LQFP64-7\*7), TQFP64, eLQFP64**

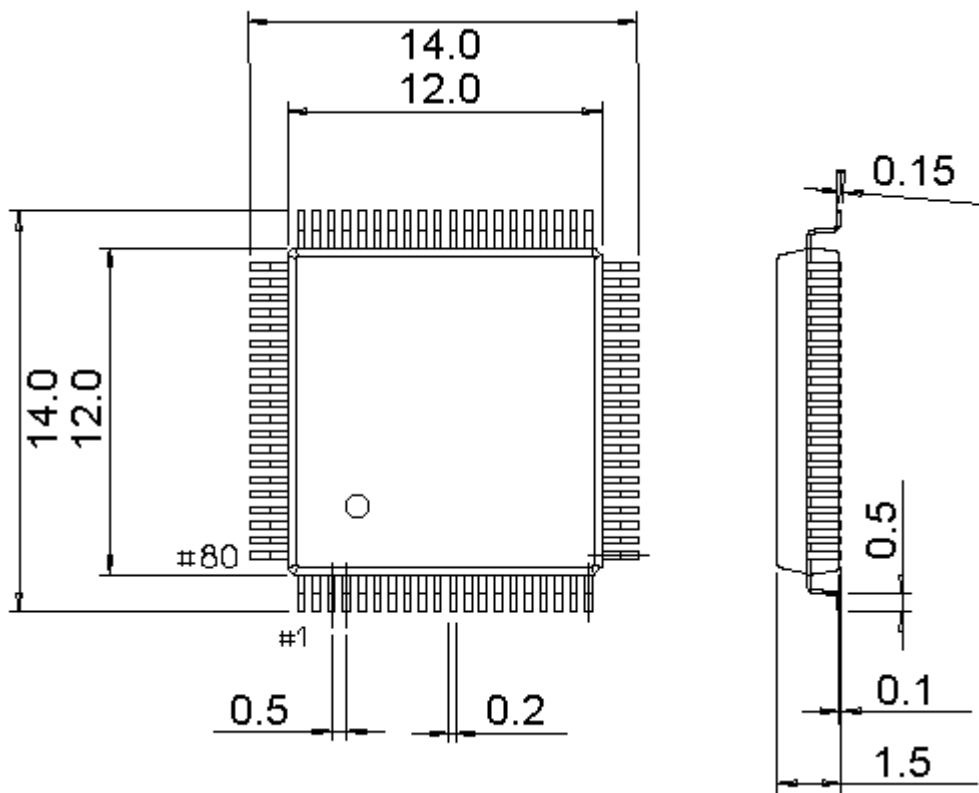
eLQFP64 Exposed Pad: 3.2\*3.2



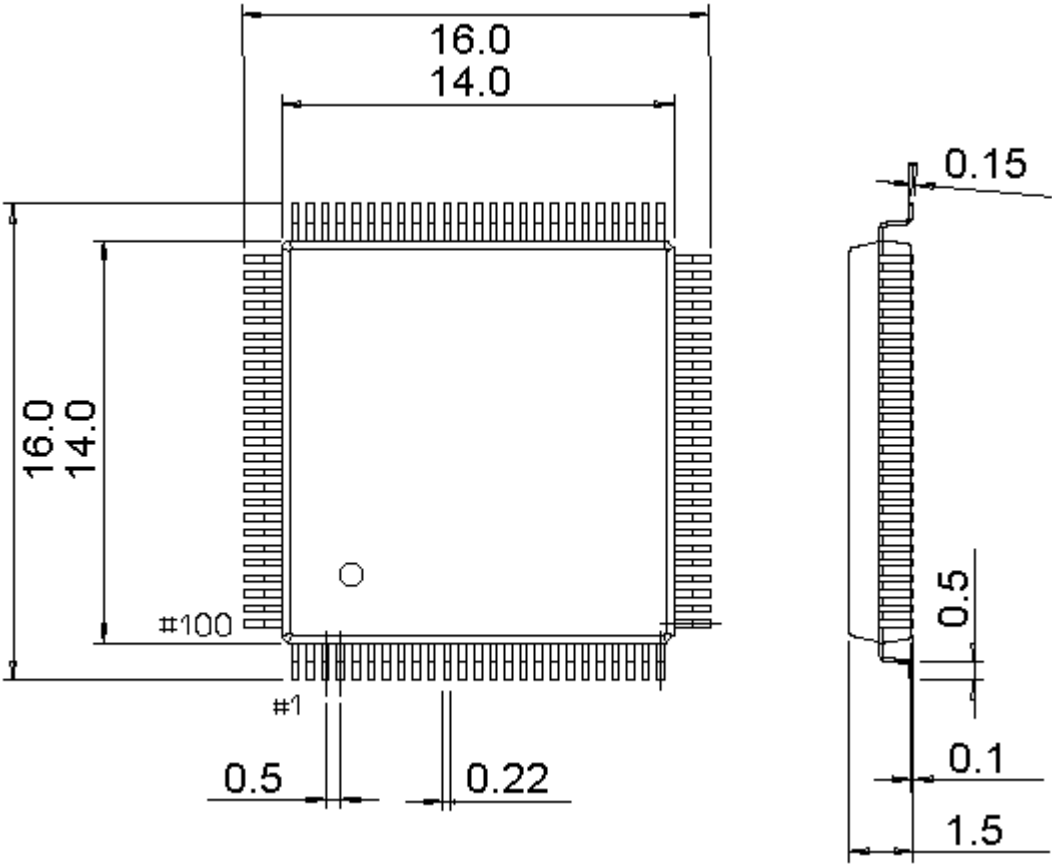
• LQFP64M (LQFP64-10\*10)



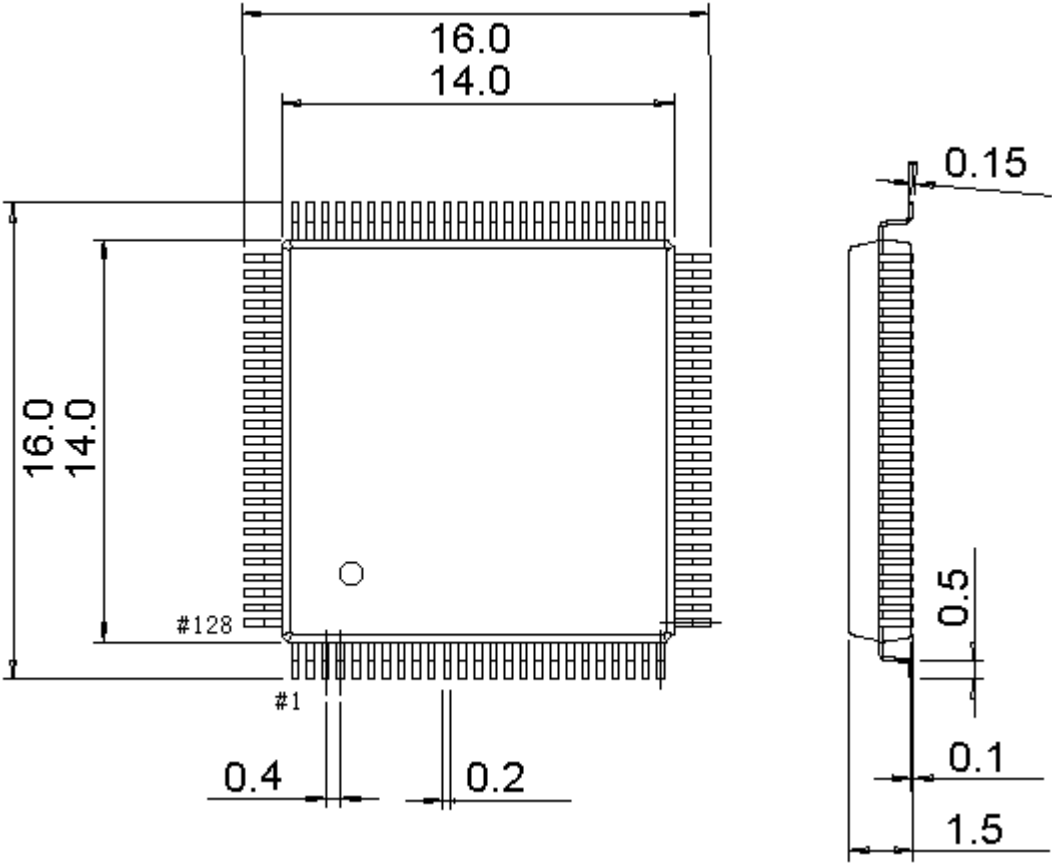
. LQFP80 (LQFP80-12\*12)



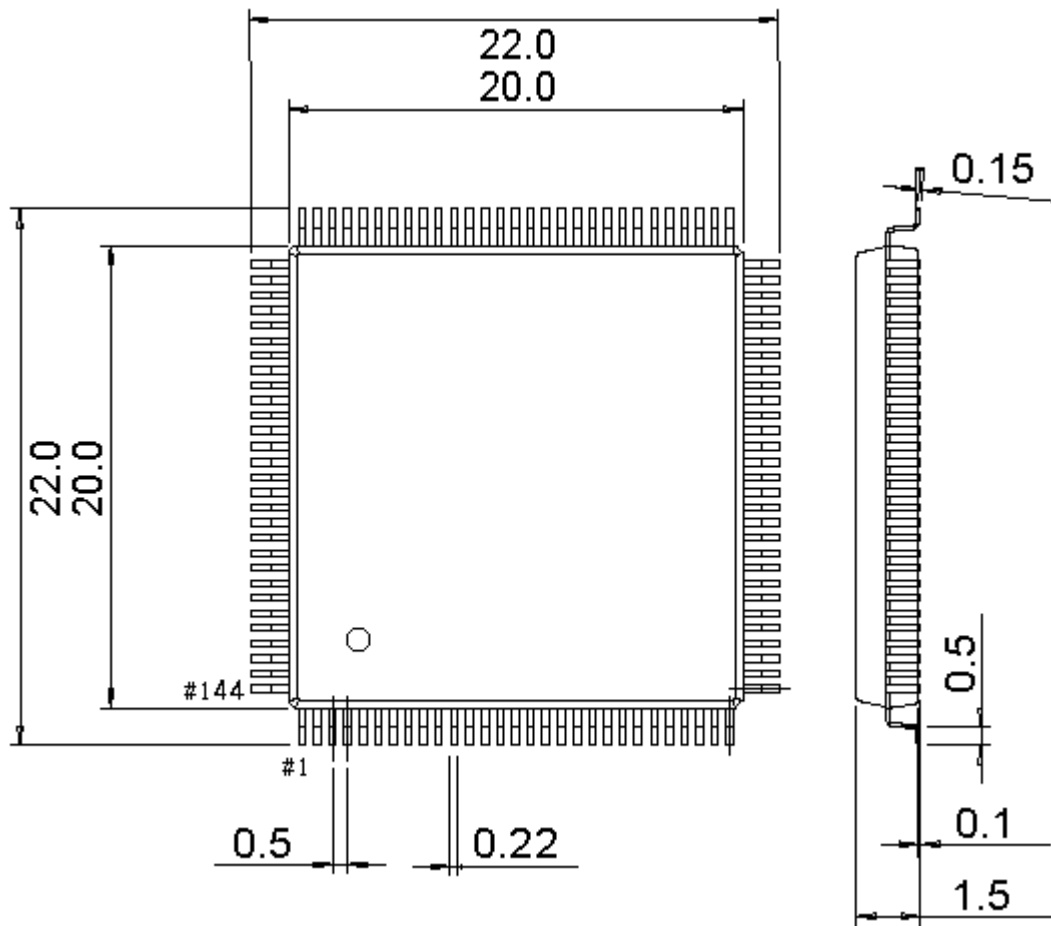
• LQFP100 (LQFP100-14\*14)



• LQFP128 (LQFP128-14\*14)



## . LQFP144 (LQFP144-20\*20)



## QFN Type

### . QFN/DFN Naming Rules Reference

#### Standard naming

QFN ab - d \* e \* t - p

Number ab: QFN package 4-side cumulative pin count, currently up to 128.

Numbers d and e: d and e represent the length and width(The side of pin 1#) of the QFN shape.

Number t: t stands for the thickness of the QFN profile. 0.75 and 0.55 are more commonly used, as well as 0.85, 0.5, 0.4 and so on.

Number p: p on behalf of the pin pitch, that is, the distance between the center of adjacent pins, if there are 2 kinds of take its small, 0.4 and 0.5 are more commonly used, and another 0.35, 0.65, 0.45, 0.3 and so on.

#### Reference simplified naming

QFN ab c d X e

Number ab: QFN package 4-side cumulative pin count, currently up to 128.

Letter c: no c represents general-purpose QFN with Exposed Pad, default size, no numbers d and e;

Letter X represents general-purpose QFN with Exposed Pad, non-default size, at least number d;

Letter L represents general-purpose QFN with Exposed Pad, the pin lengths L (or EPAD size) are different;

Letter C represents WCH-designed general-purpose QFN with Exposed Pad, at least number d;

Letter F represents general-purpose QFN without Exposed Pad, or WCH-customized special QFN without Exposed Pad, at least number d;

Letter E represents WCH-customized special QFN with Exposed Pad, at least number d.

Numbers d and e: d and e represent the length and width(The side of pin 1#) of the QFN profile;

No d no e represents the default size, which is also the common size;

With d without e means the size is d\*d square;

If the number exceeds 13 then divide by 10, e.g. 45 means 4.5.

The default size is usually the smallest square size when pitch=0.4, which is also the common size:

QFN12, QFN14 defaults to 2\*2 square;

QFN16, QFN20 defaults to 3\*3 square;

QFN24, QFN28, QFN32 defaults to 4\*4 square;

QFN36, QFN40, QFN44 defaults to 5\*5 square.

#### Take QFN20 for example:

QFN20: General-purpose QFN with Exposed Pad, default size (QFN20 is 3\*3 by default);

QFN20X4: General-purpose QFN with Exposed Pad, size 4\*4;

QFN20X25X45: General-purpose QFN with Exposed Pad, size 2.5\*4.5;

QFN20L: General-purpose QFN with Exposed Pad, default size, different pin lengths;

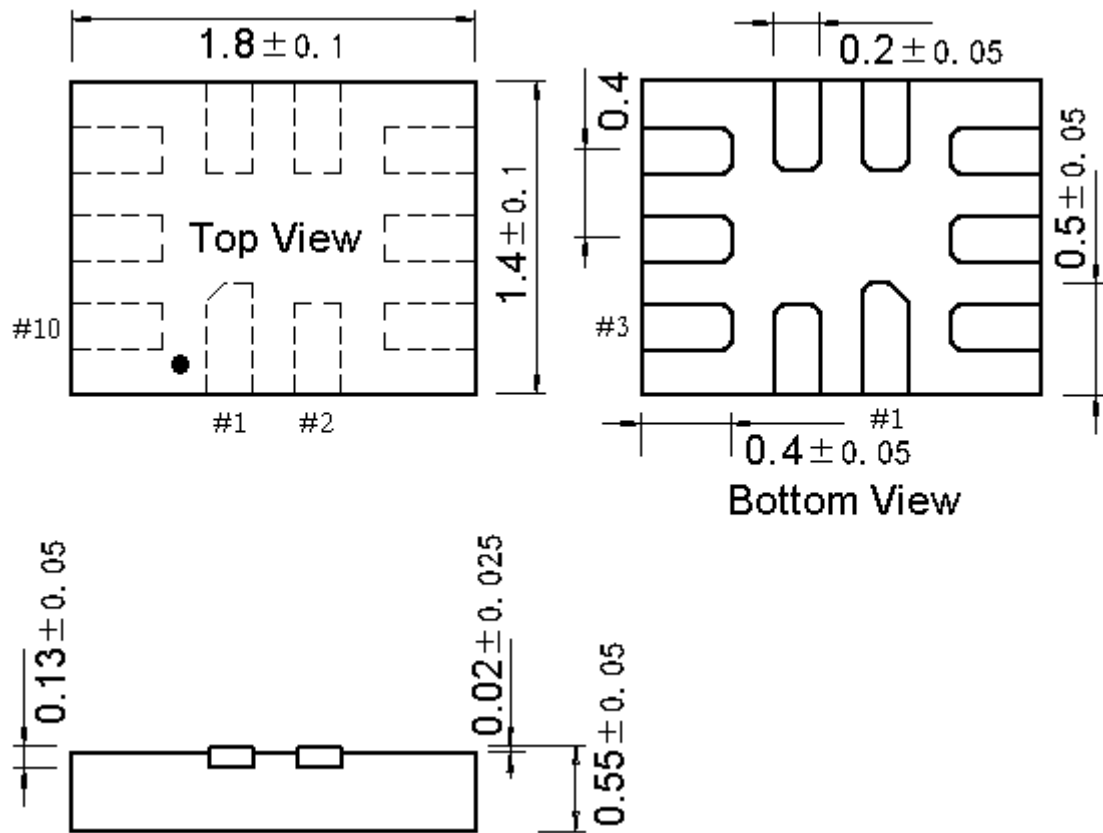
QFN20F: General-purpose QFN without Exposed Pad, default size;

QFN20C2: WCH-designed general-purpose QFN with Exposed Pad, size 2\*2;

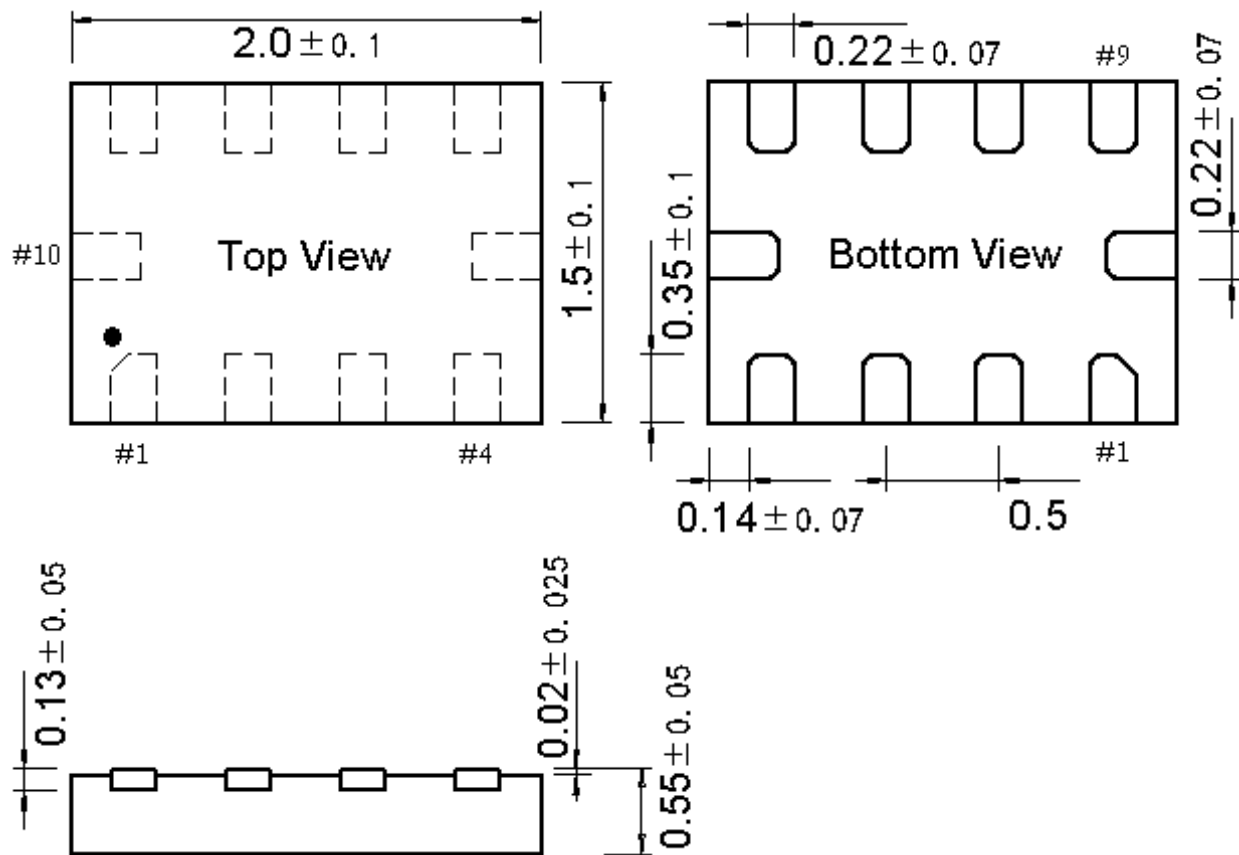
QFN20F3: WCH-customized special QFN, without Exposed Pad, size 3\*3;

QFN20E3X4: WCH-customized special QFN with Exposed Pad, size 3\*4.

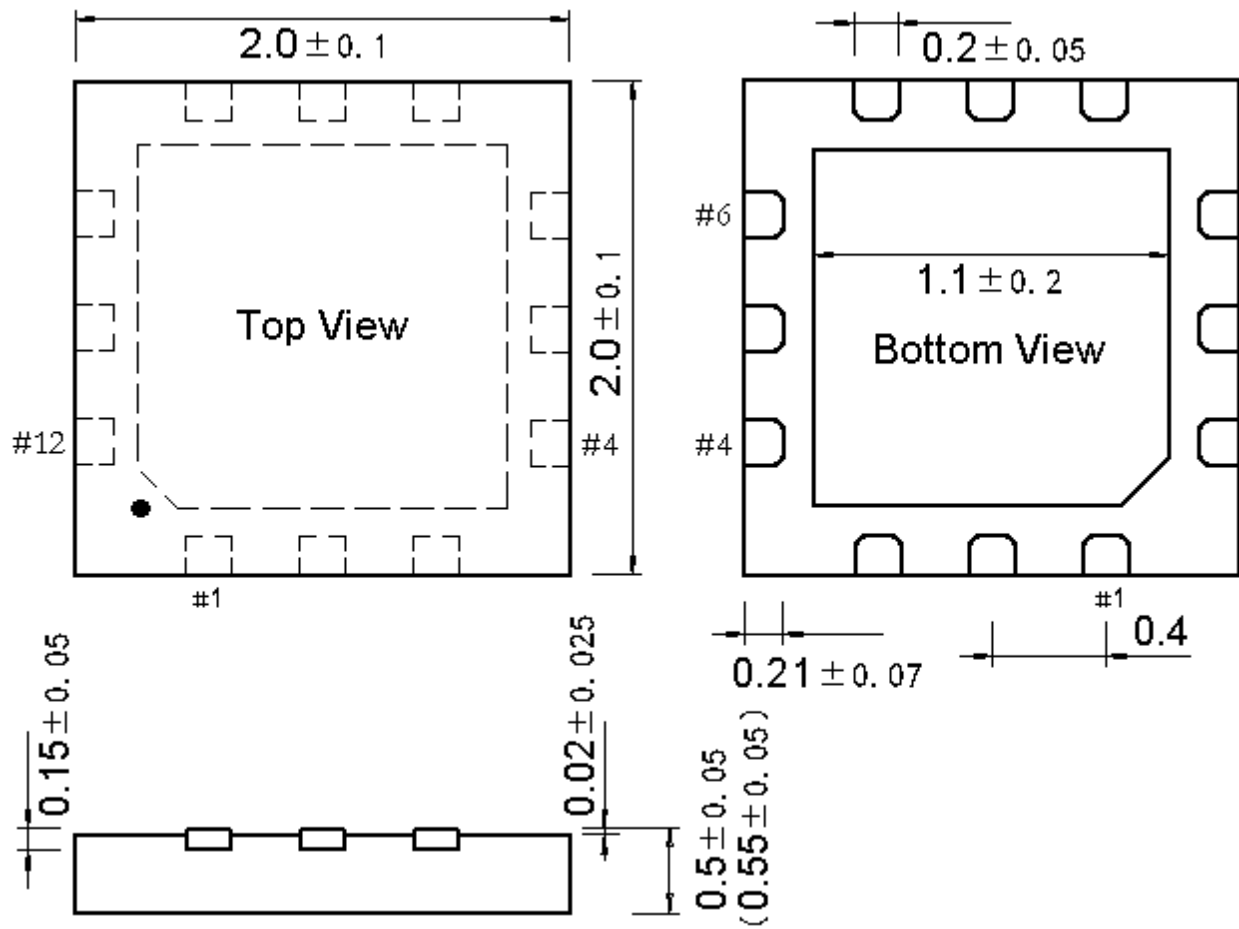
## . QFN10 (QFN10-1.4\*1.8\*0.55-0.4)



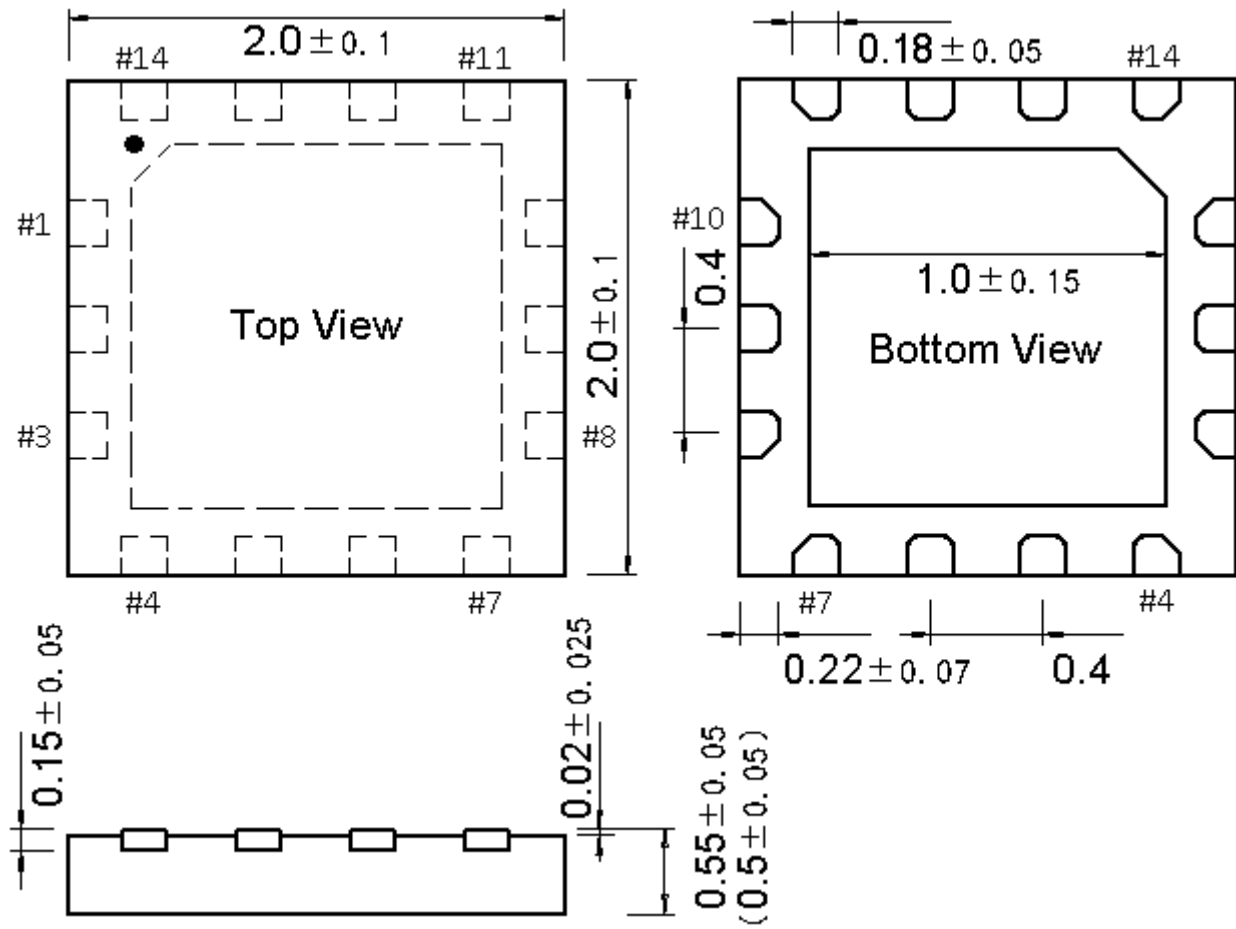
## . QFN10F15X2 (QFN10-1.5\*2\*0.55-0.4)



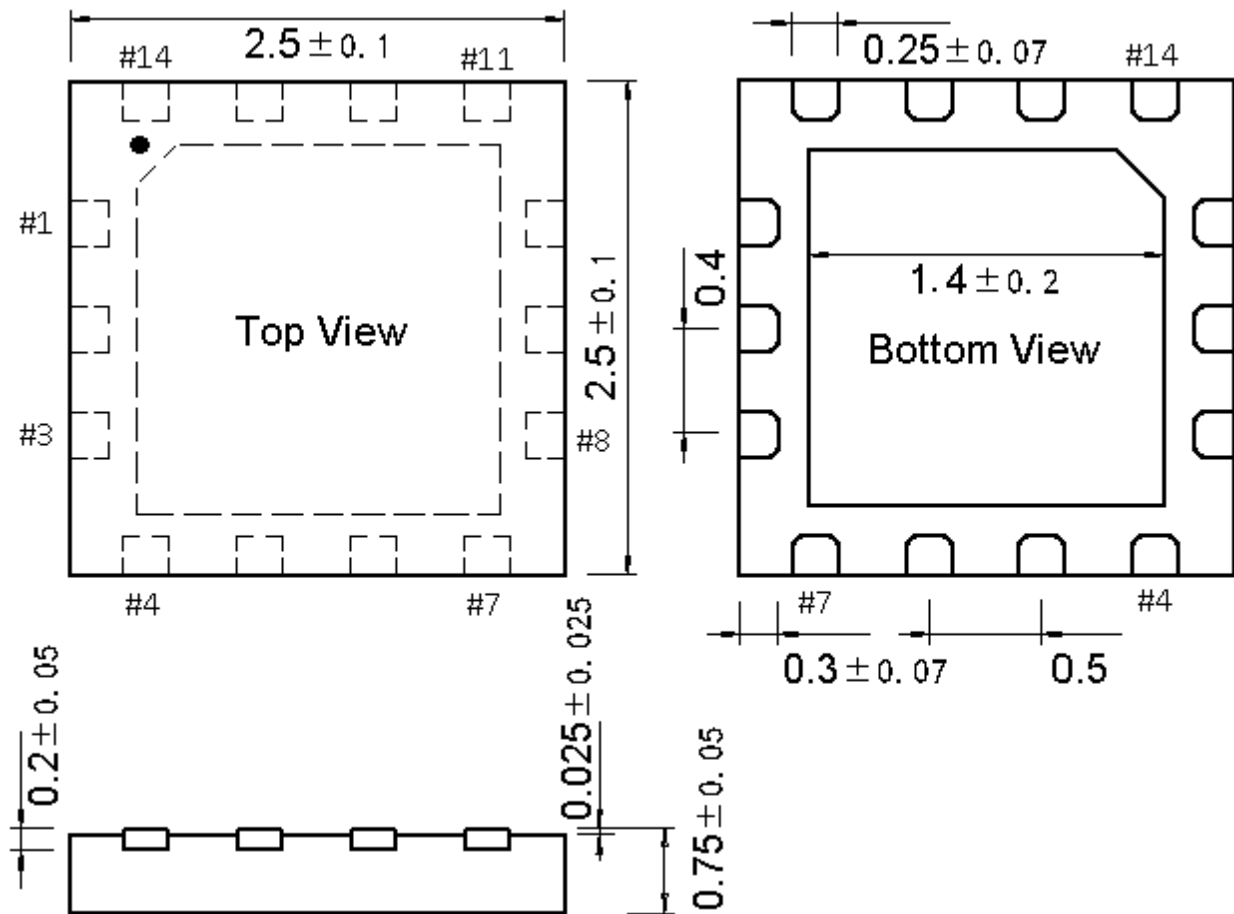
• QFN12 (QFN12-2\*2\*0.5-0.4)



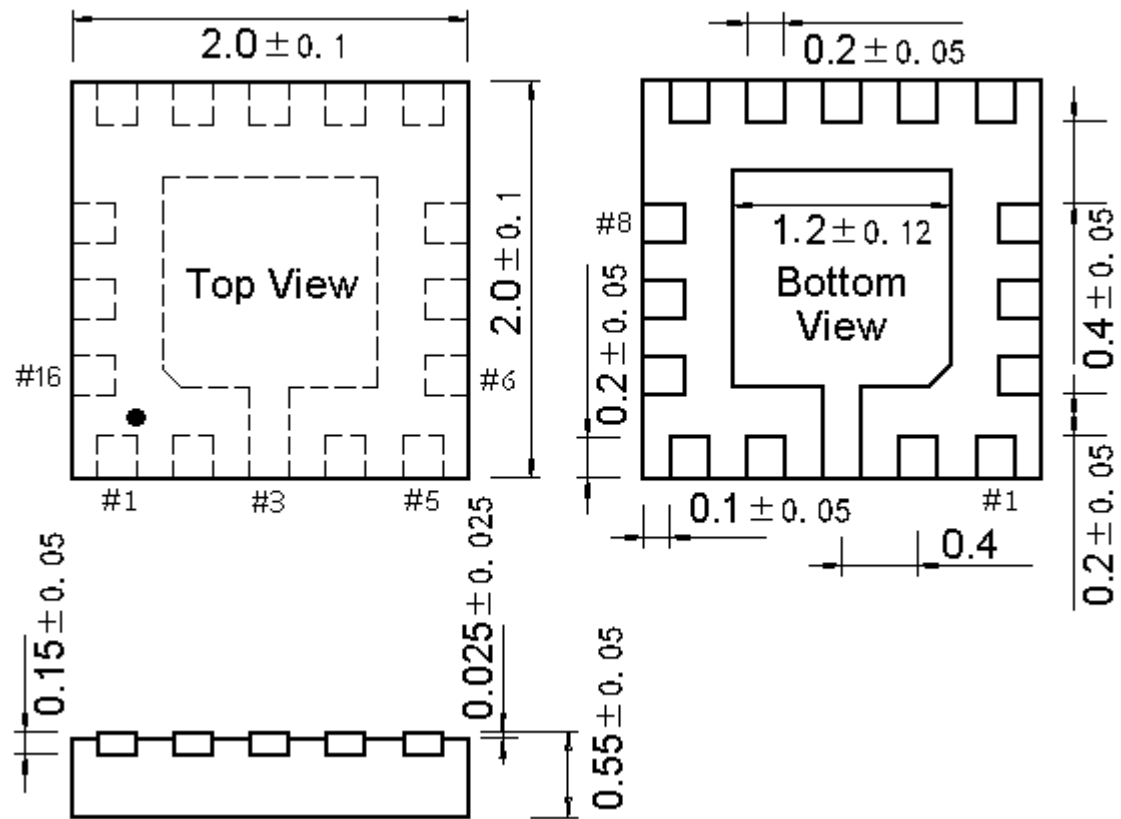
• QFN14 (QFN14-2\*2\*0.55-0.4)



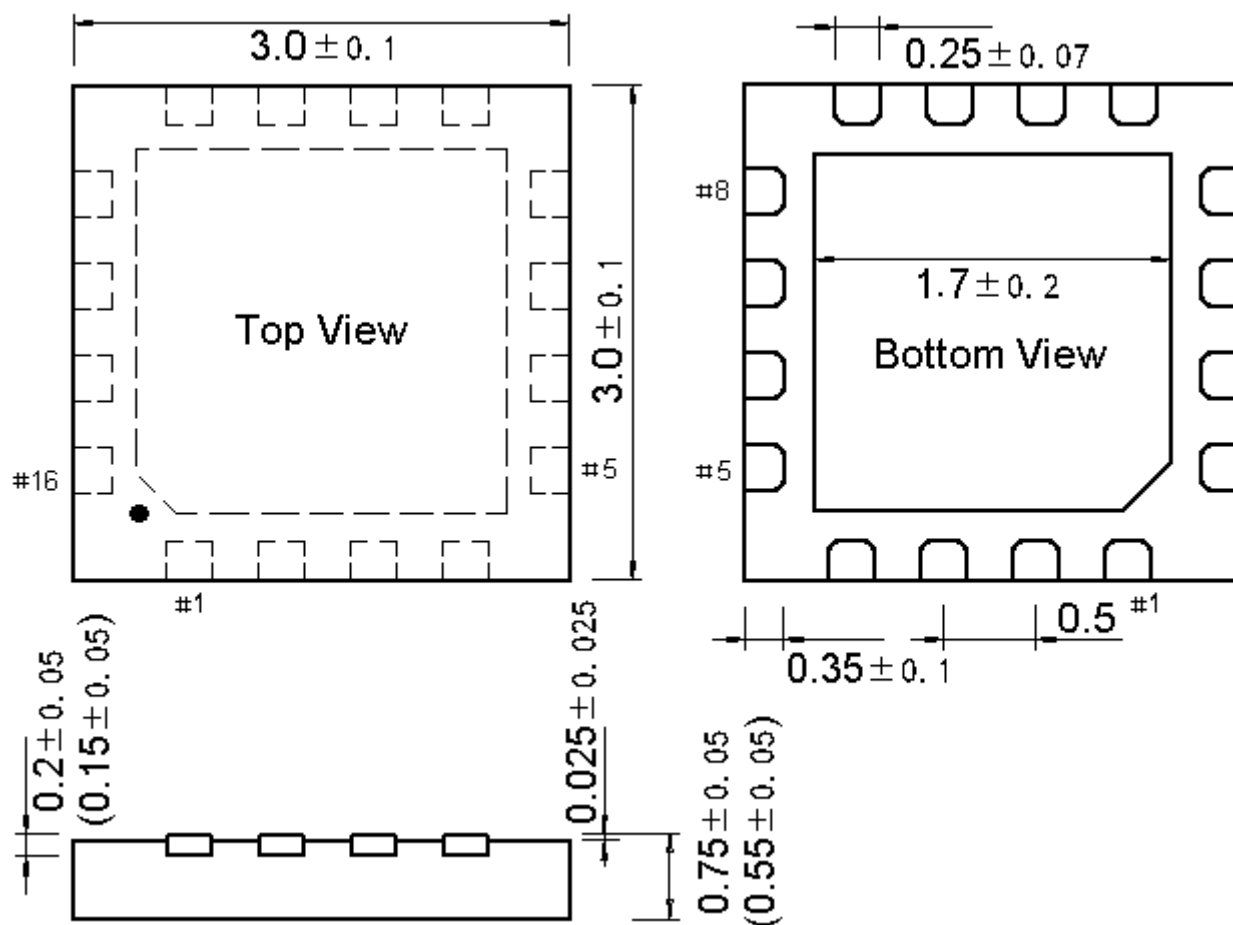
• QFN14X25 (QFN14-2.5\*2.5\*0.75-0.5)



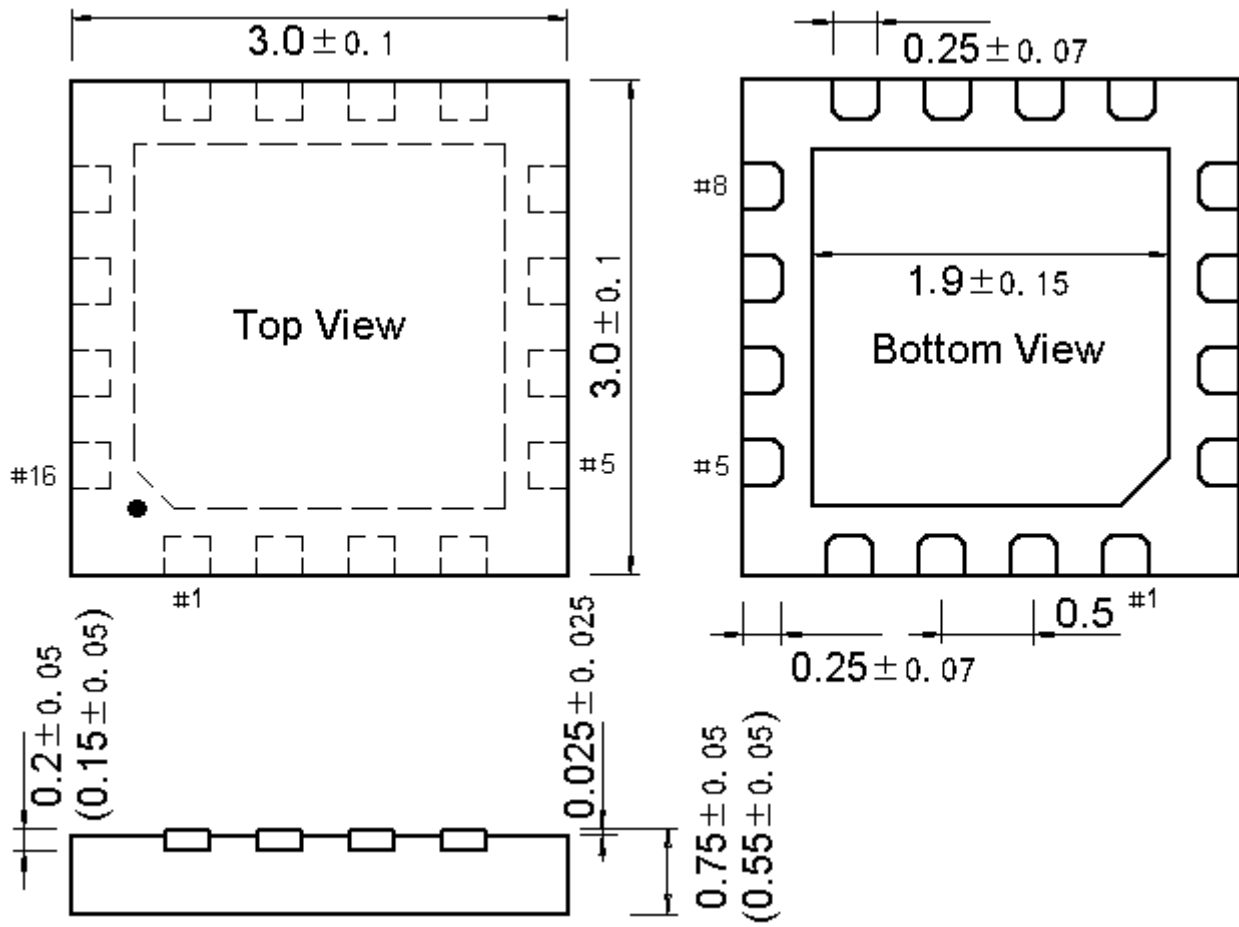
• QFN16C2 (QFN16C-2\*2\*0.55-0.4)



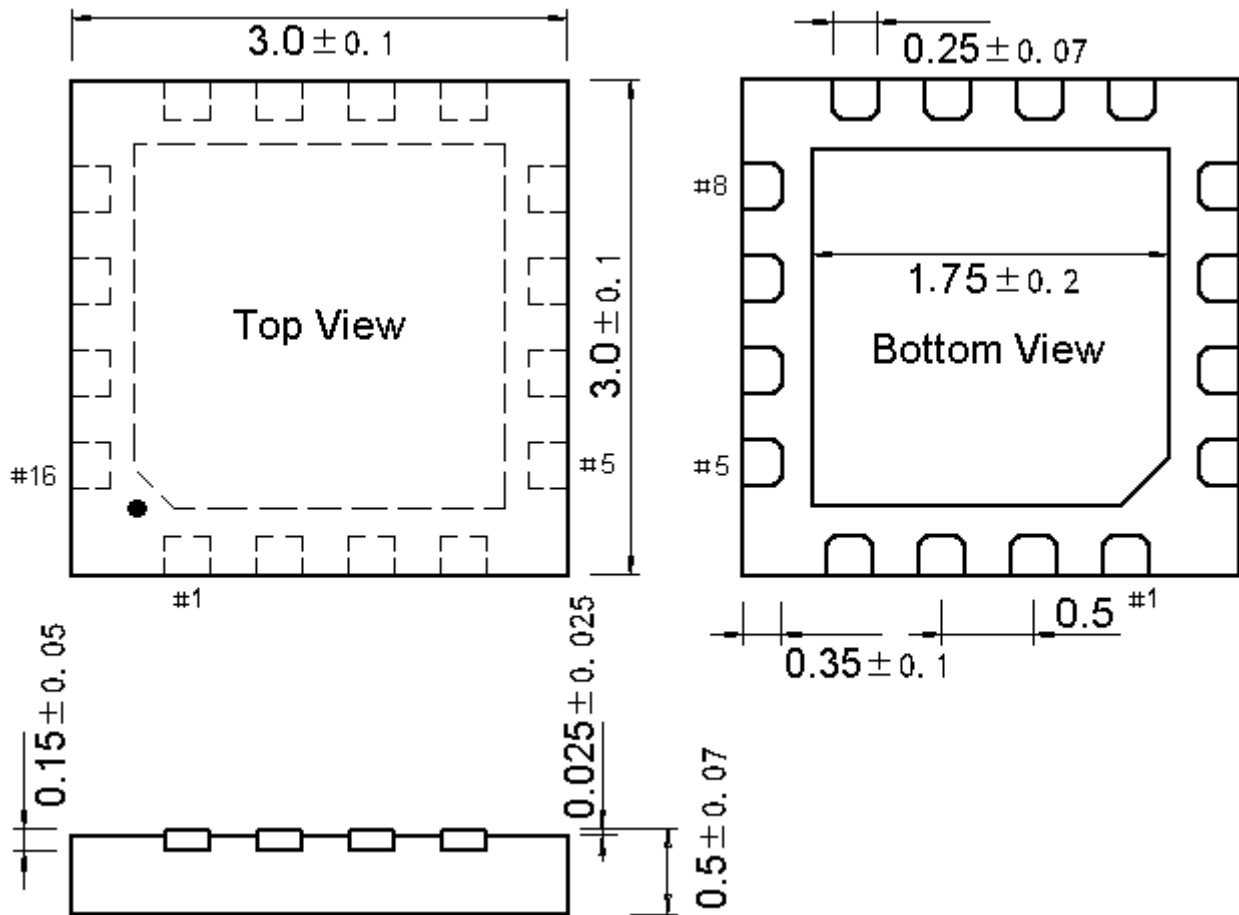
**. QFN16 (QFN16-3\*3-0.5)**



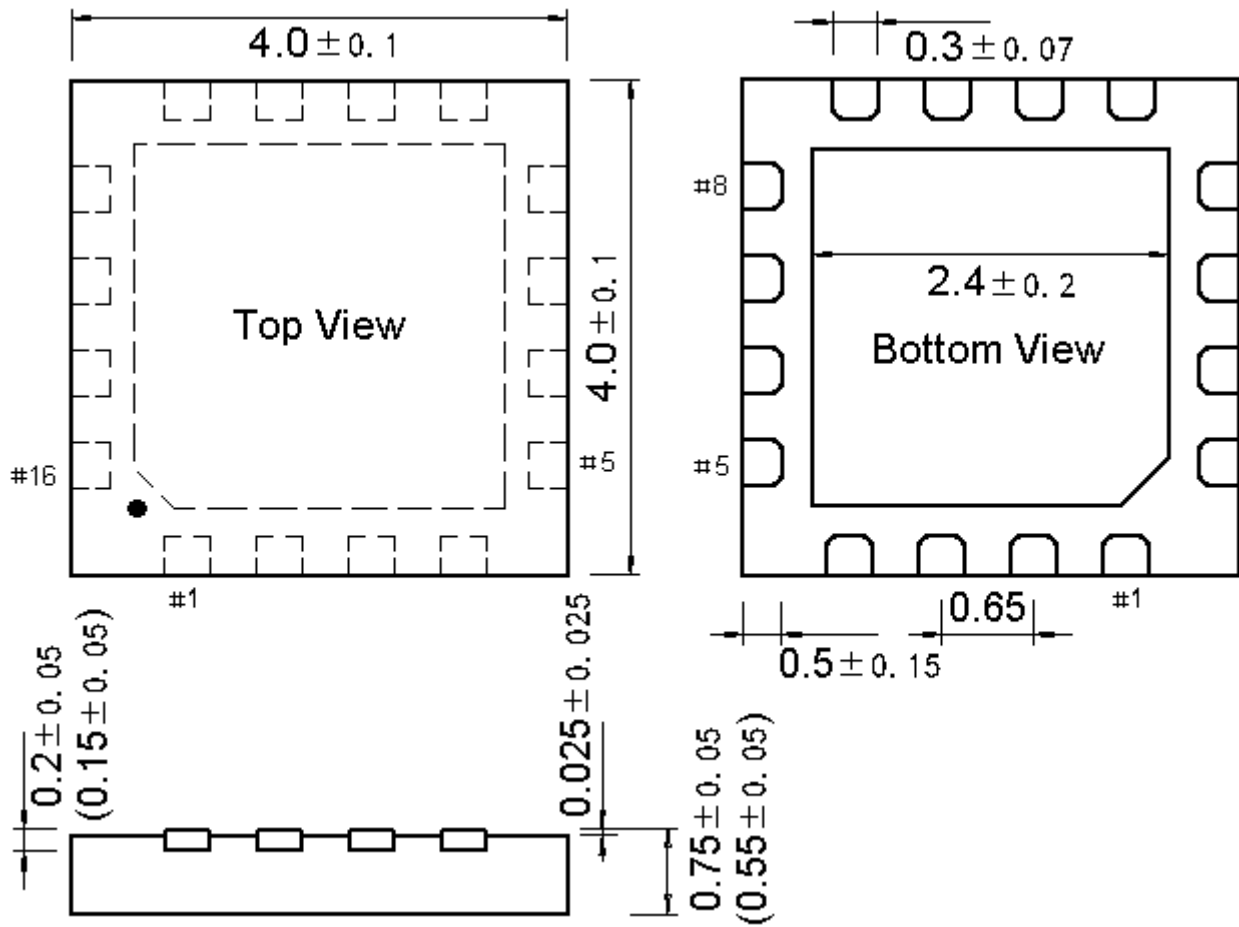
• QFN16L (QFN16L-3\*3-0.5)



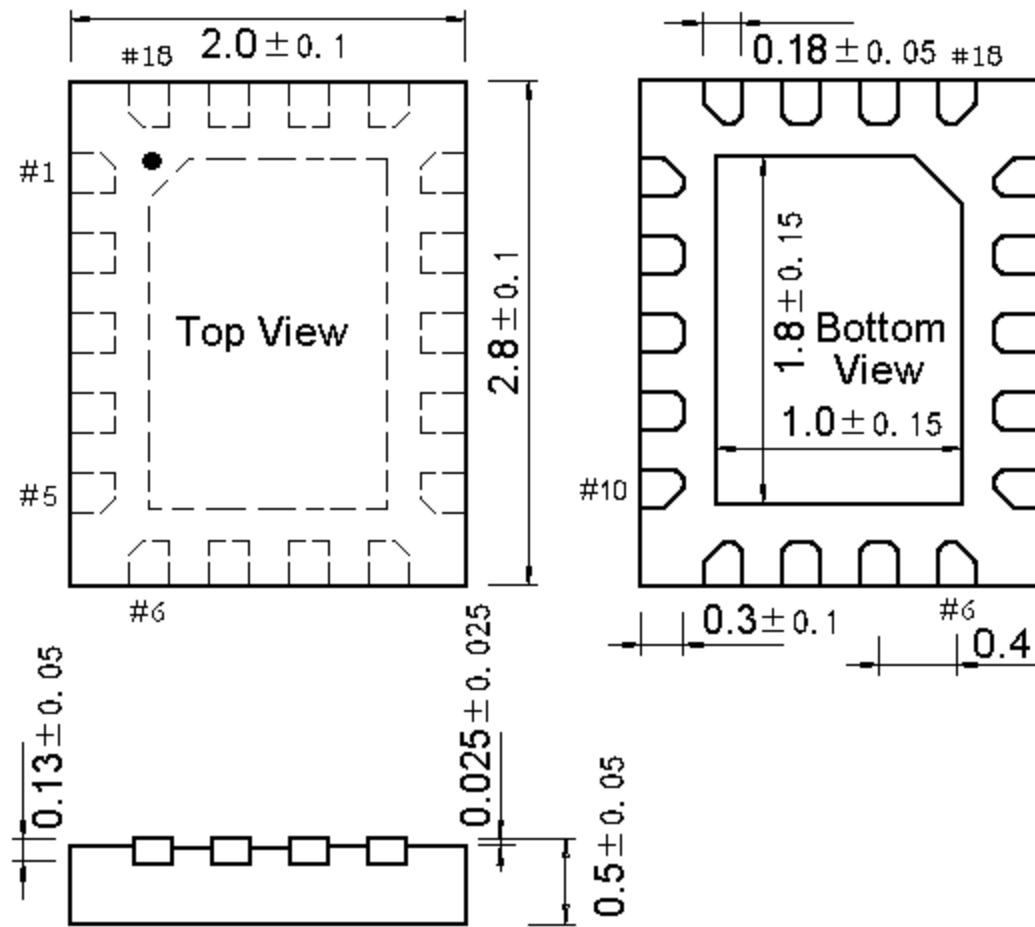
## . QFN16-3\*3\*0.5-0.4



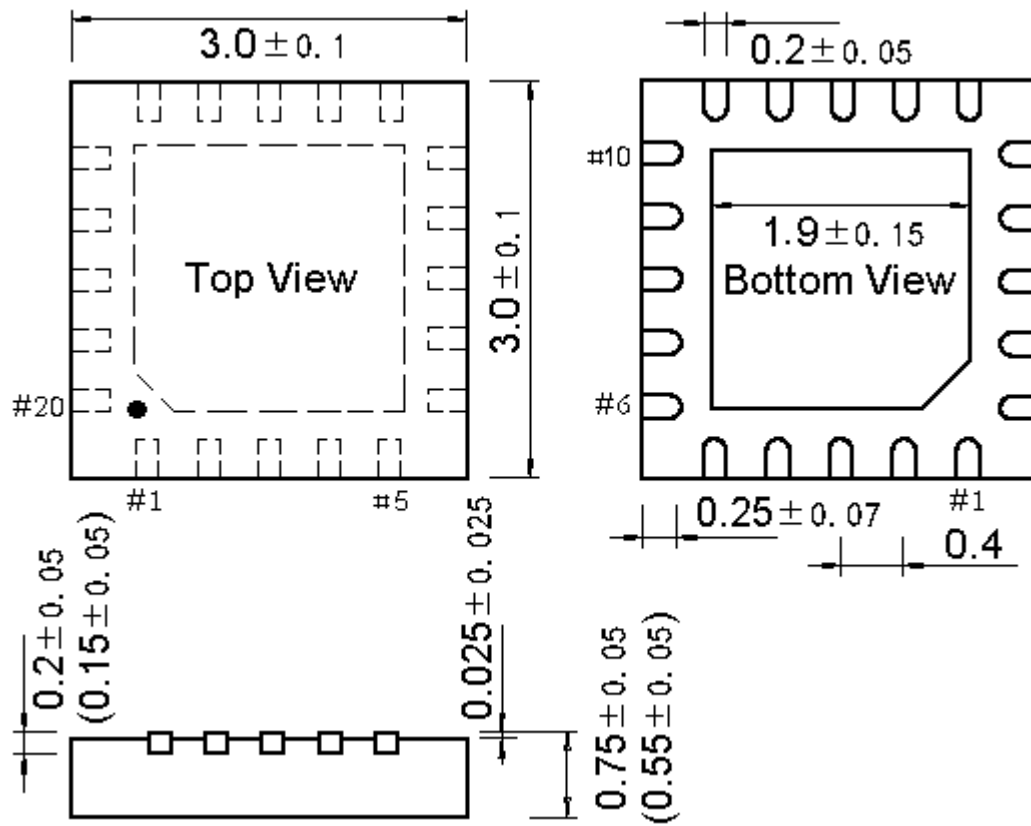
• QFN16X4 (QFN16-4\*4-0.65)



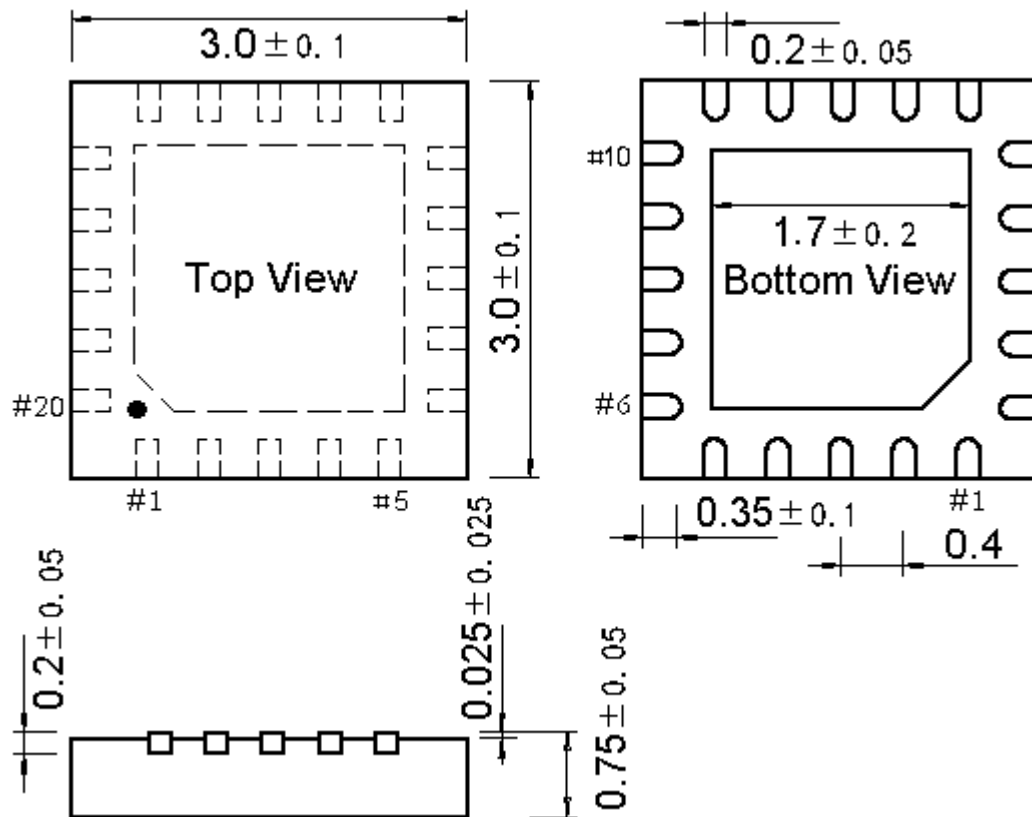
## . QFN18X2X28 (QFN18-2\*2.8\*0.5-0.4)



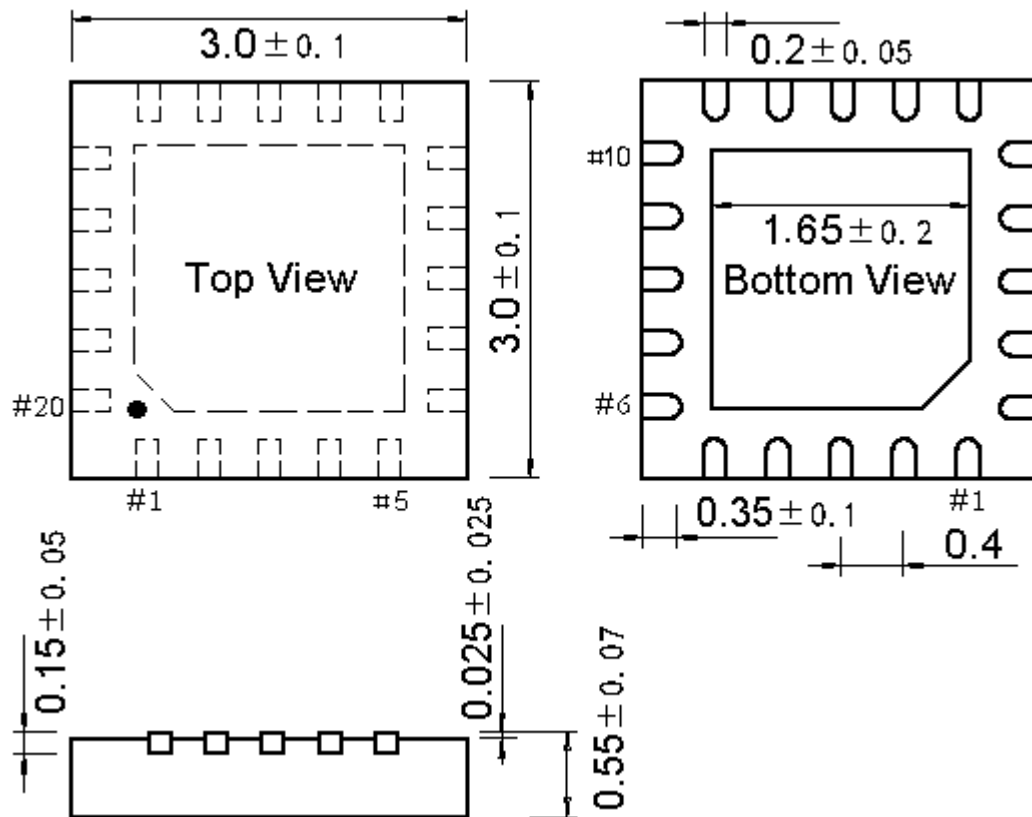
• QFN20 (QFN20-3\*3-0.4)



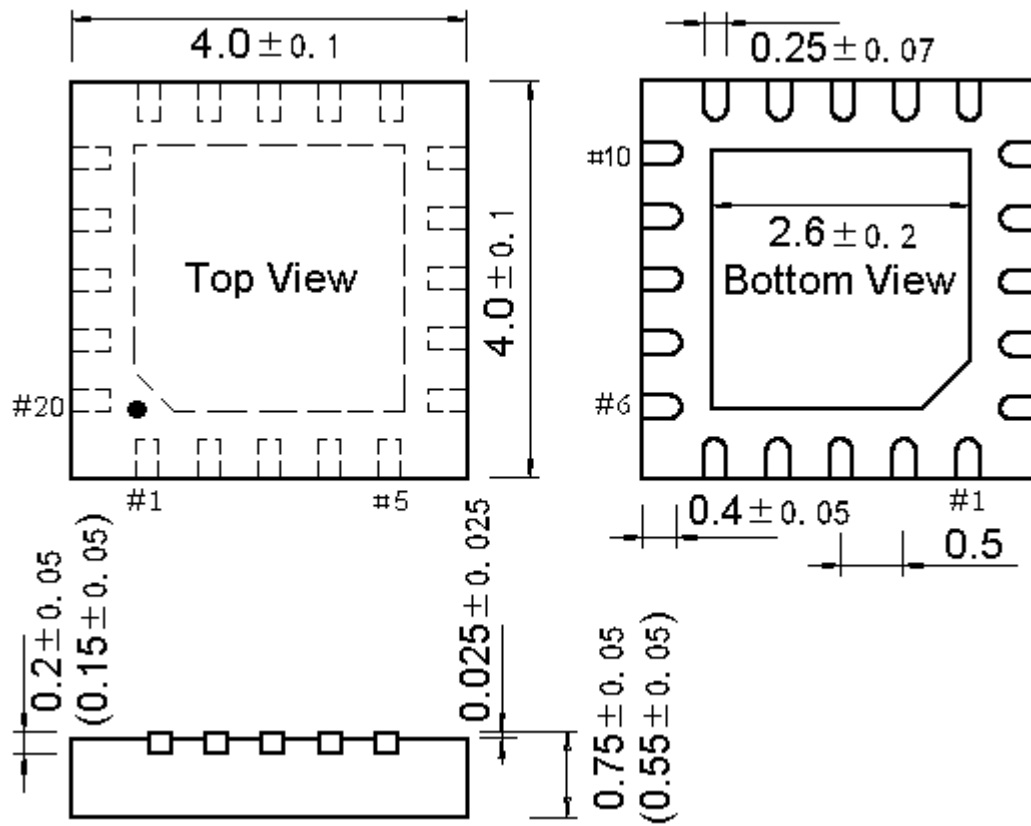
## . QFN20L (QFN20L-3\*3-0.4)



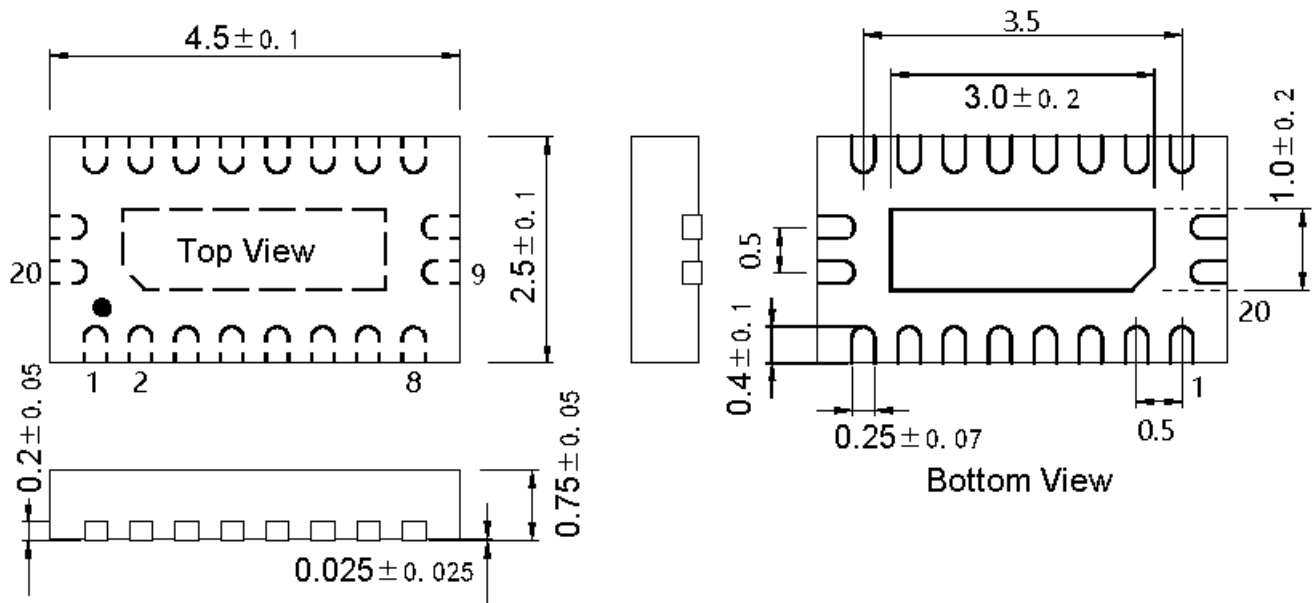
## . QFN20-3\*3\*0.55-0.4



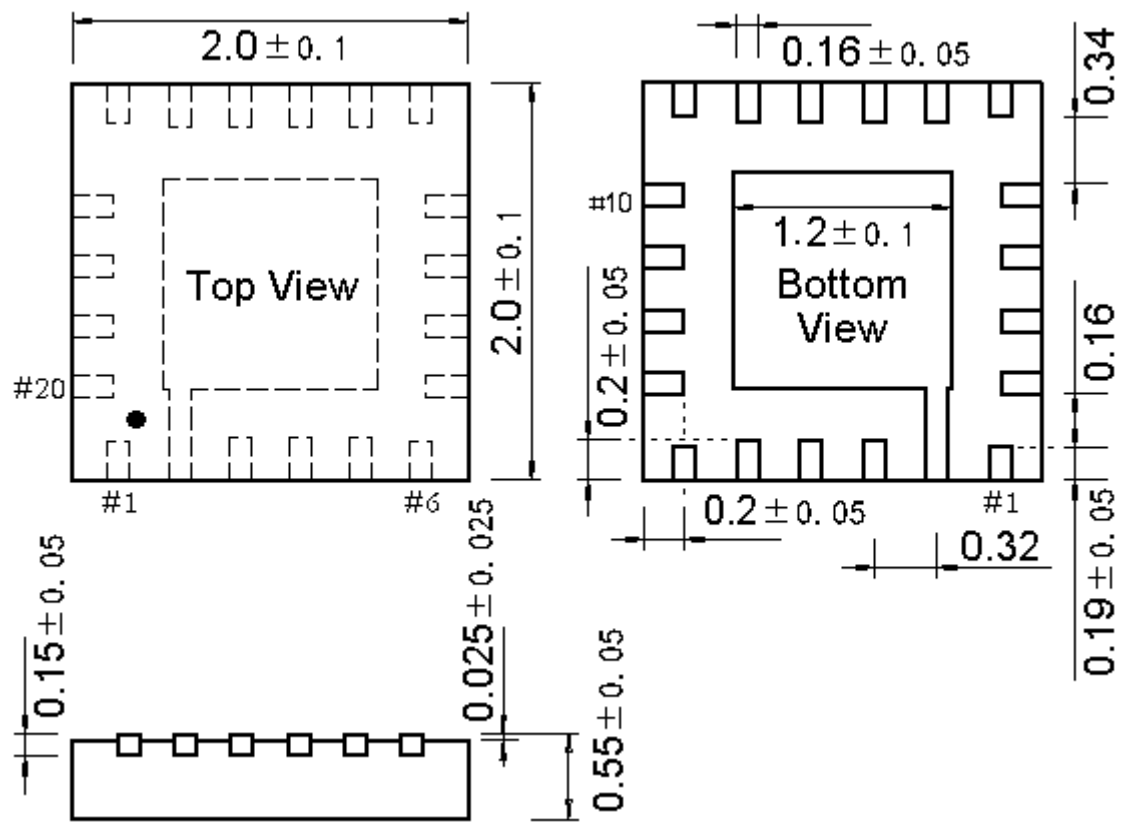
• QFN20X4 (QFN20-4\*4-0.5)



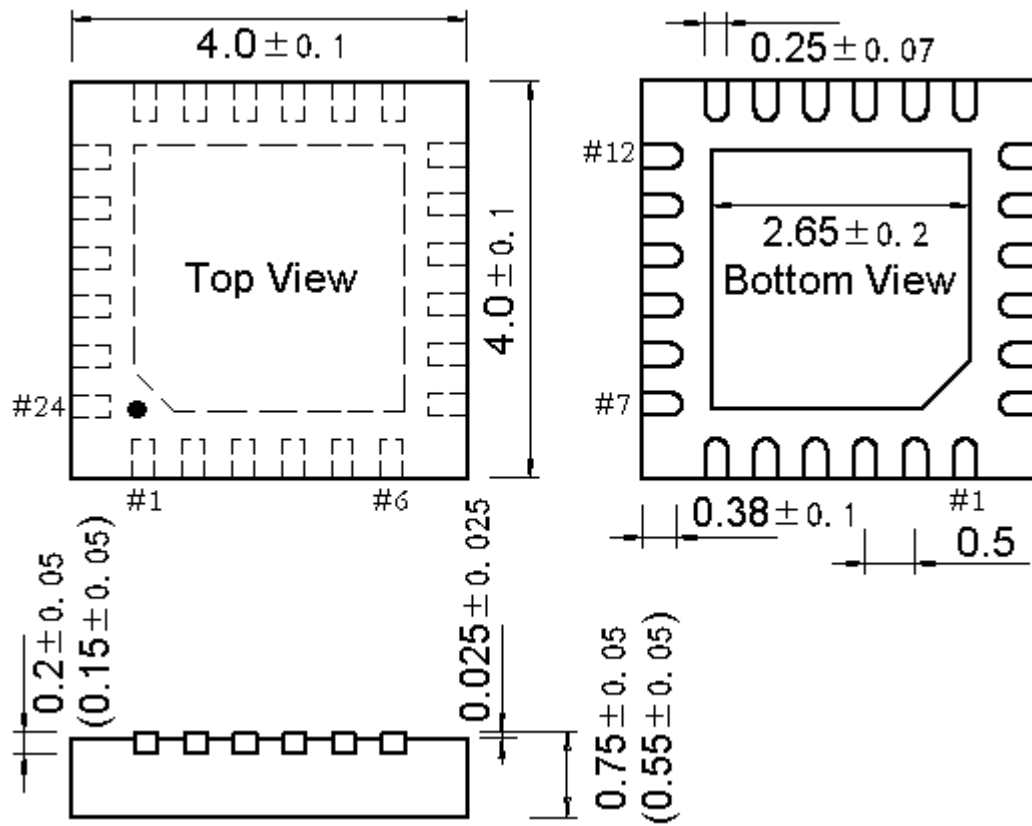
## . QFN20X25X45 (QFN20-2.5\*4.5\*0.75-0.5)



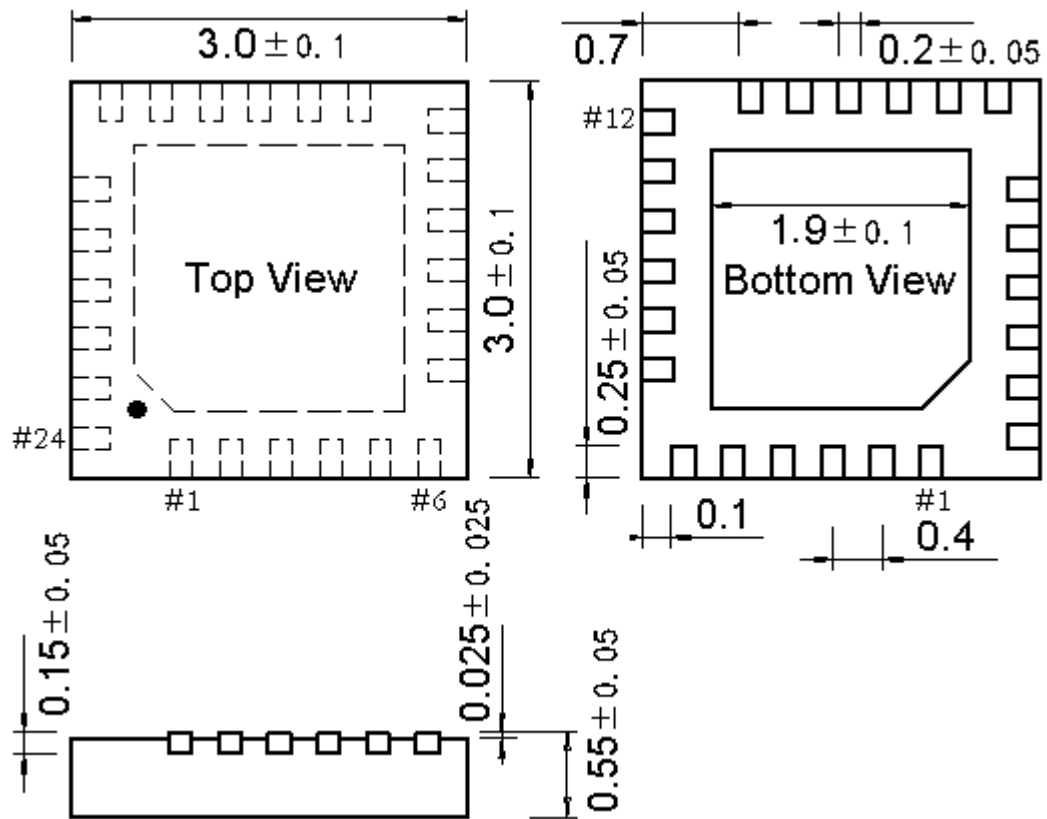
• QFN20C2 (QFN20C-2\*2\*0.55-0.32)



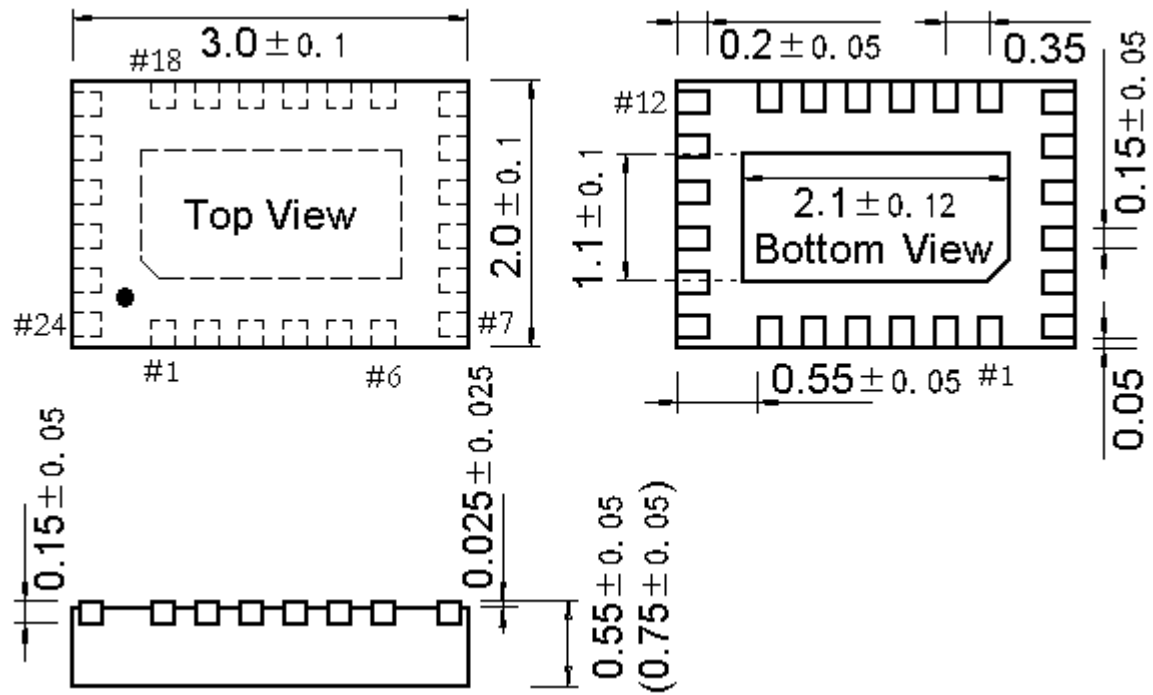
## . QFN24 (QFN24-4\*4-0.5)



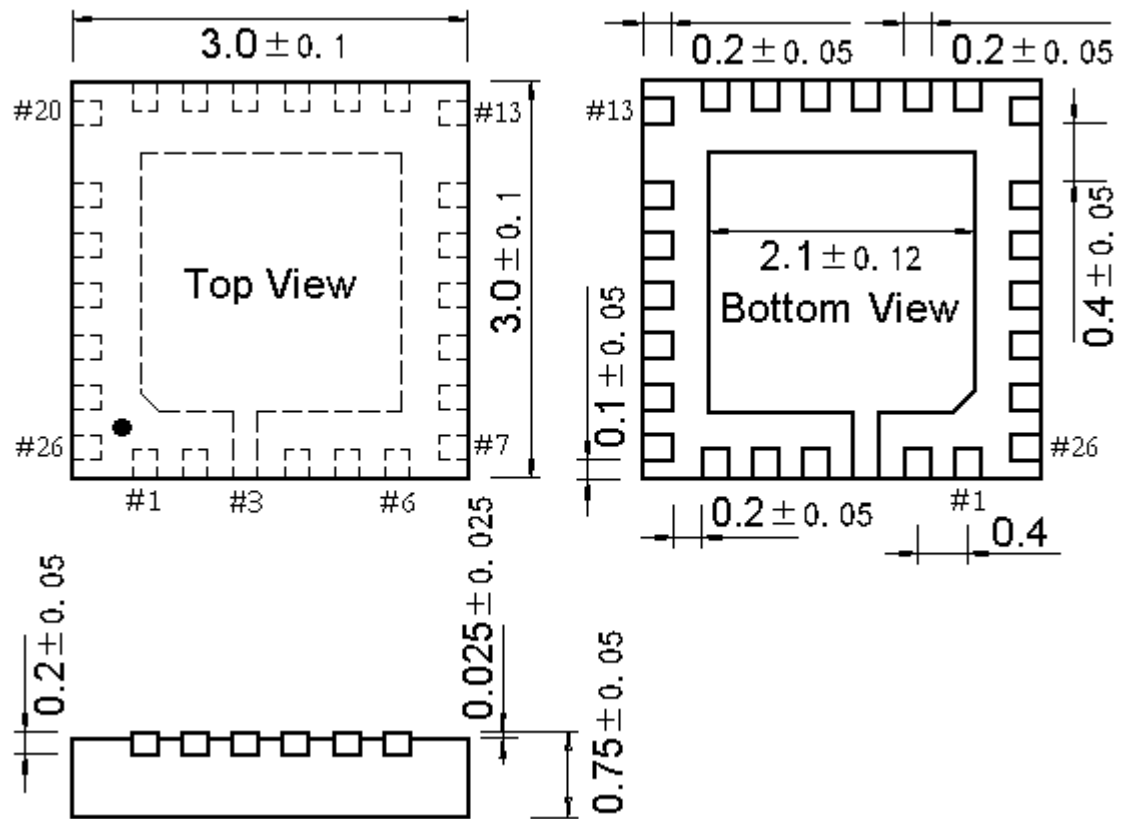
## . QFN24X3 (QFN24-3\*3\*0.55-0.4)



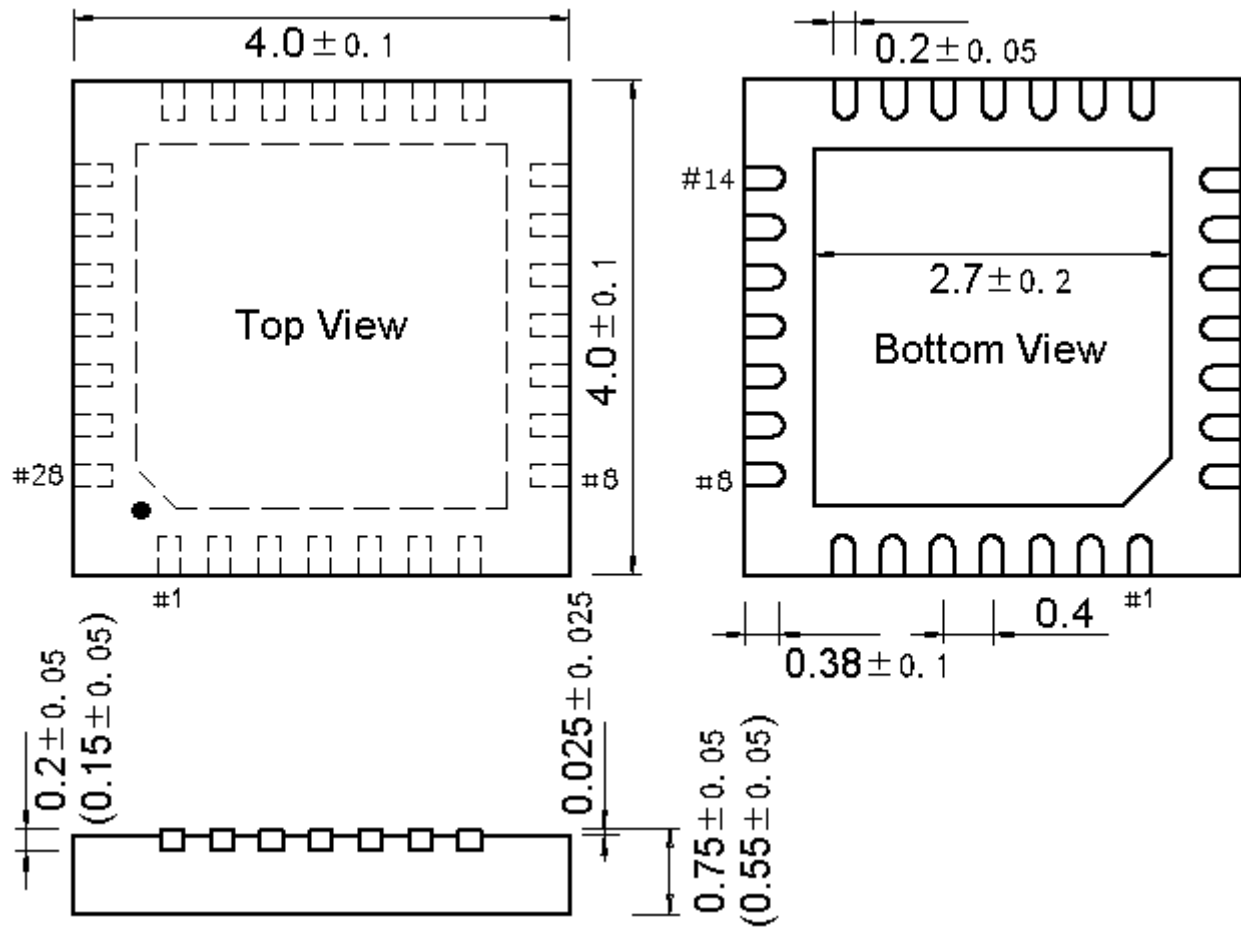
• QFN24C2X3 (QFN24C-2\*3-0.35)



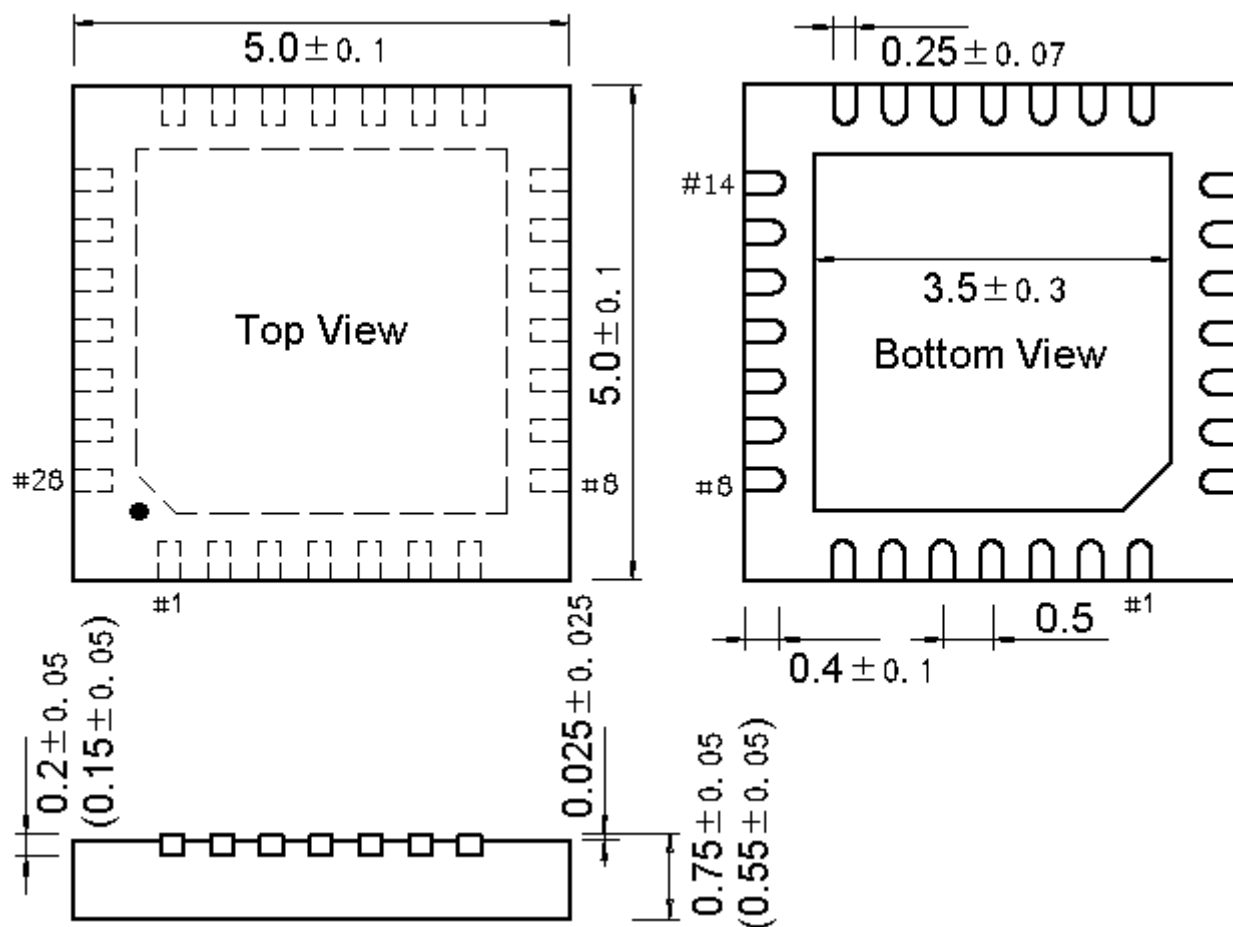
## . QFN26C3 (QFN26C-3\*3\*0.75-0.4)



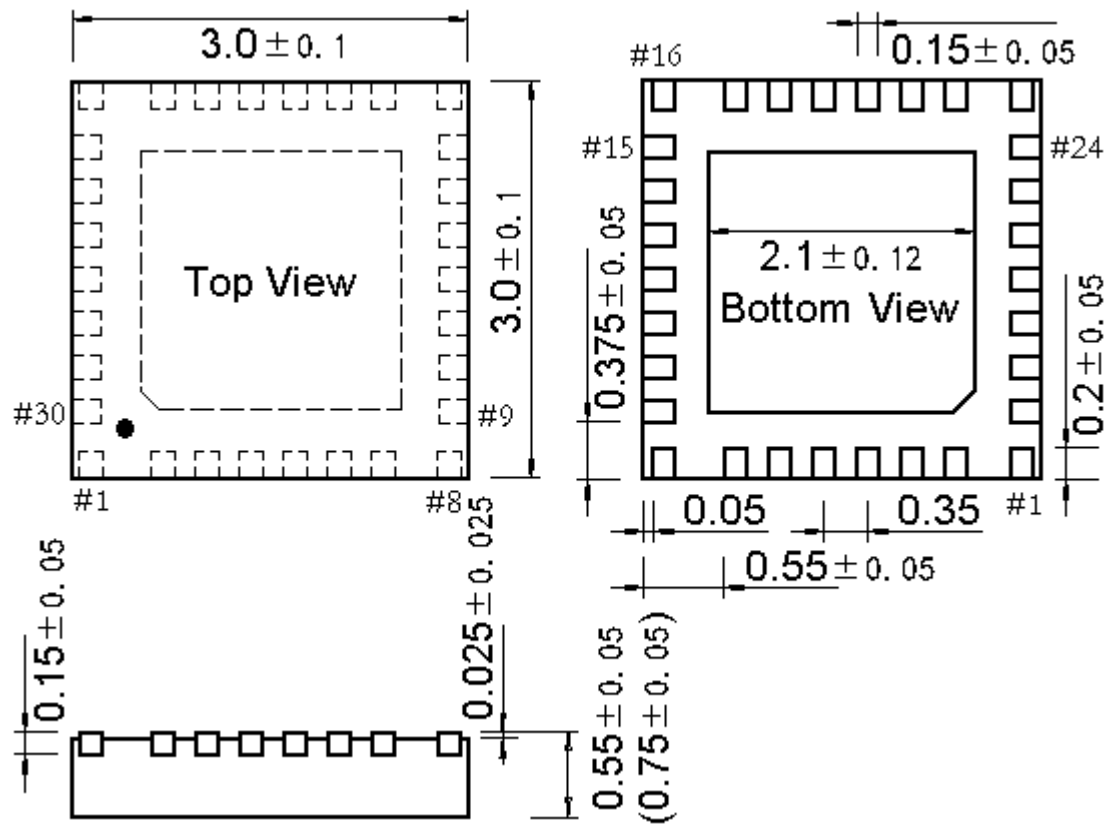
• QFN28 (QFN28-4\*4-0.4)



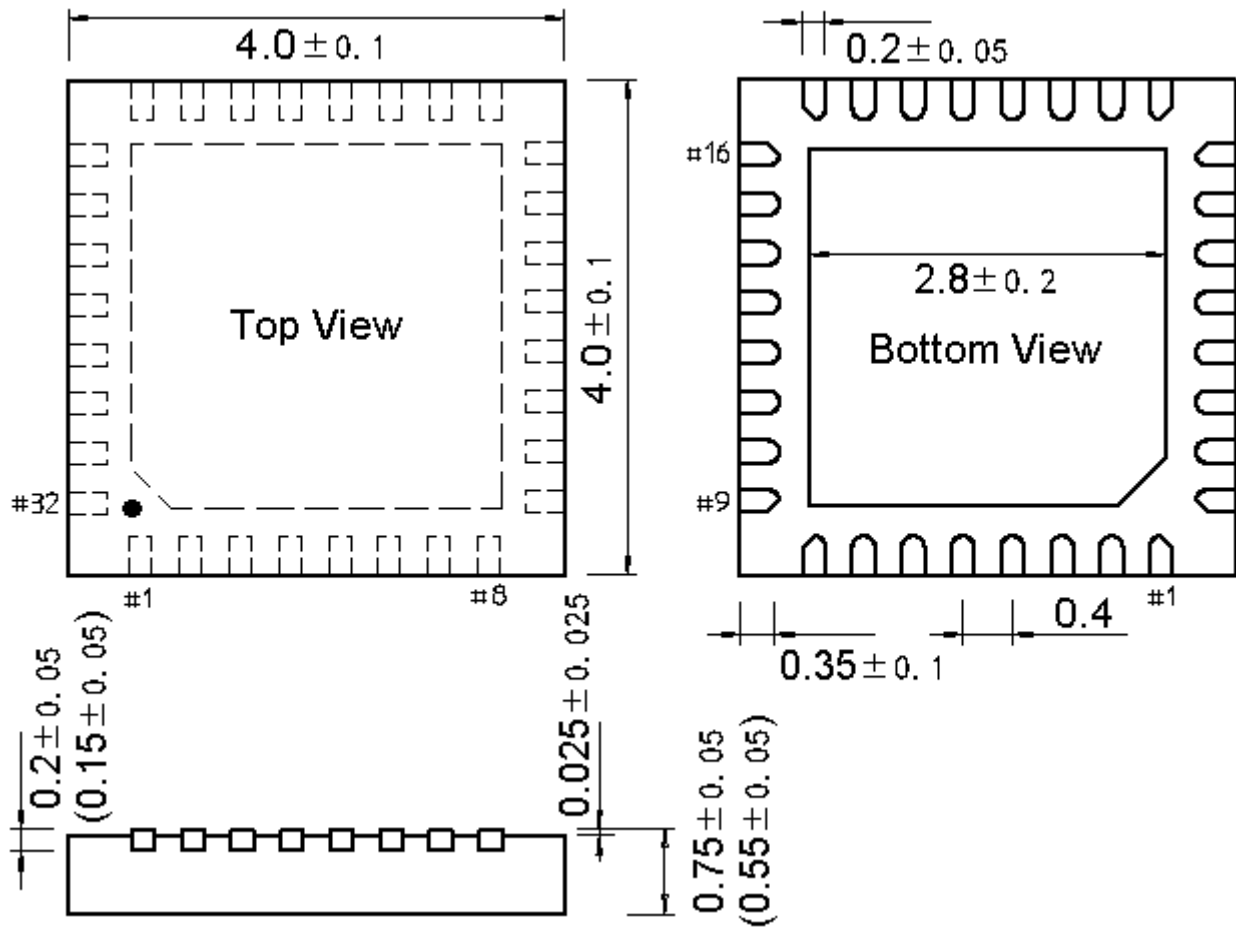
**. QFN28X5 (QFN28-5\*5-0.5)**



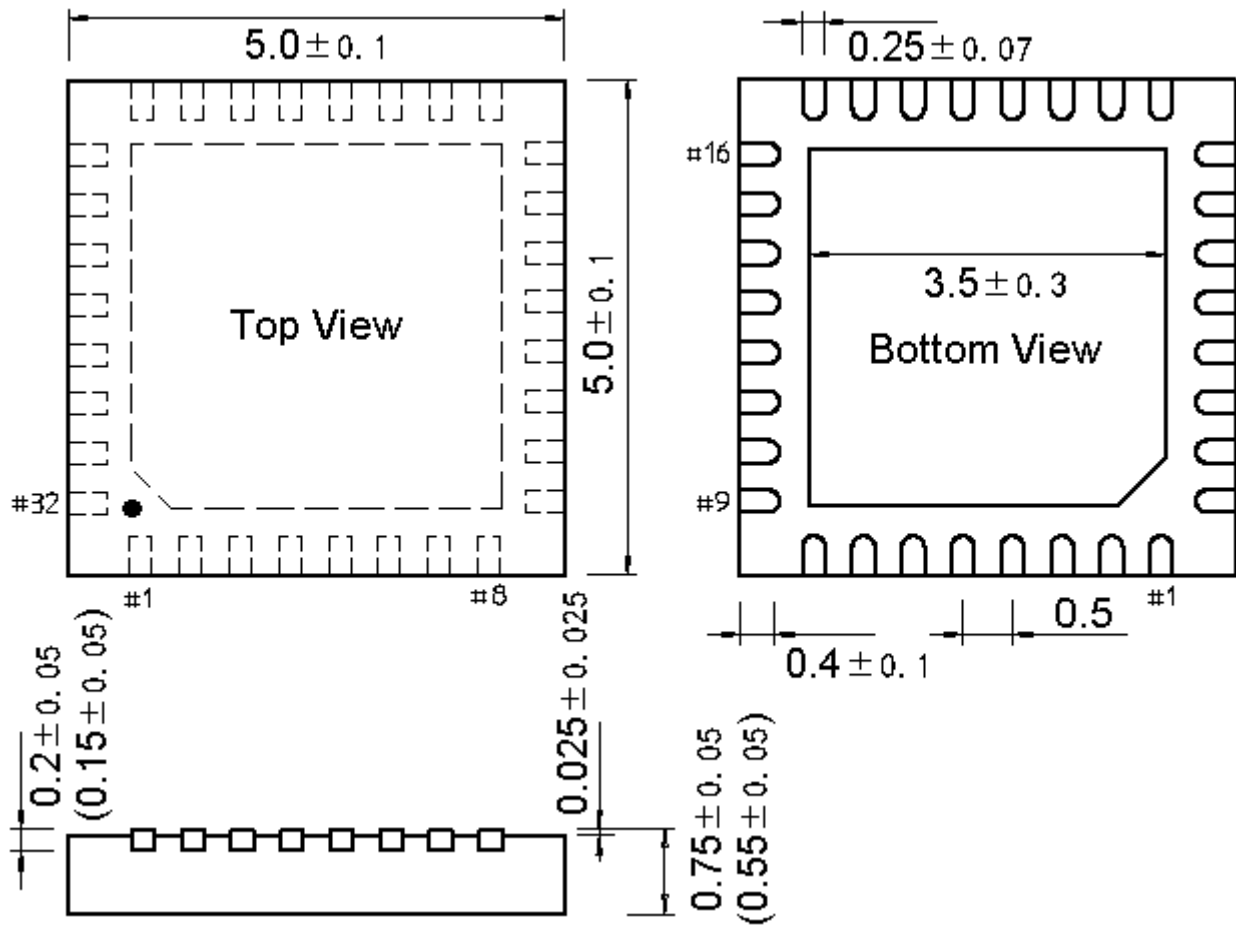
• QFN30C3 (QFN30C-3\*3-0.35)



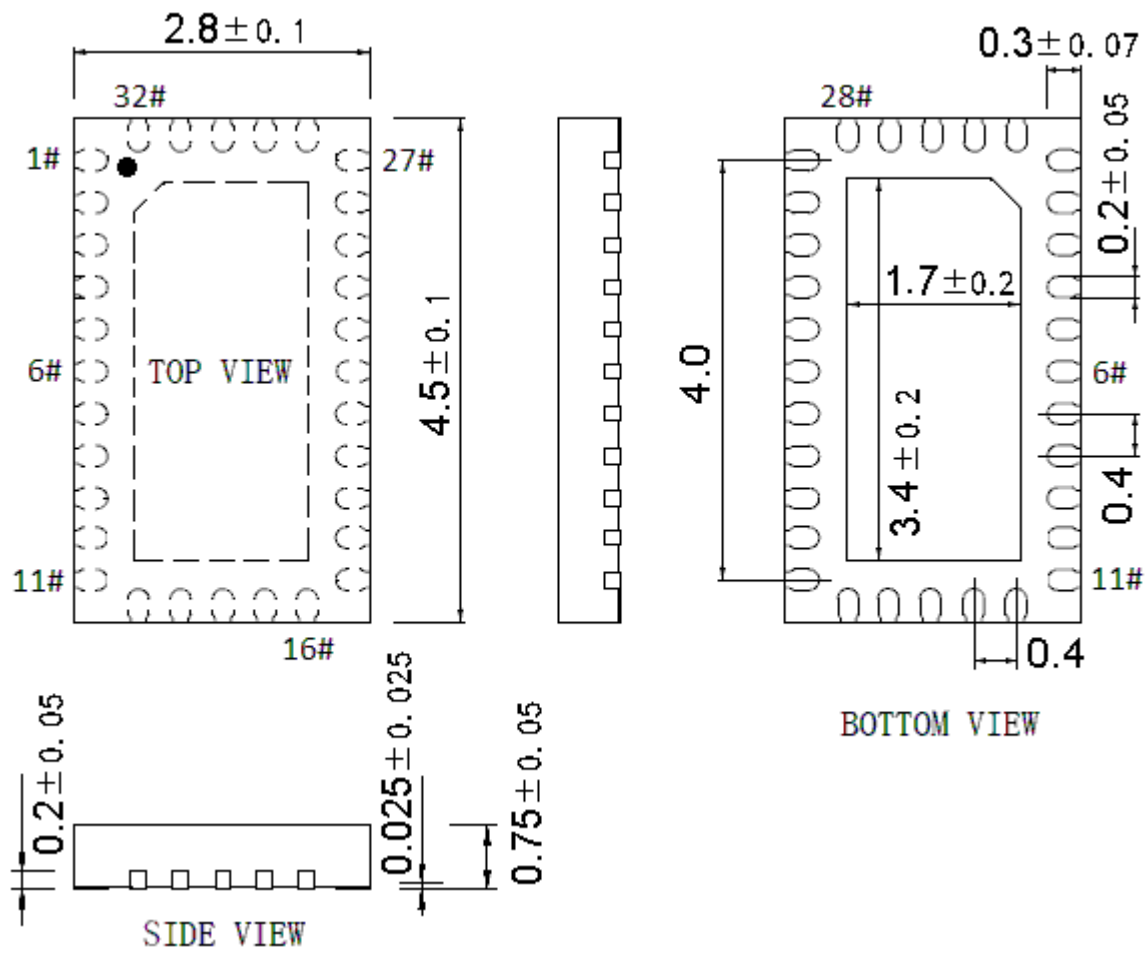
• QFN32 (QFN32-4\*4-0.4)



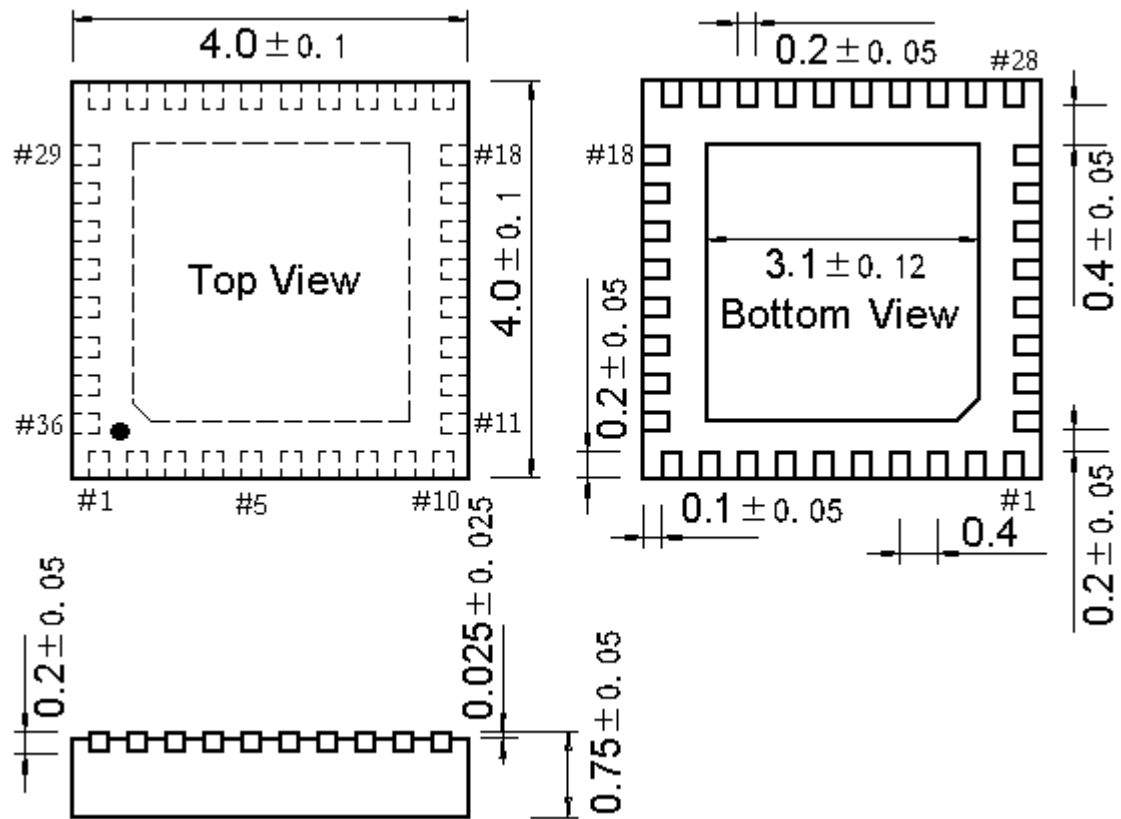
• QFN32X5 (QFN32-5\*5-0.5)



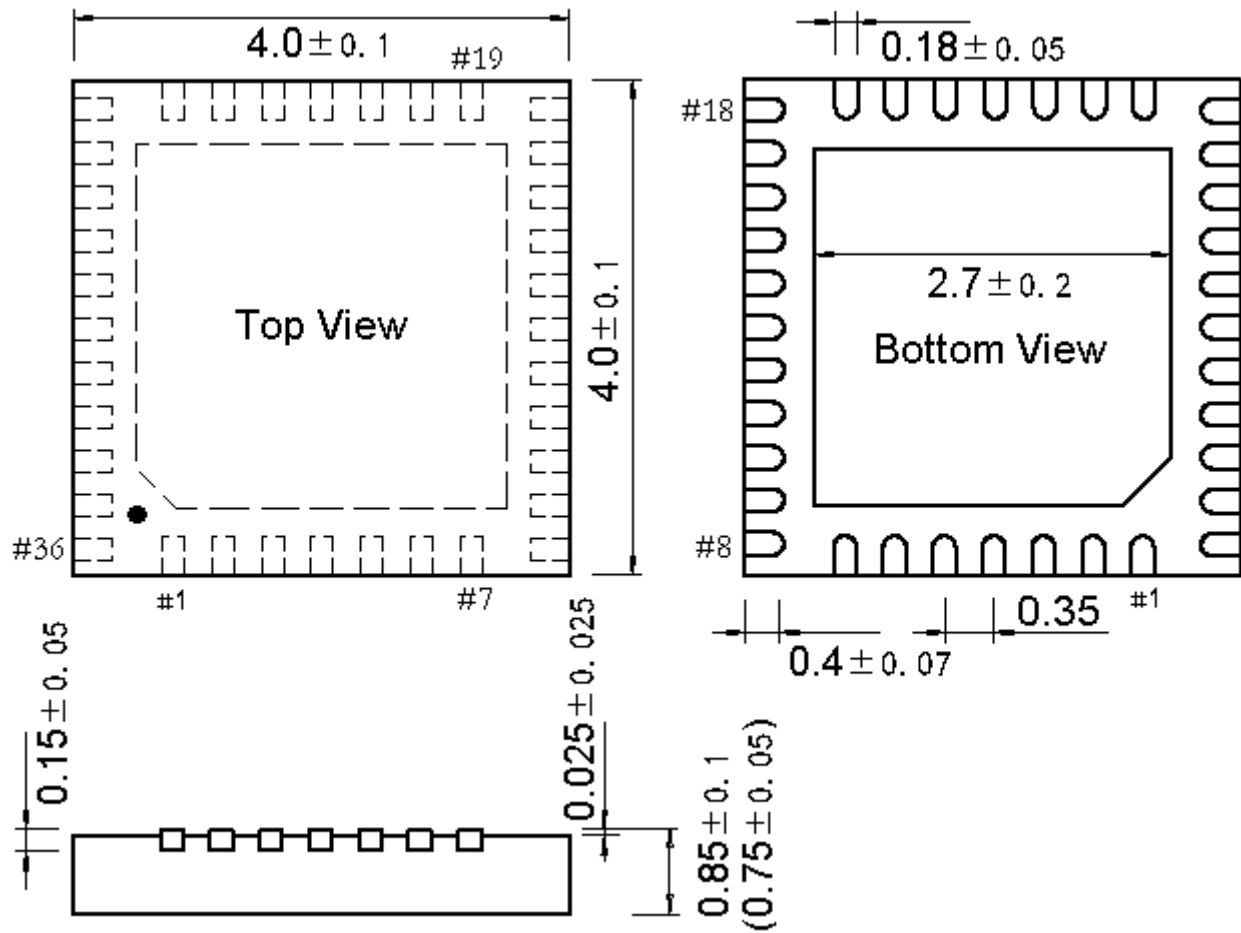
• QFN32C28X45 (QFN32C-2.8\*4.5\*0.75-0.4)



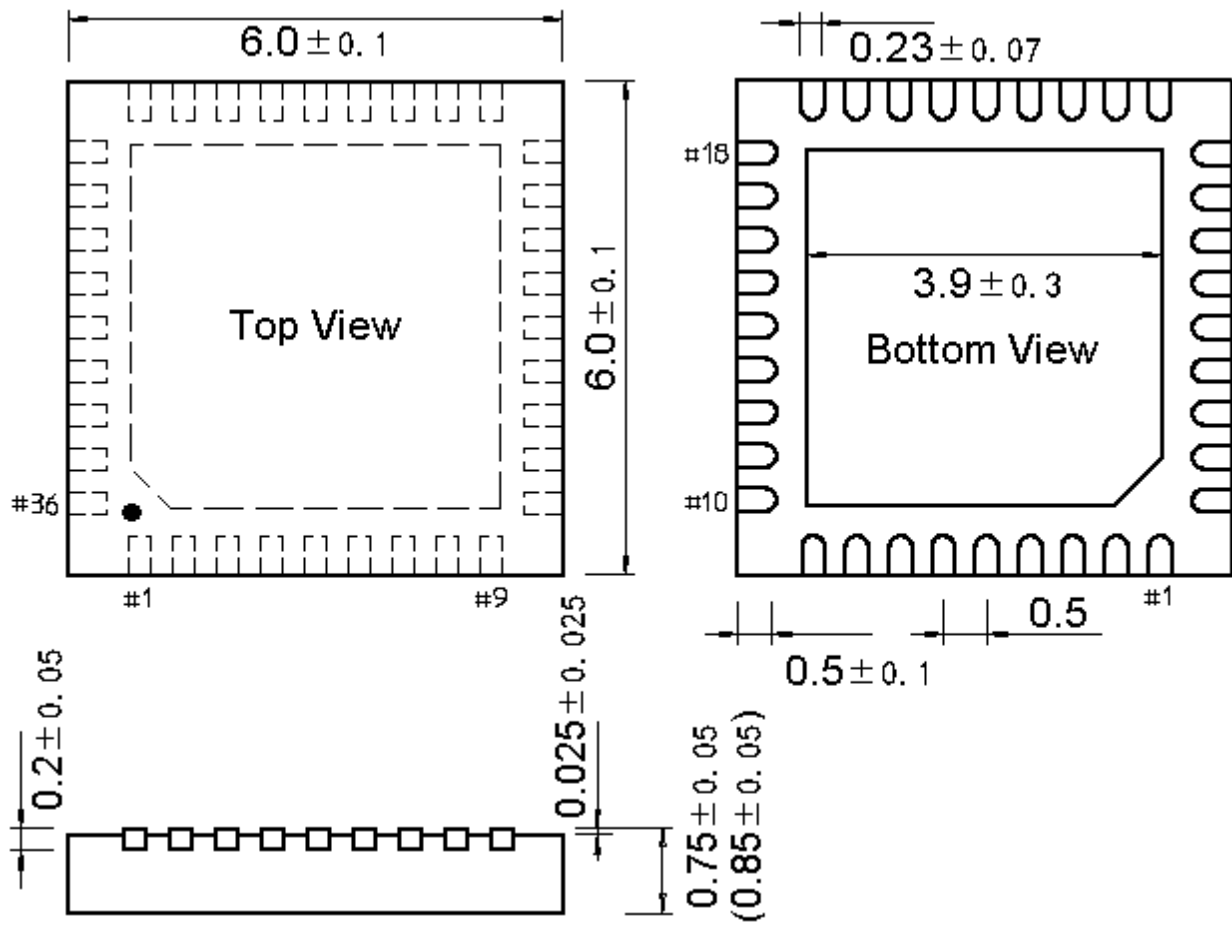
• QFN36C4 (QFN36C-4\*4\*0.75-0.4)



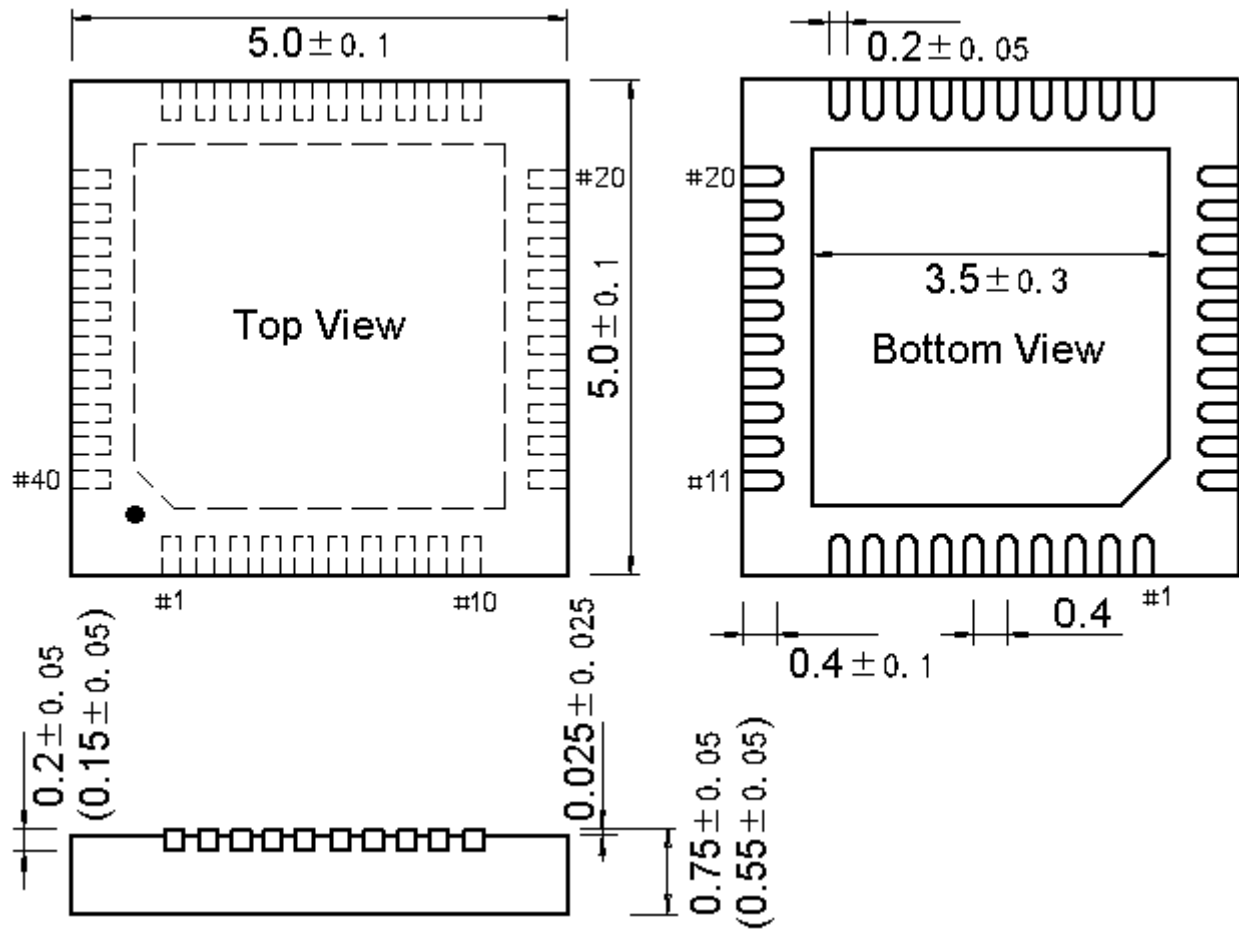
• QFN36X4 (QFN36-4\*4-0.35)



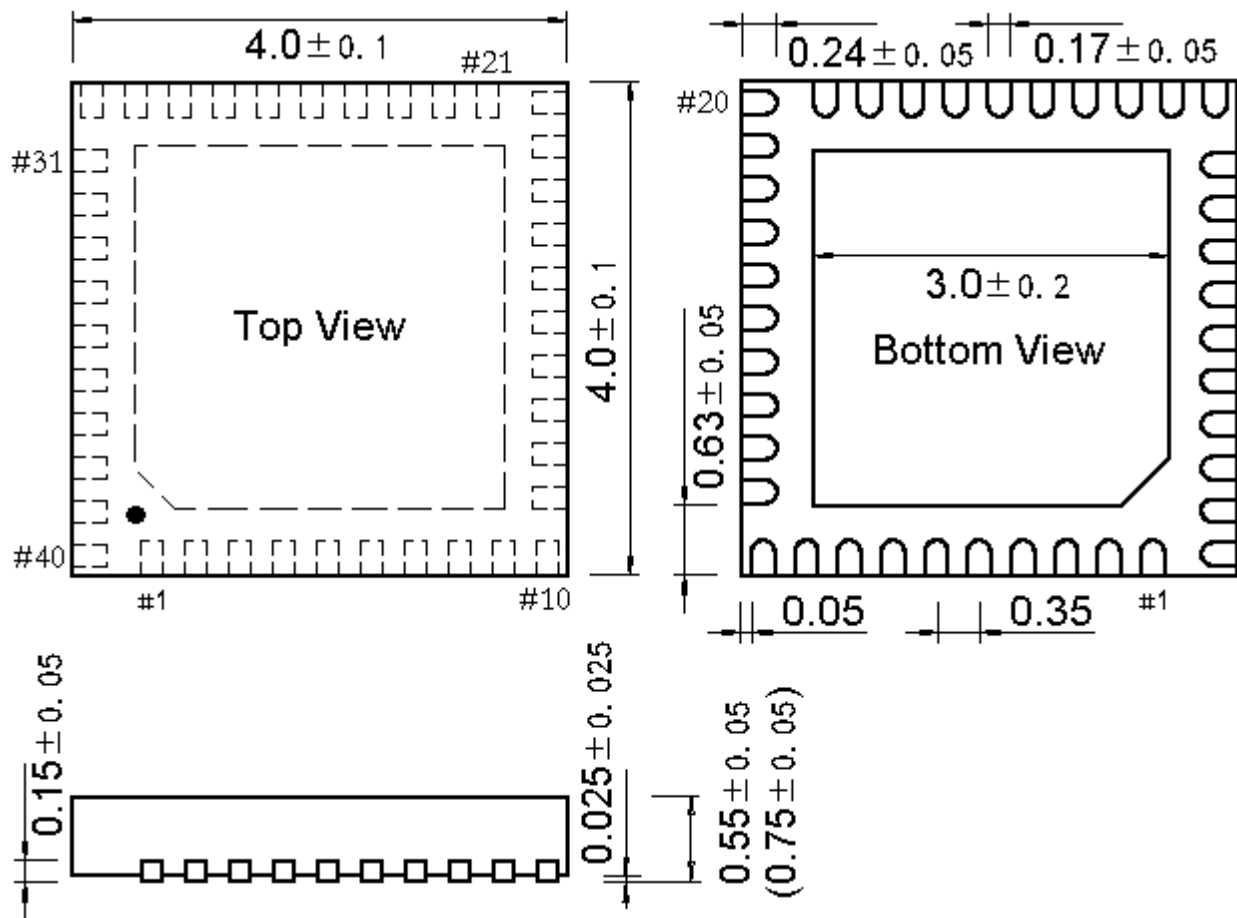
• QFN36X6 (QFN36-6\*6-0.5)



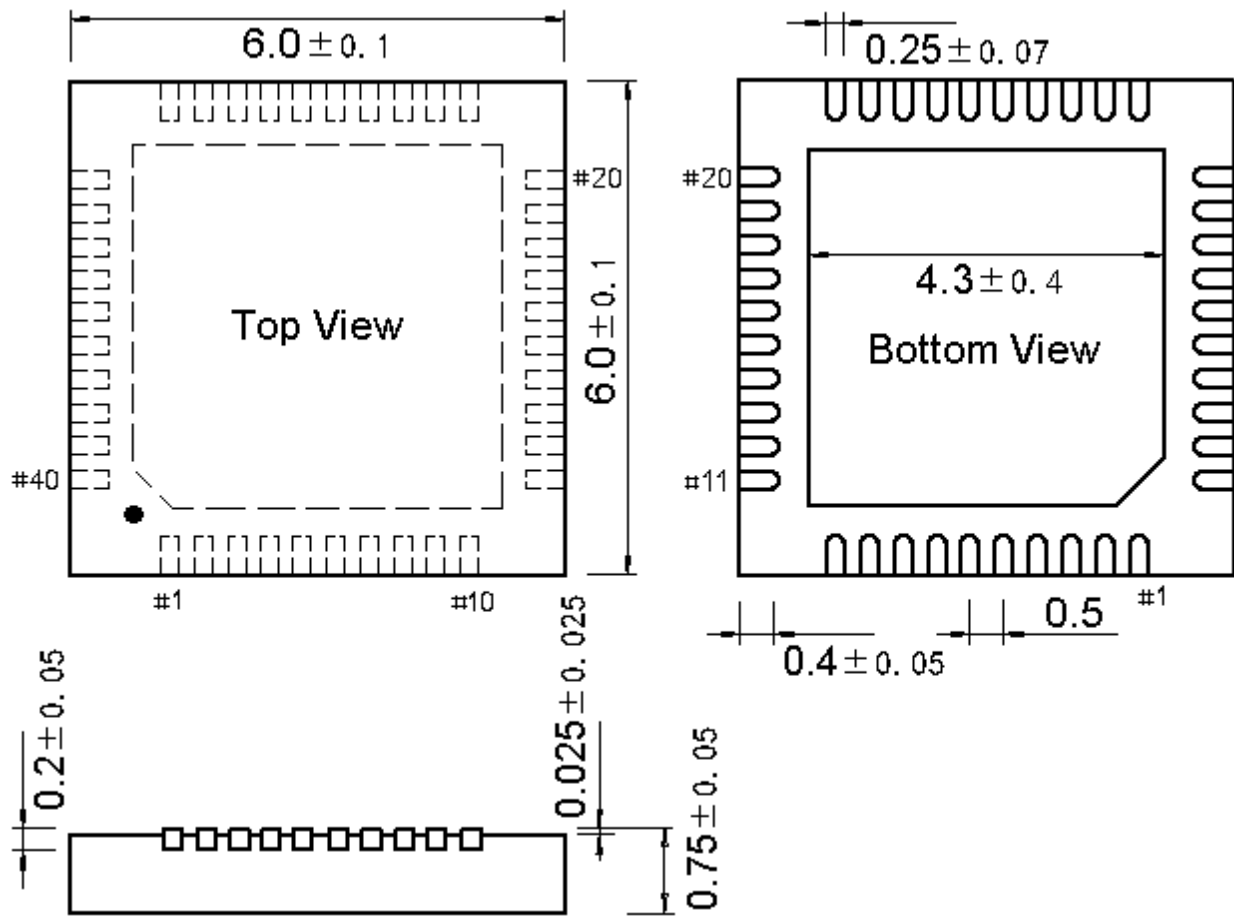
• QFN40 (QFN40-5\*5-0.4)



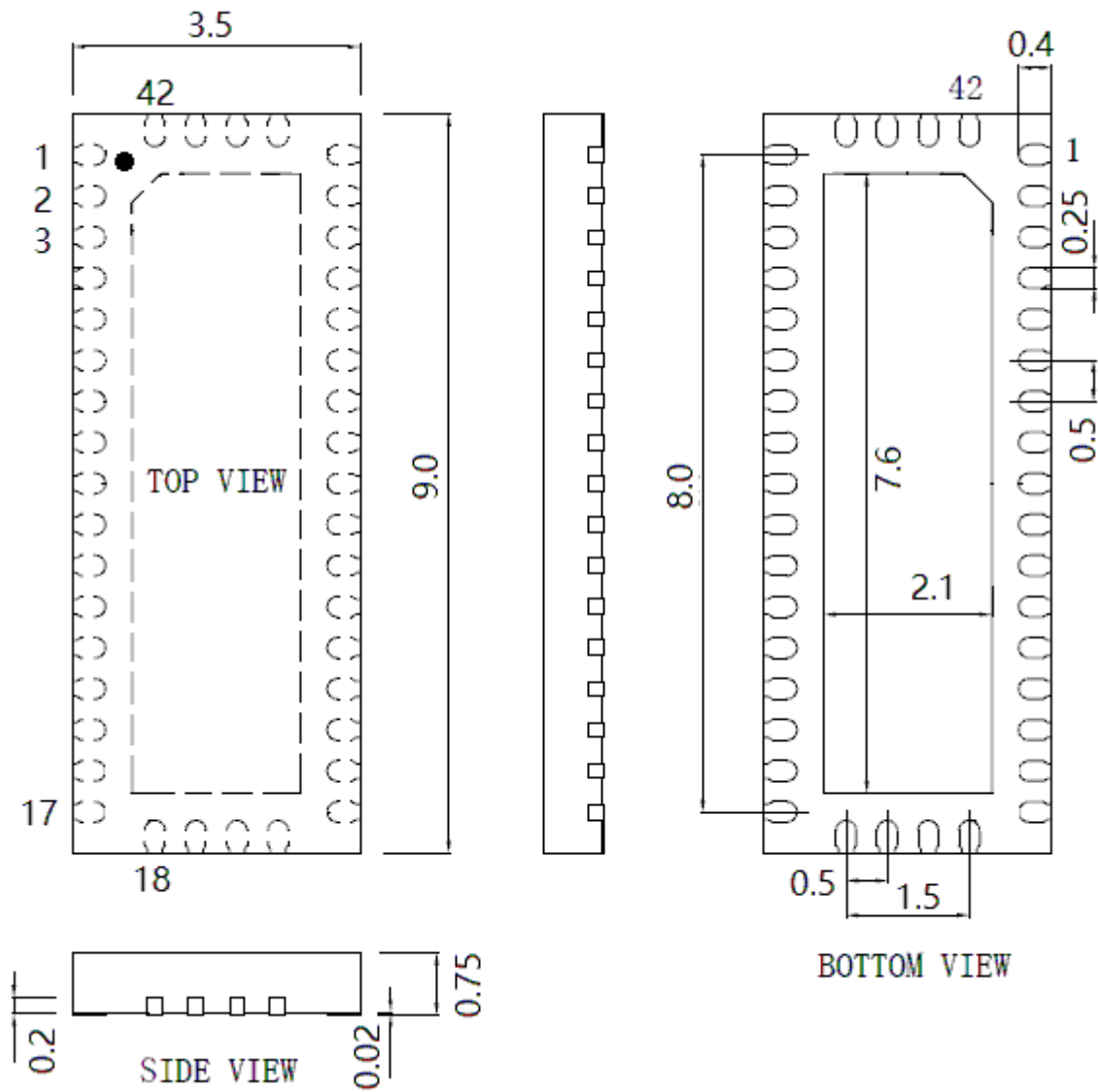
• QFN40C4 (QFN40C-4\*4-0.35)



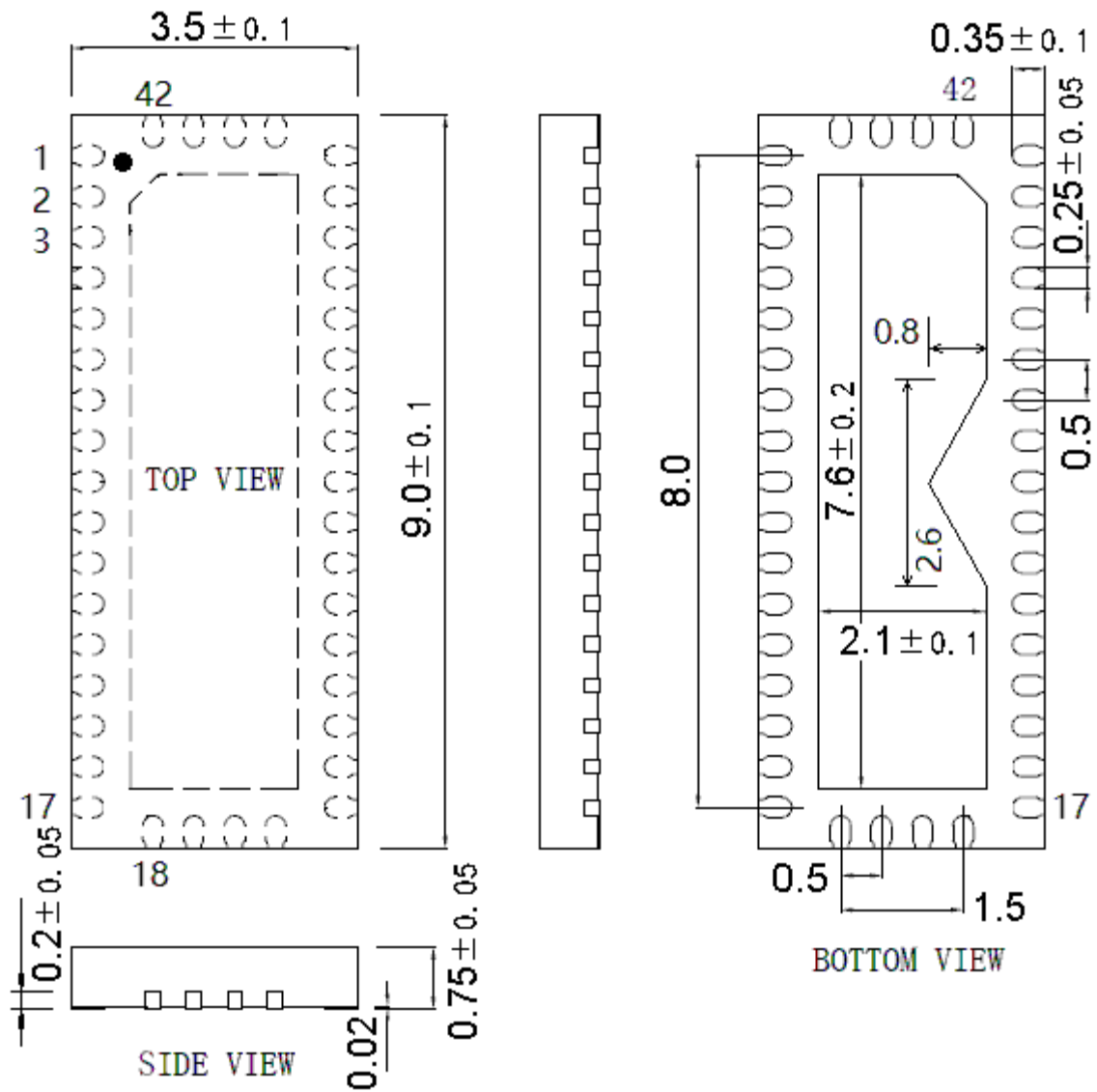
• QFN40X6 (QFN40-6\*6-0.5)



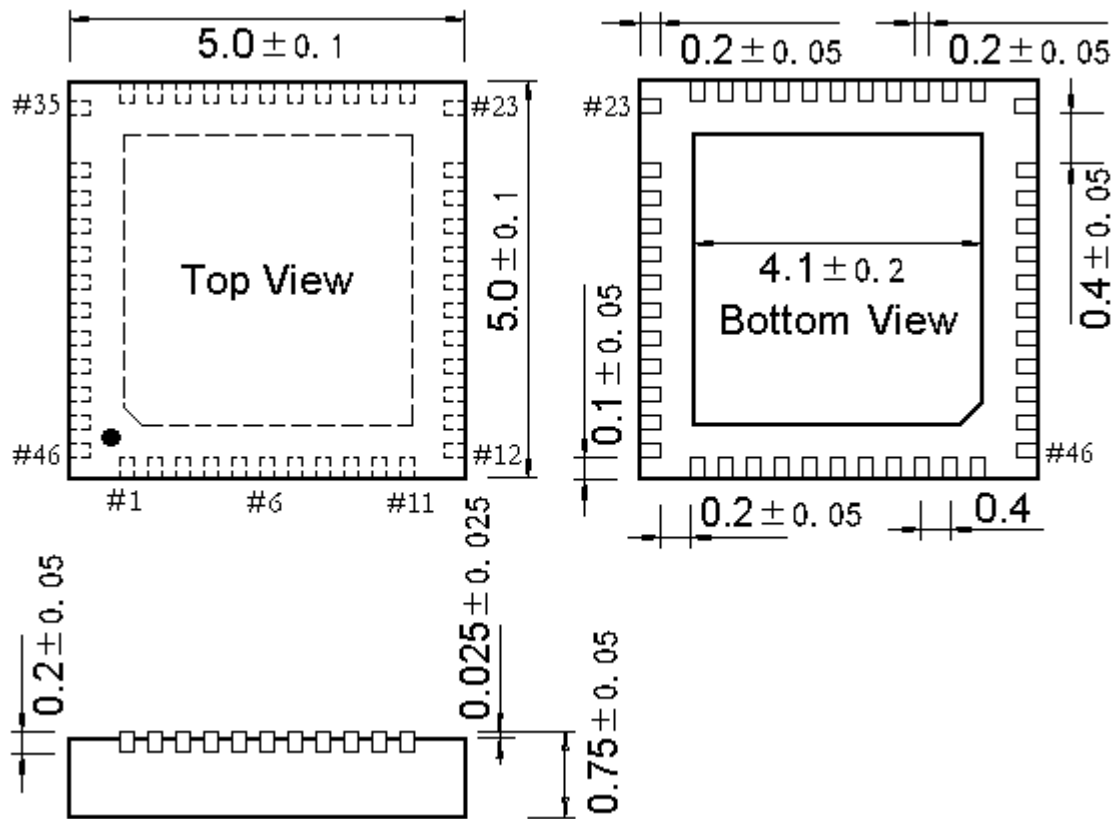
## . QFN42X35X9 (QFN42-3.5\*9\*0.75-0.5)



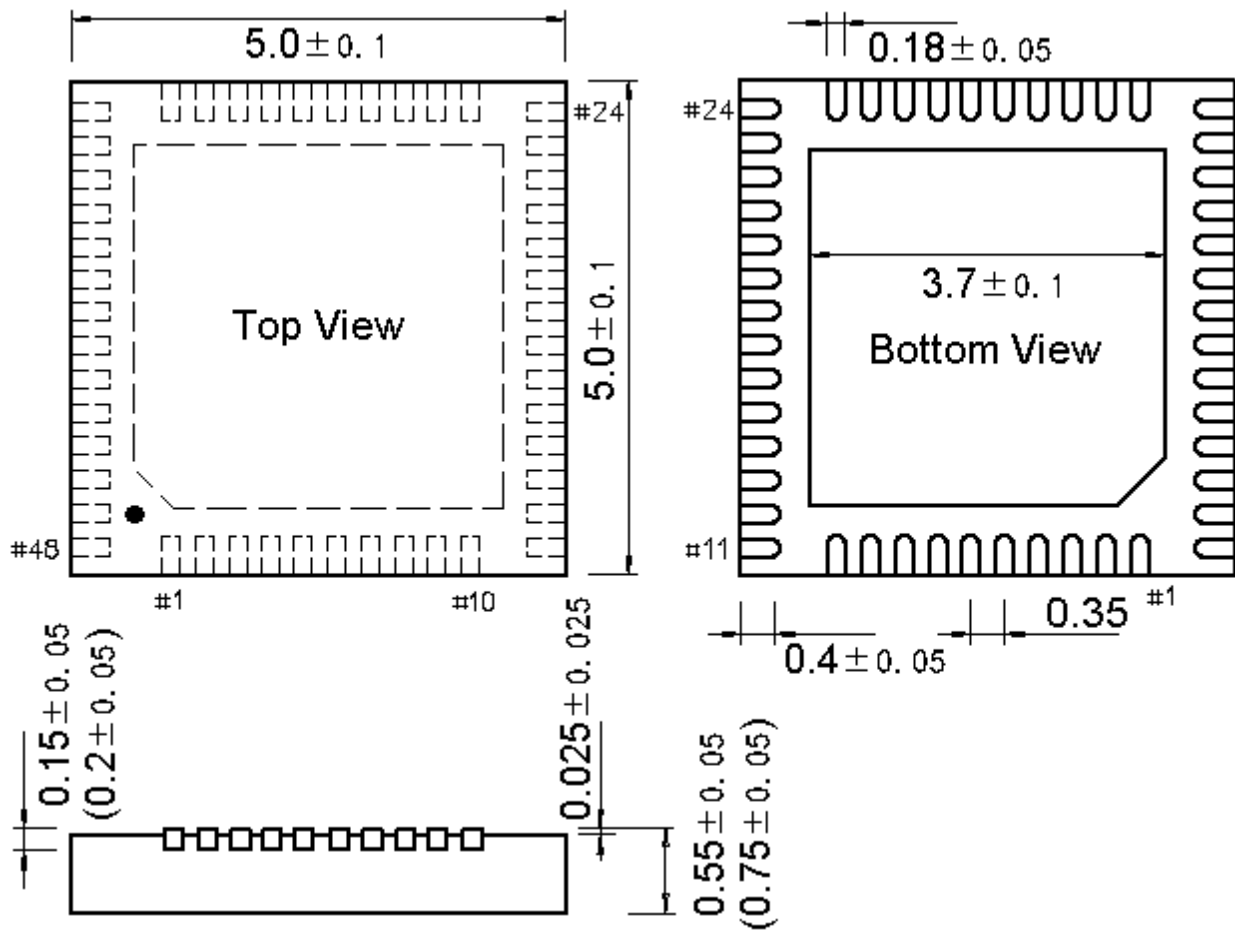
• QFN42C35X9 (QFN42C-3.5\*9\*0.75-0.5)



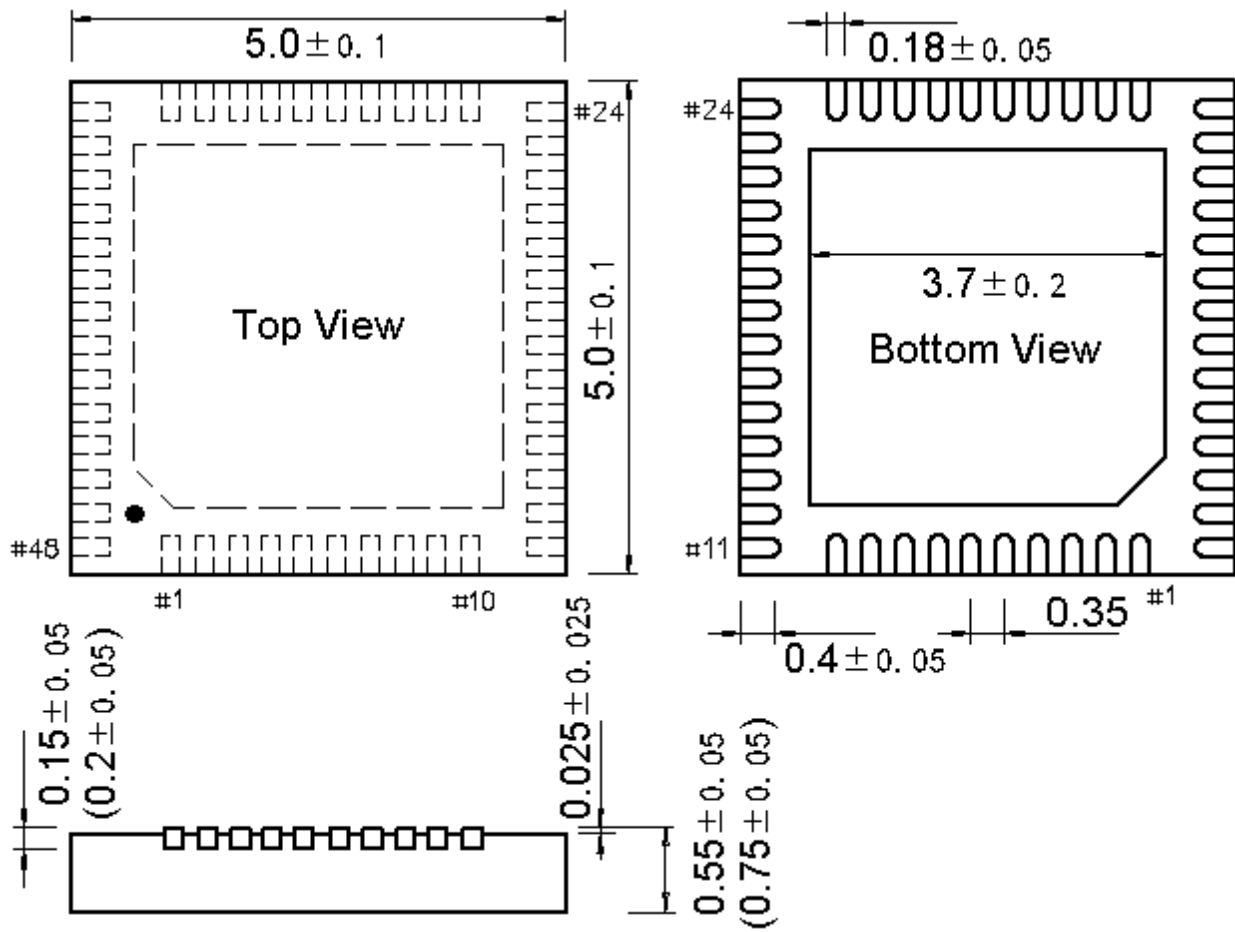
• QFN46C5 (QFN46C-5\*5\*0.75-0.4)



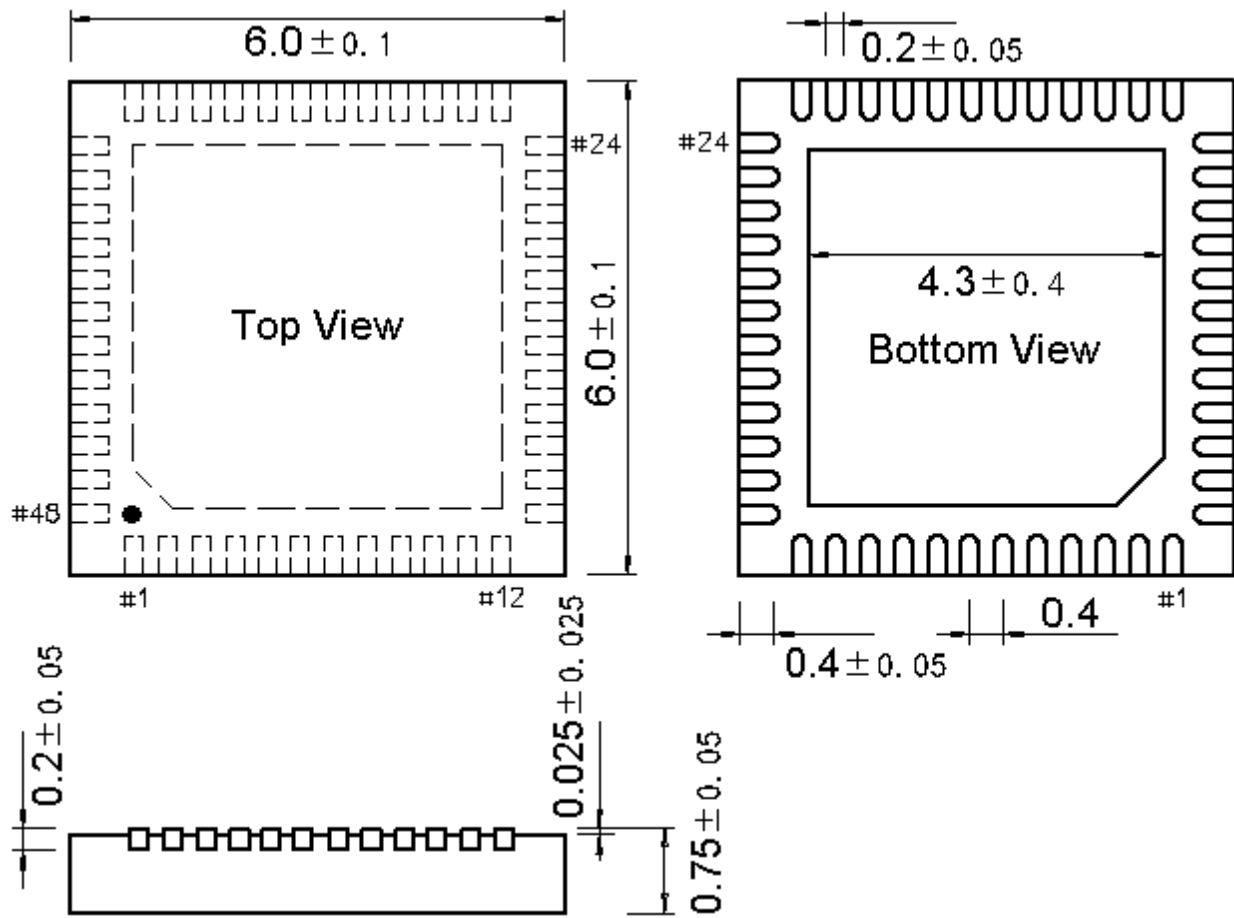
• QFN48 (QFN48-5\*5\*0.75-0.35)



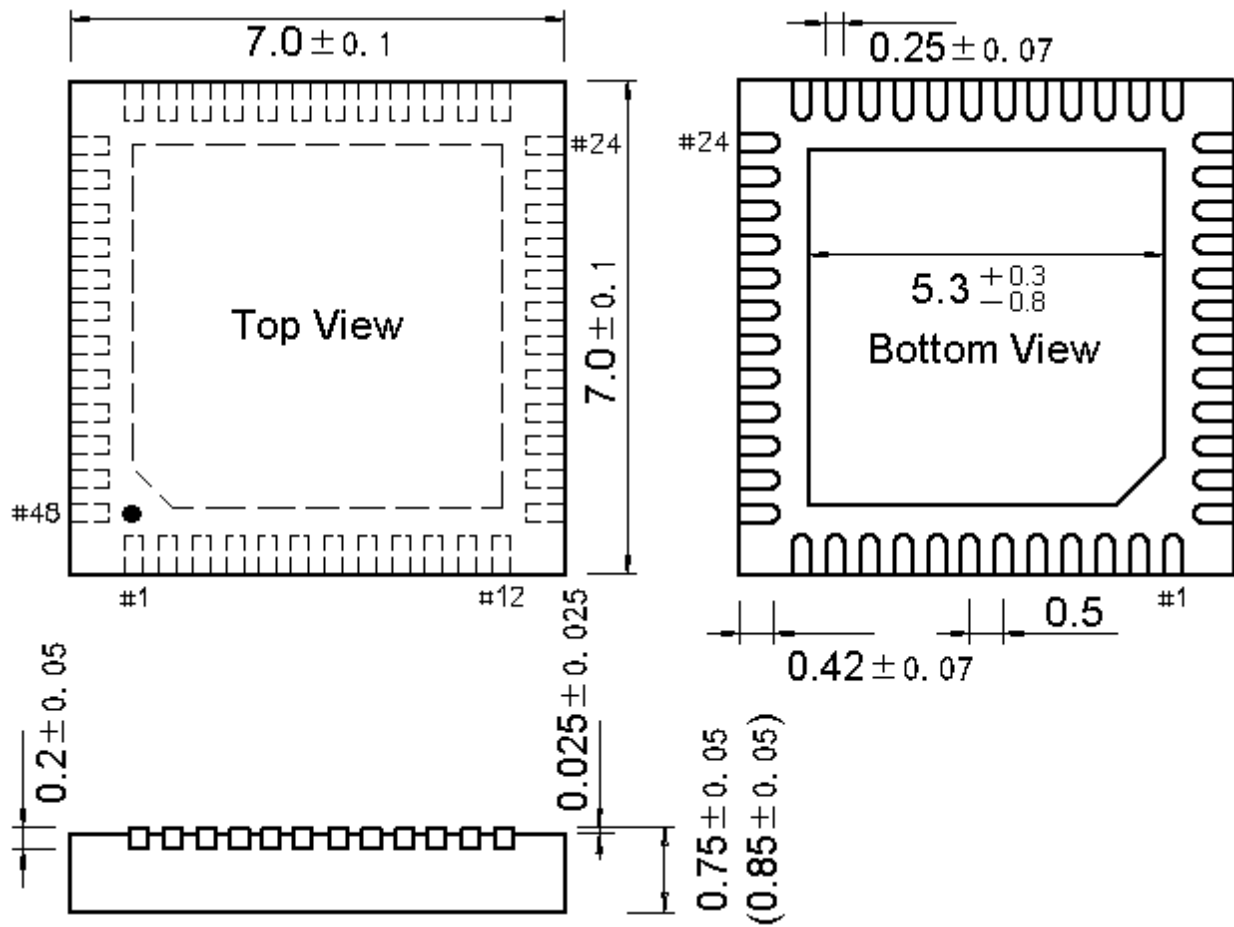
• QFN48-5\*5\*0.55-0.35



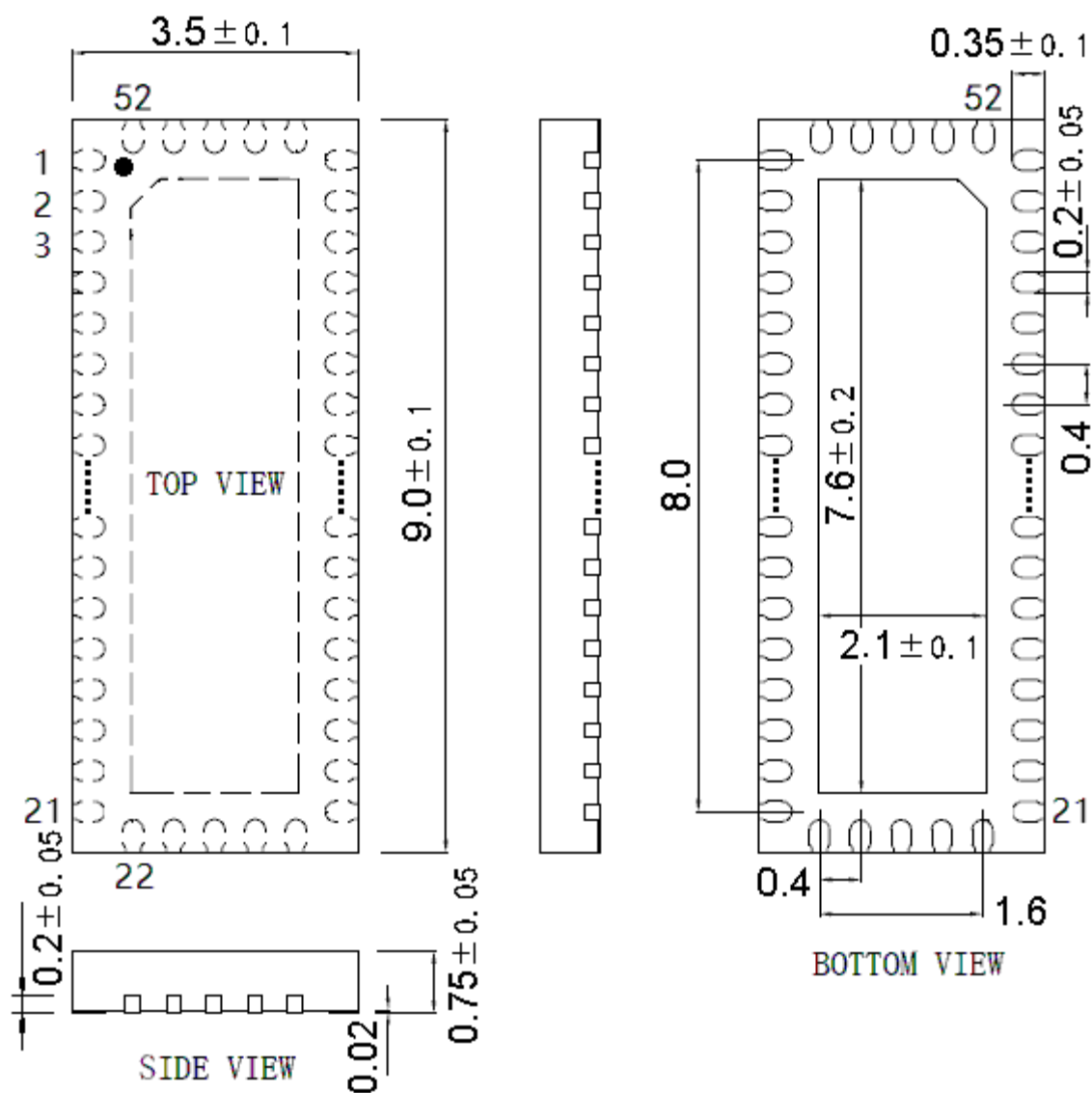
• QFN48X6 (QFN48-6\*6-0.4)



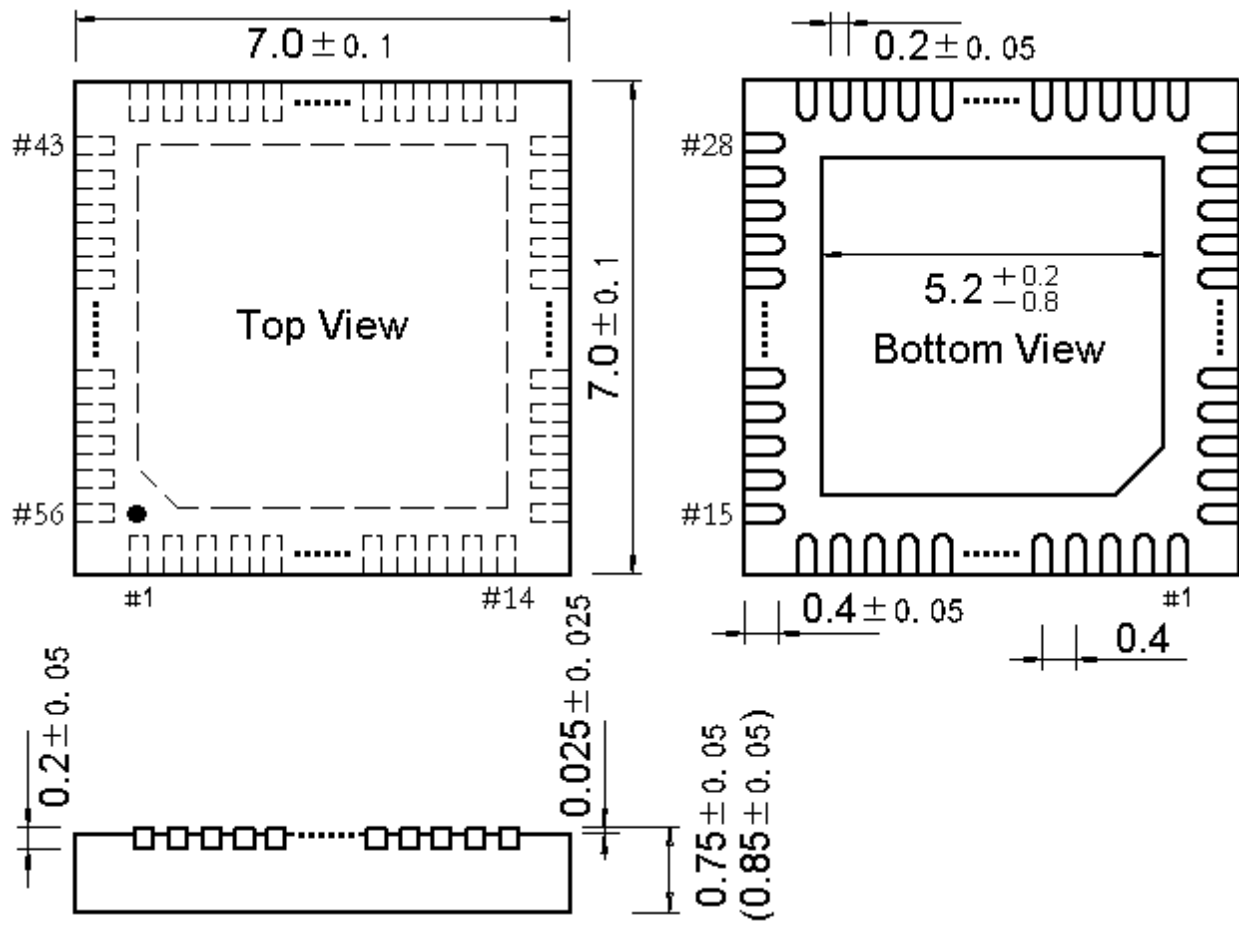
• QFN48X7 (QFN48-7\*7-0.5)



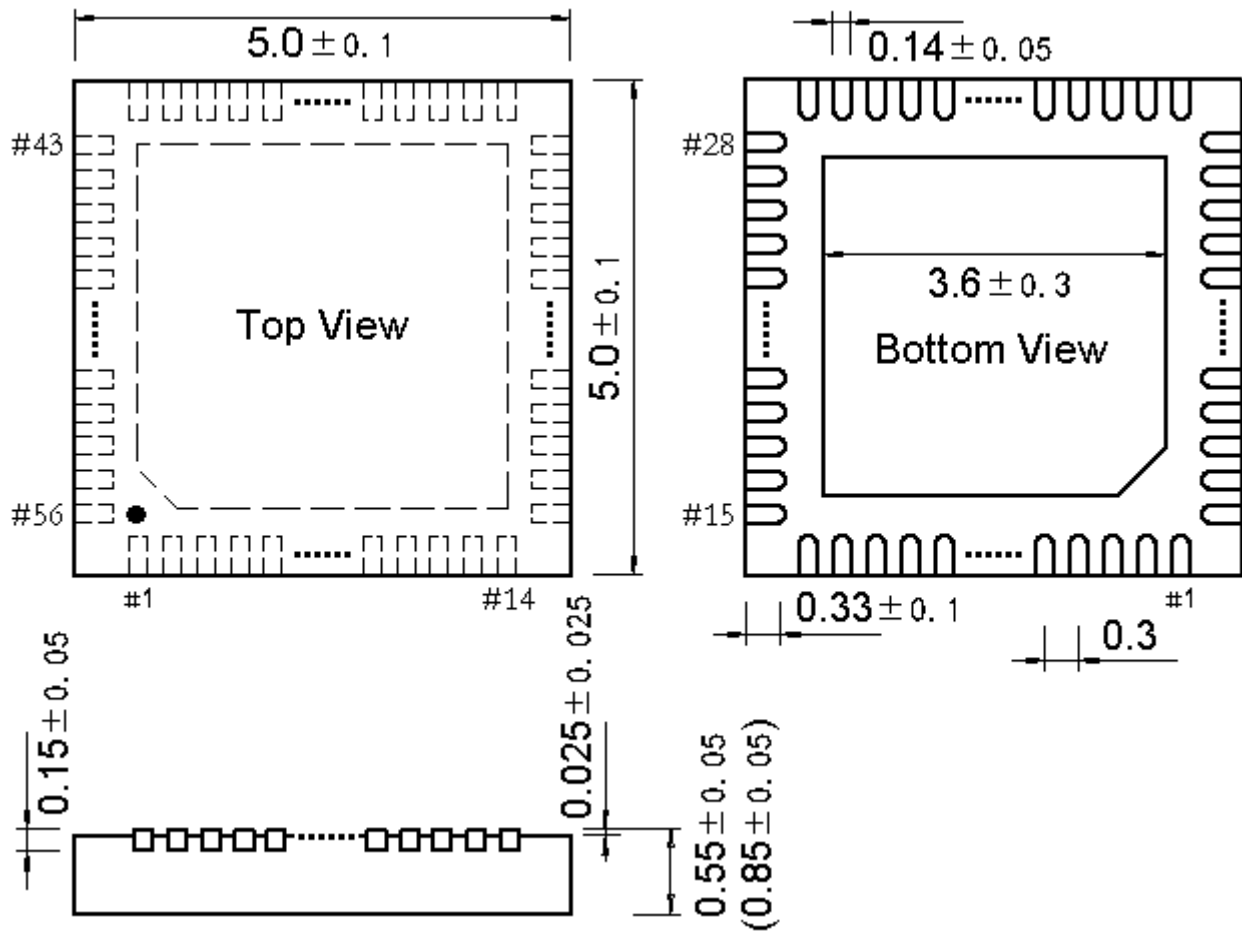
**. QFN52X35X9 (QFN52-3.5\*9\*0.75-0.4)**



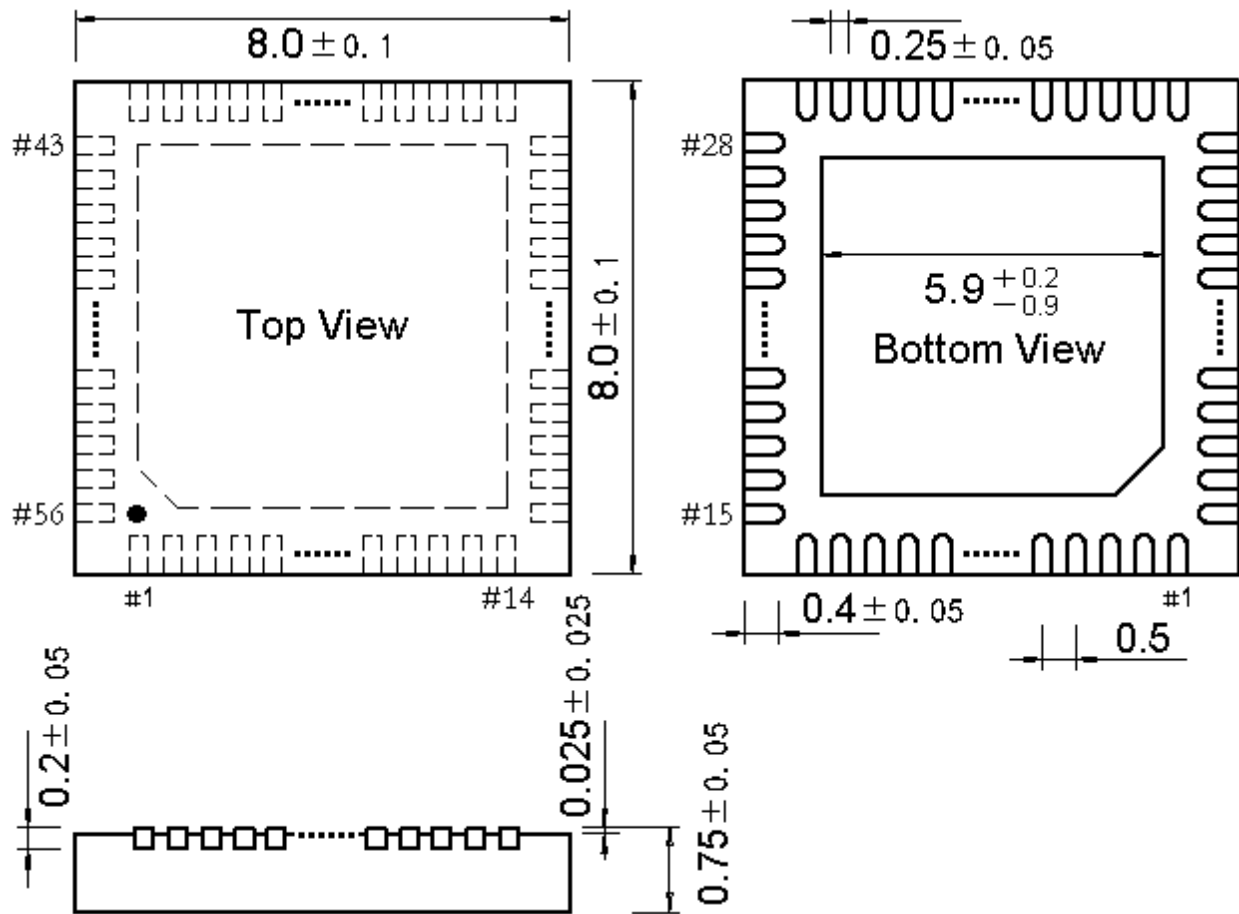
• QFN56 (QFN56-7\*7-0.4)



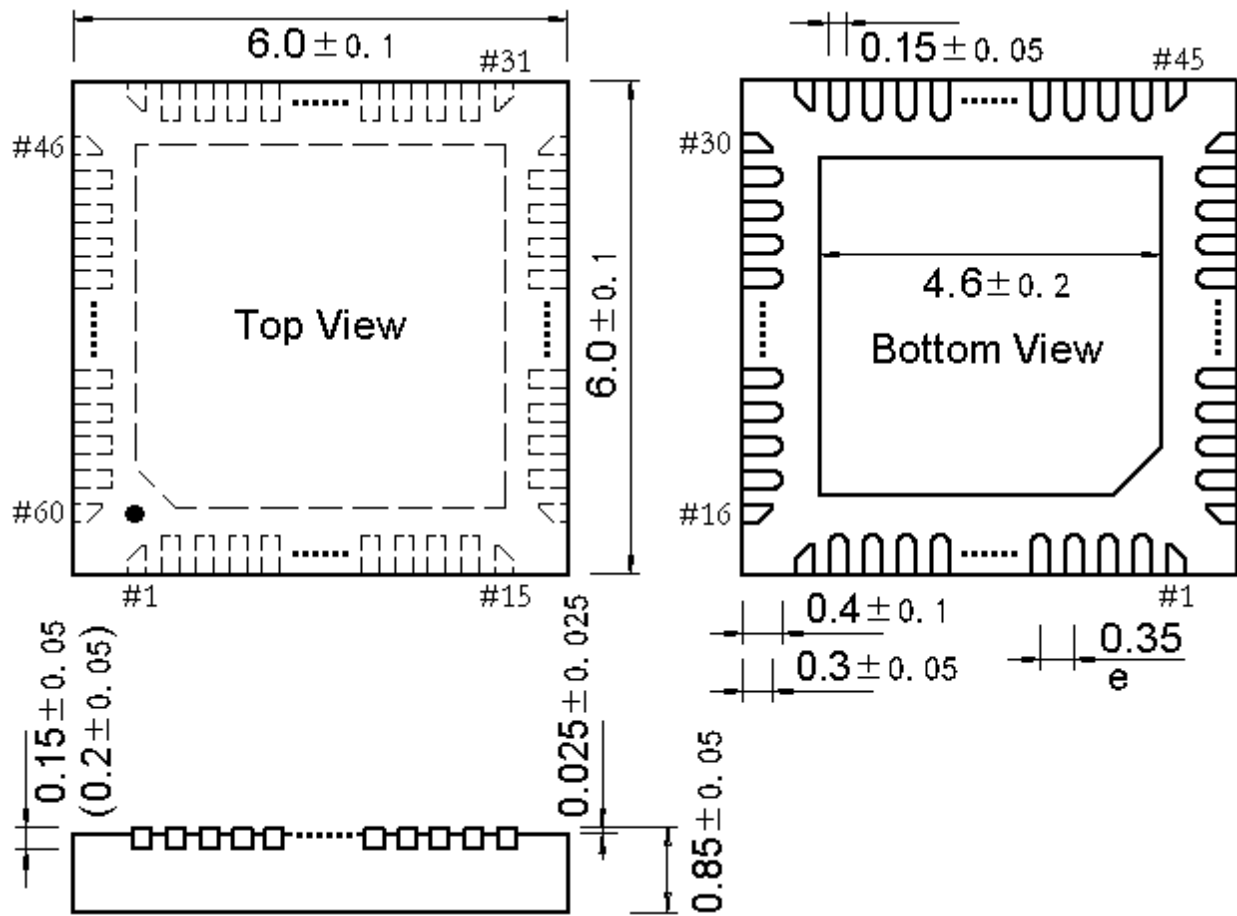
• QFN56X5 (QFN56-5\*5-0.3)



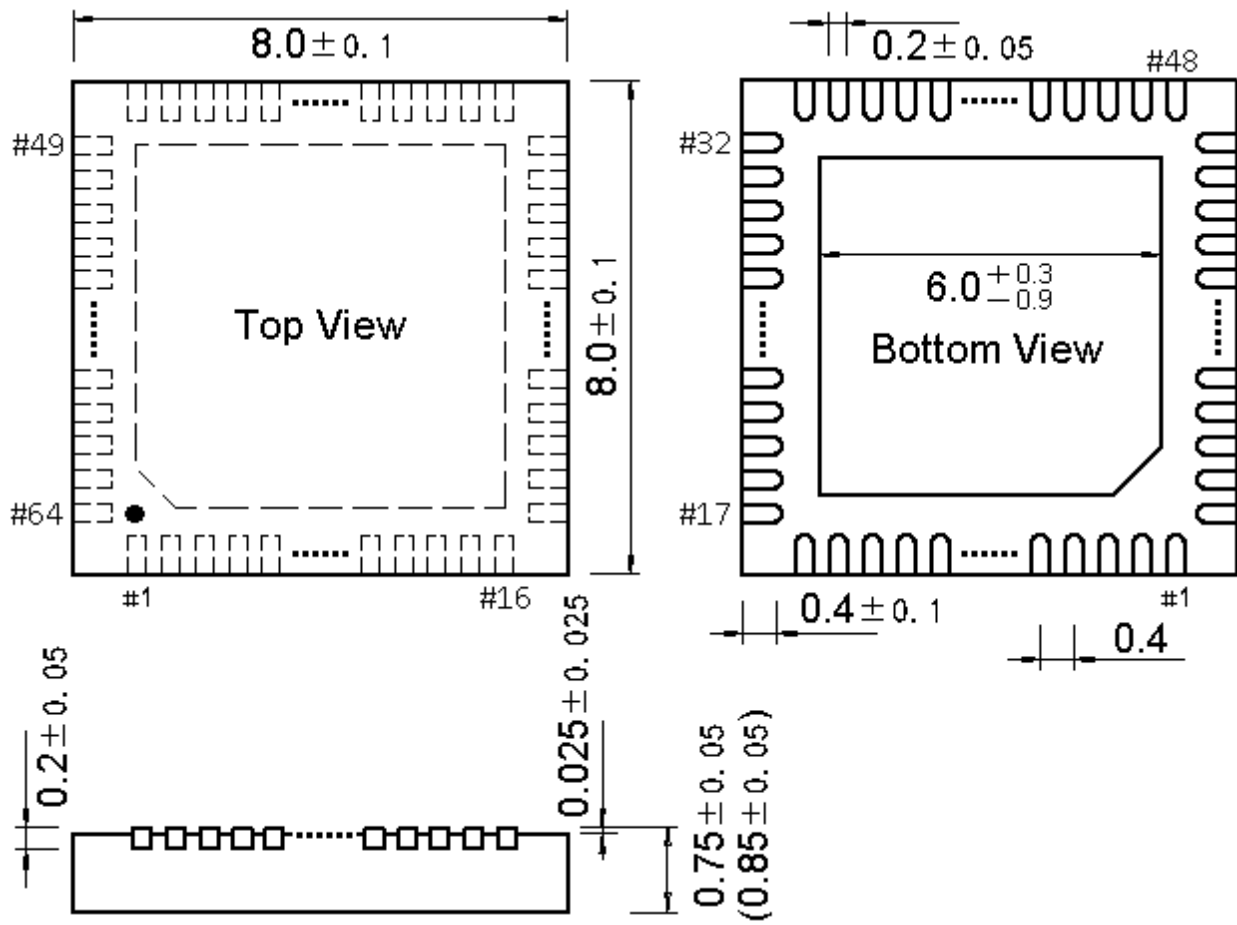
• QFN56X8 (QFN56-8\*8-0.5)



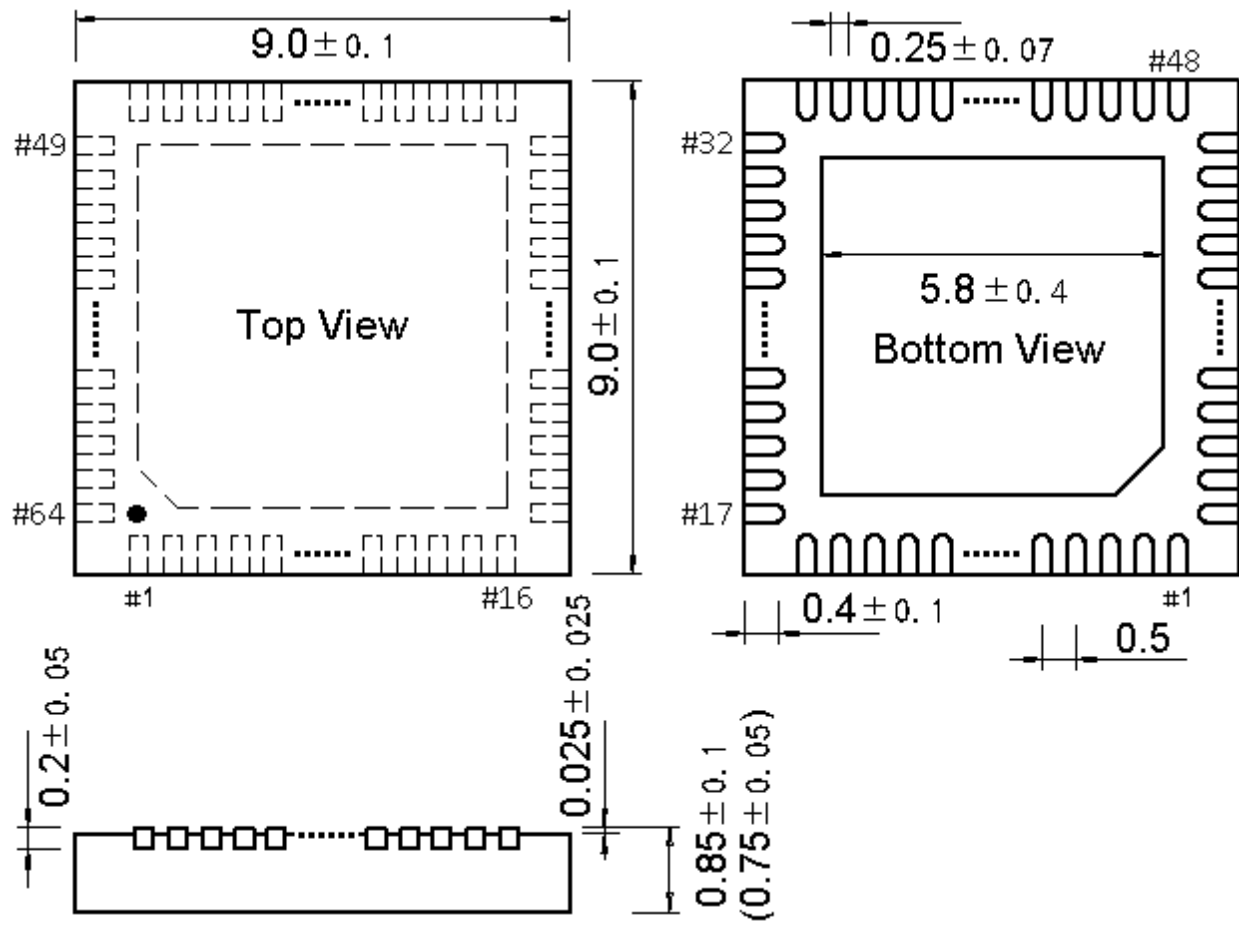
• QFN60X6 (QFN60-6\*6-0.35)



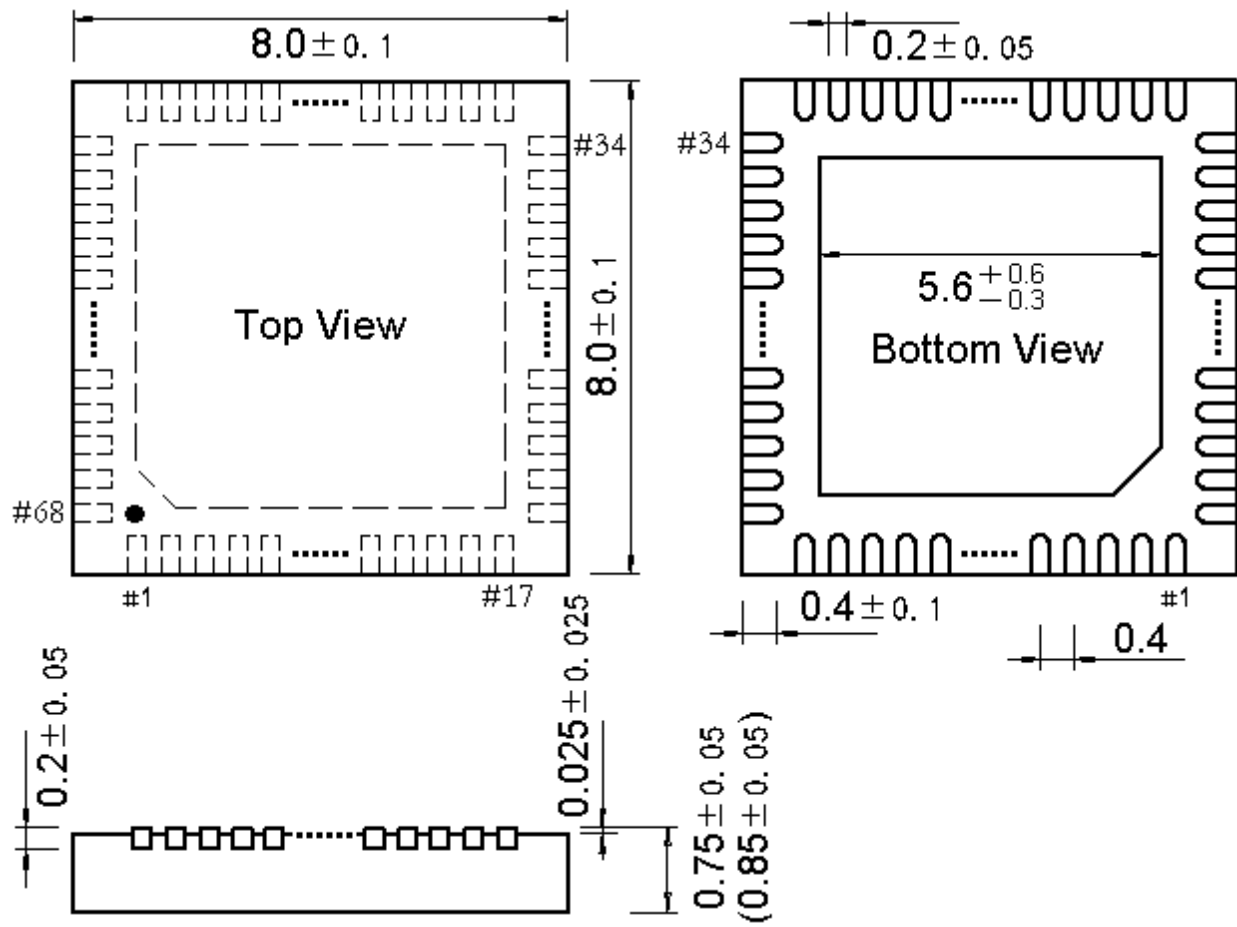
• QFN64 (QFN64-8\*8-0.4)



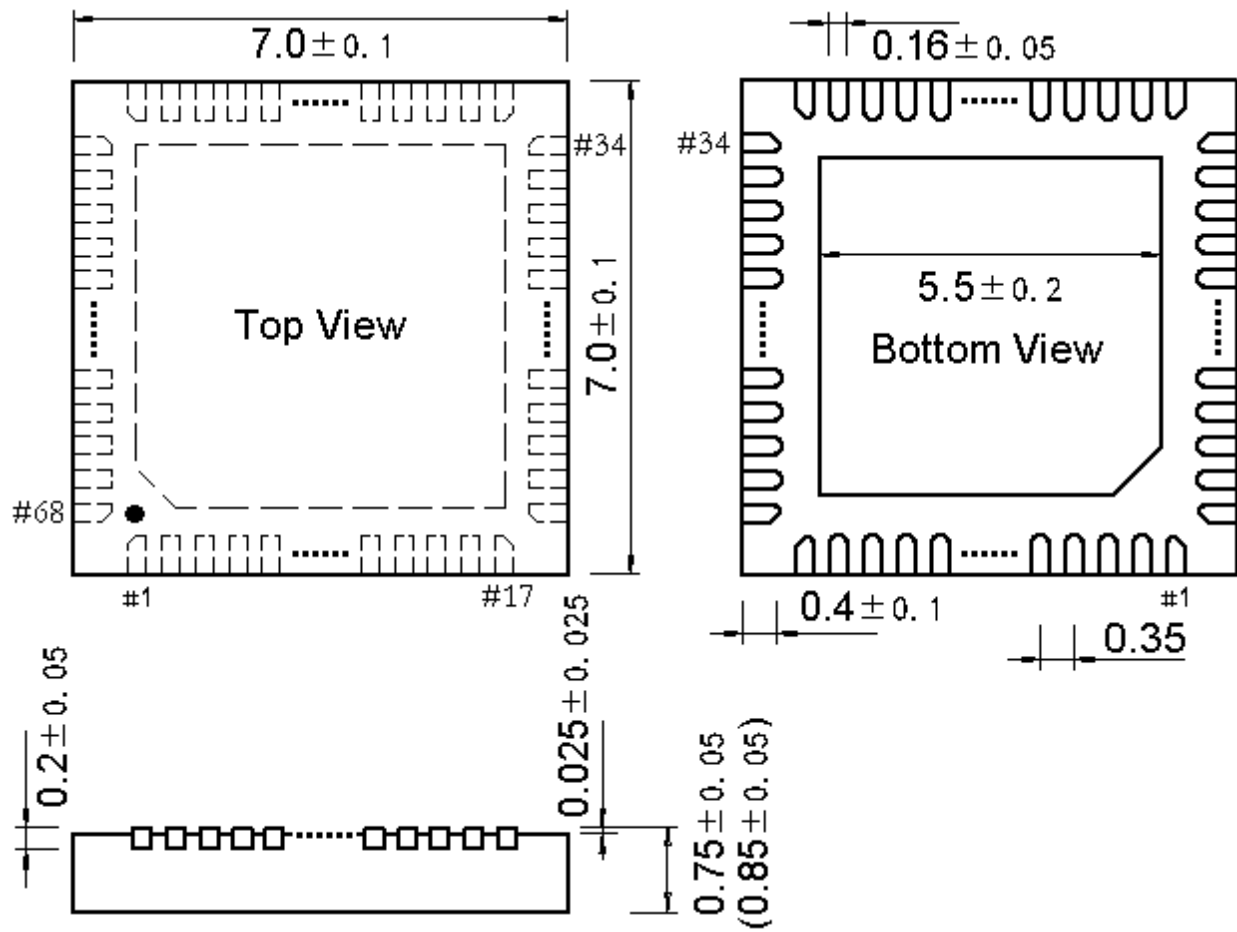
• QFN64X9 (QFN64-9\*9-0.5)



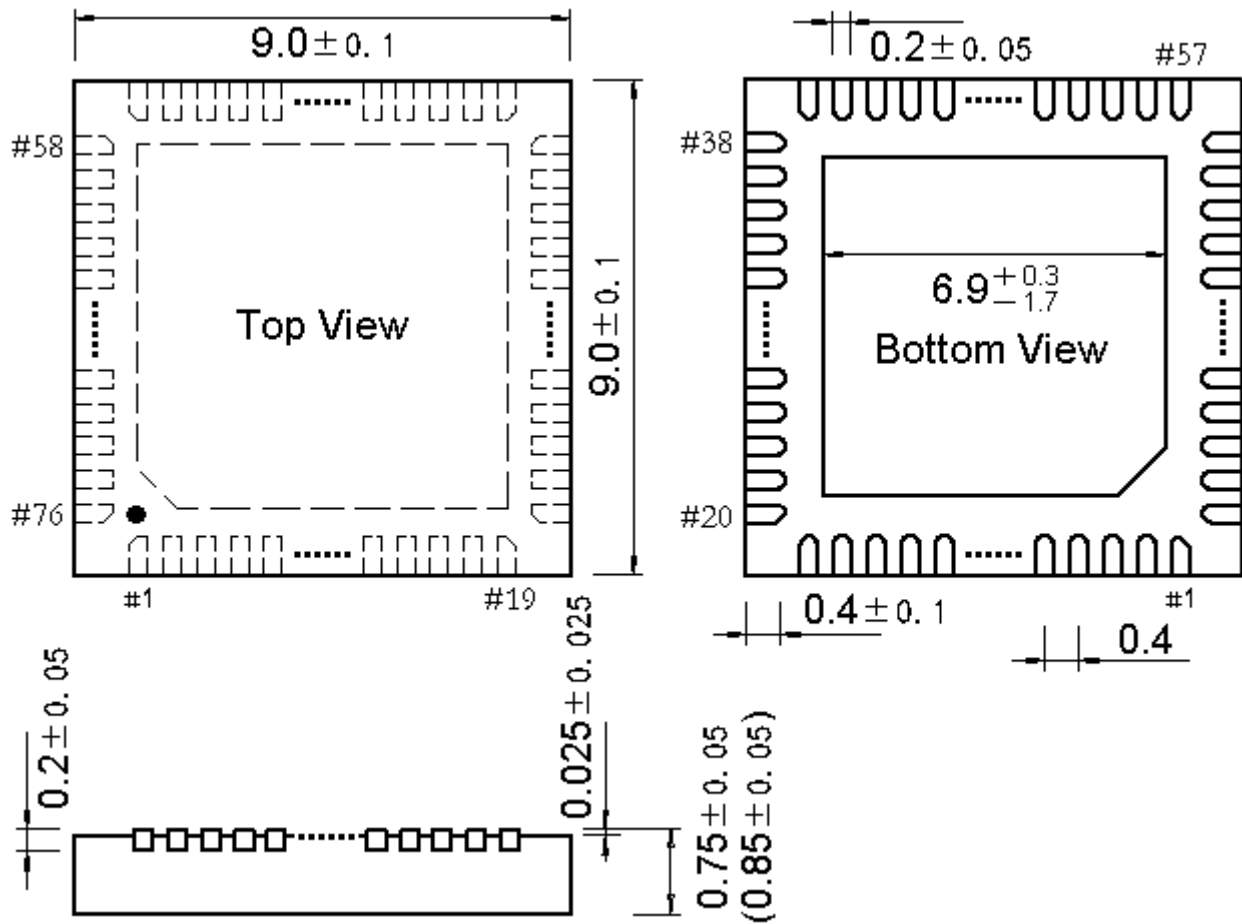
• QFN68 (QFN68-8\*8-0.4)



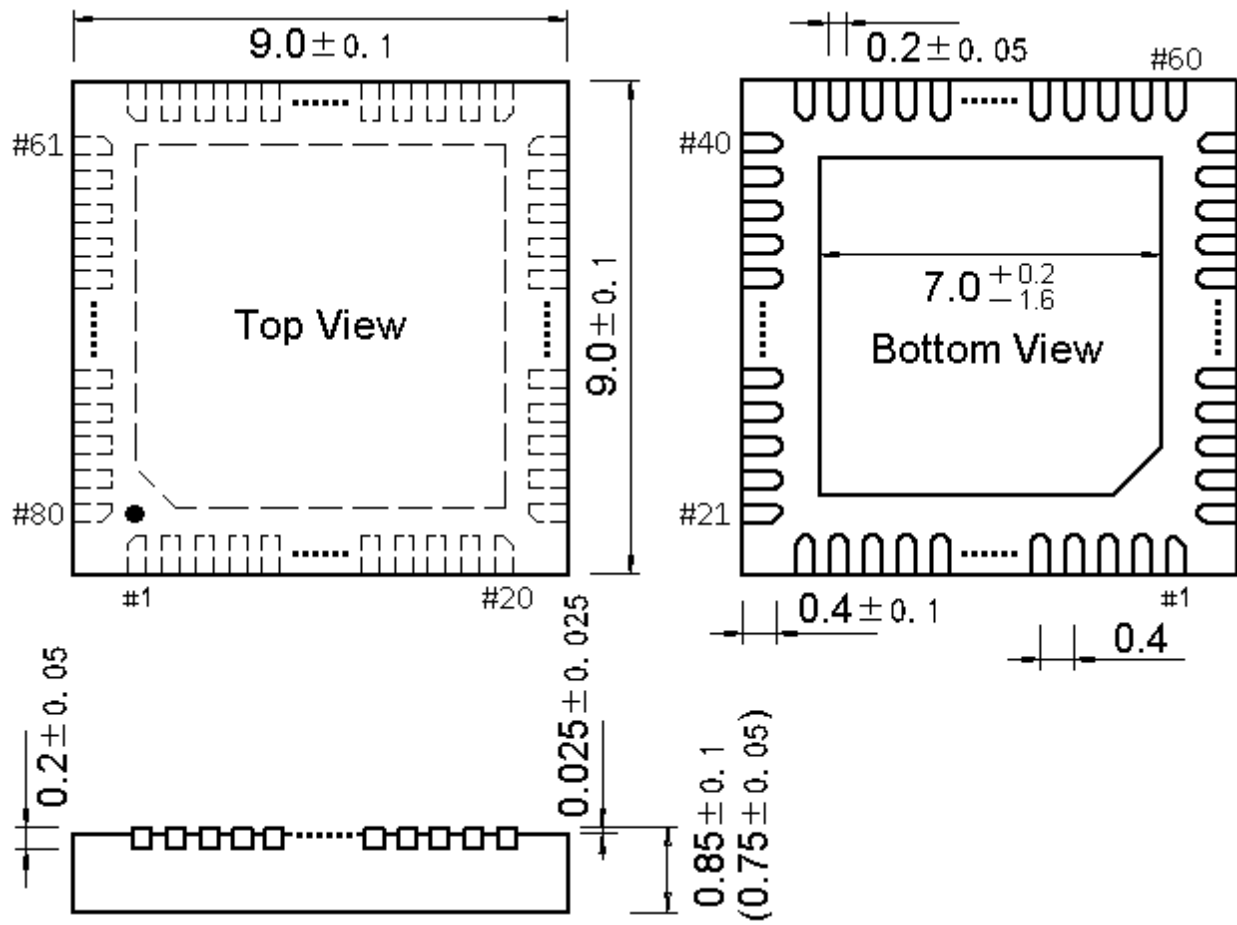
• QFN68X7 (QFN68-7\*7-0.35)



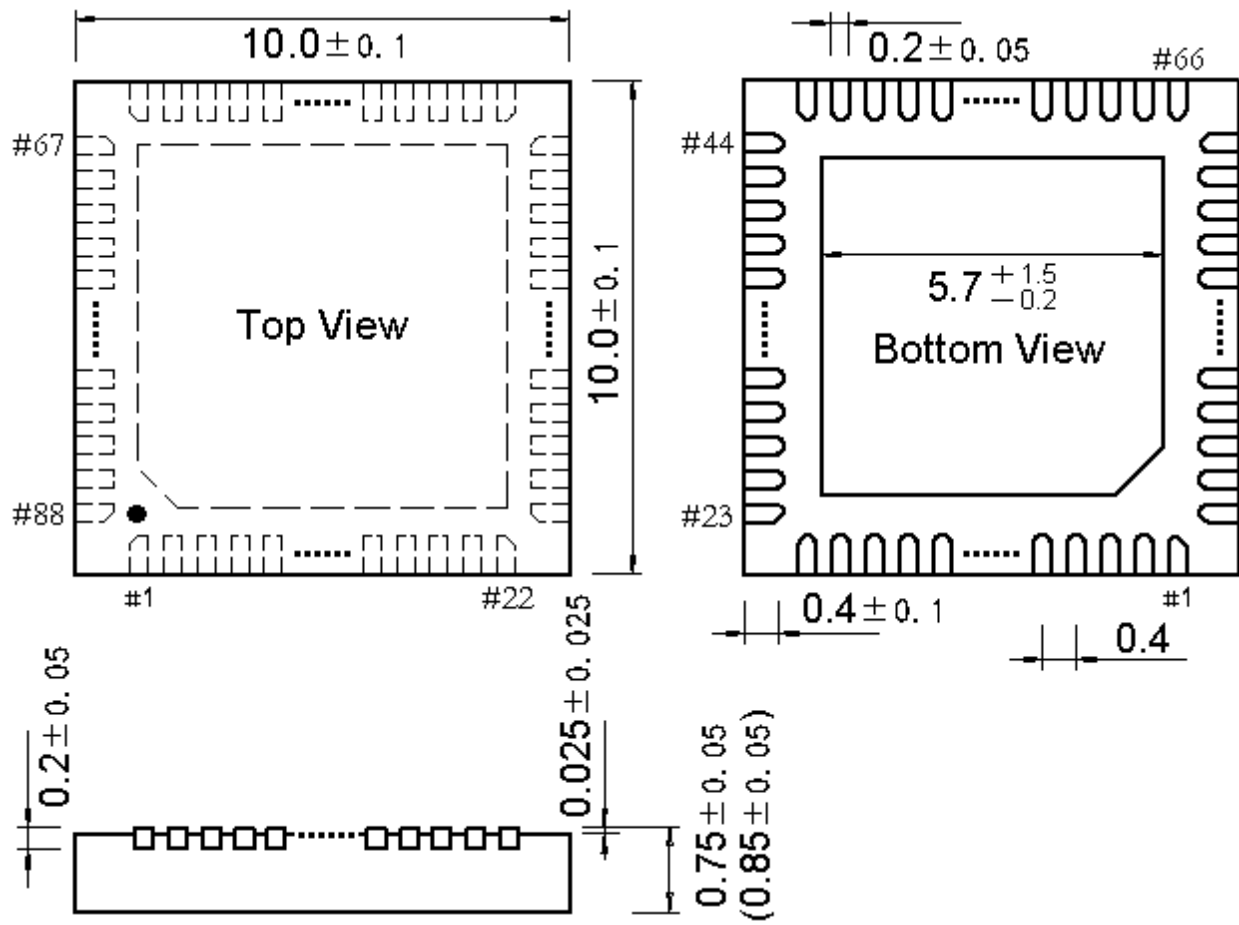
• QFN76 (QFN76-9\*9-0.4)



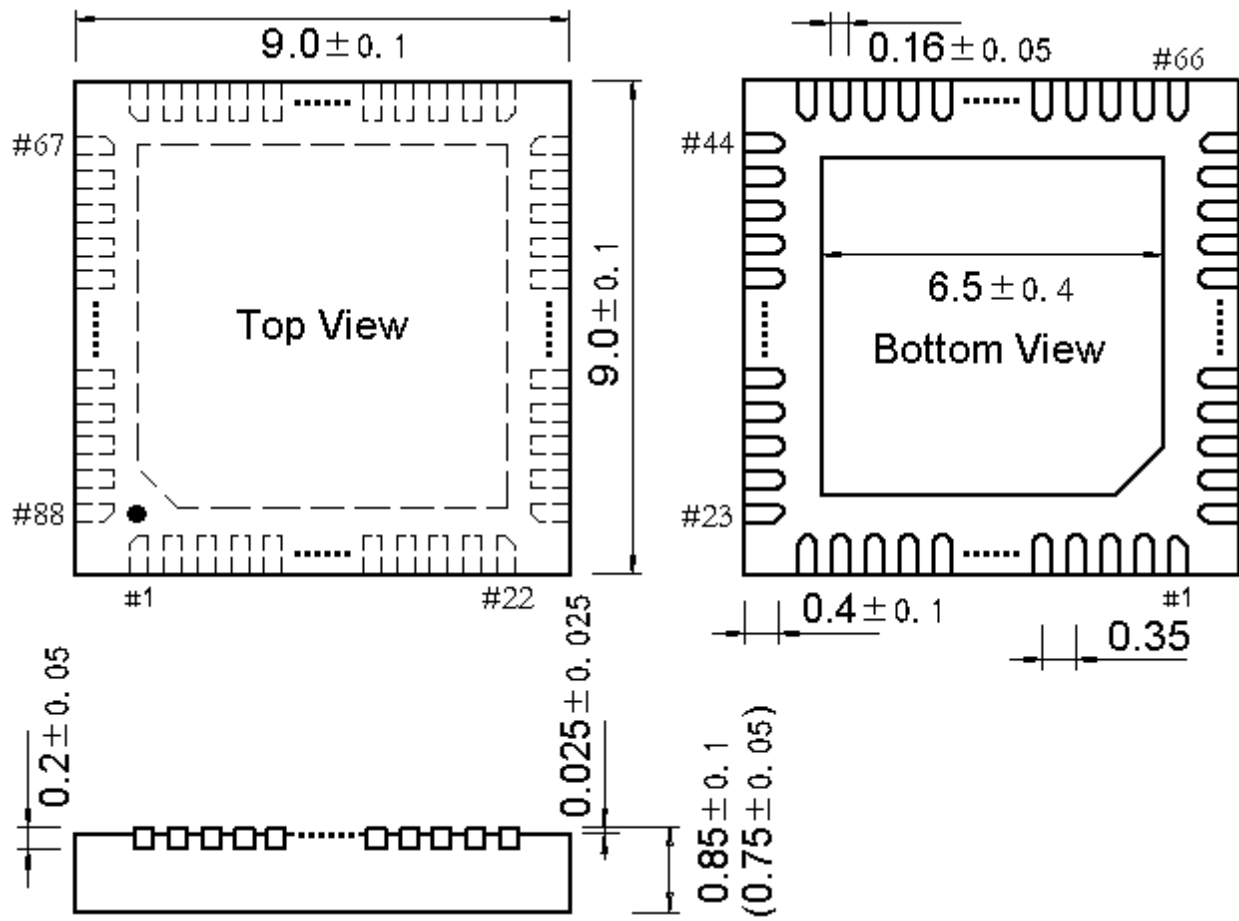
• QFN80 (QFN80-9\*9)



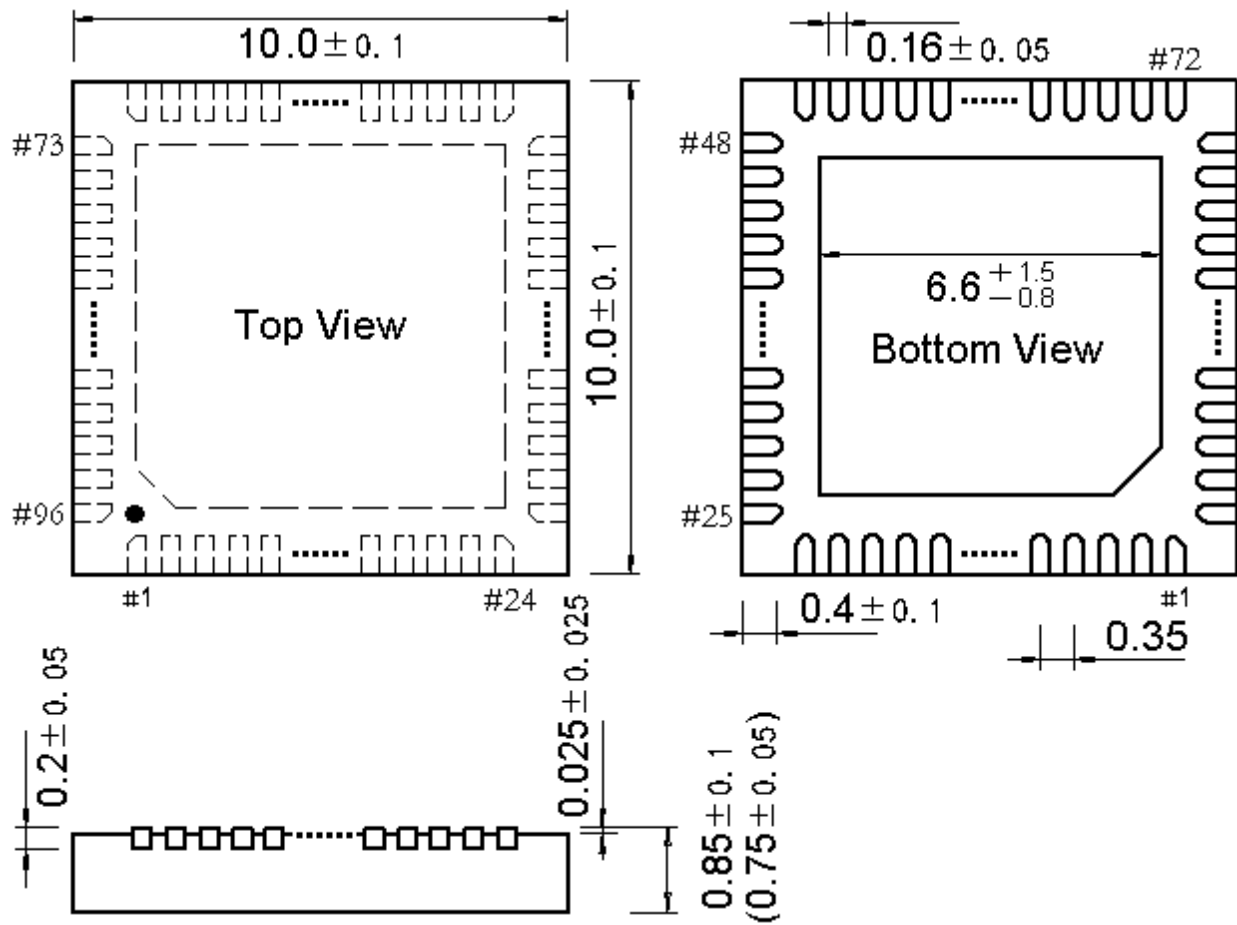
• QFN88 (QFN88-10\*10-0.4)



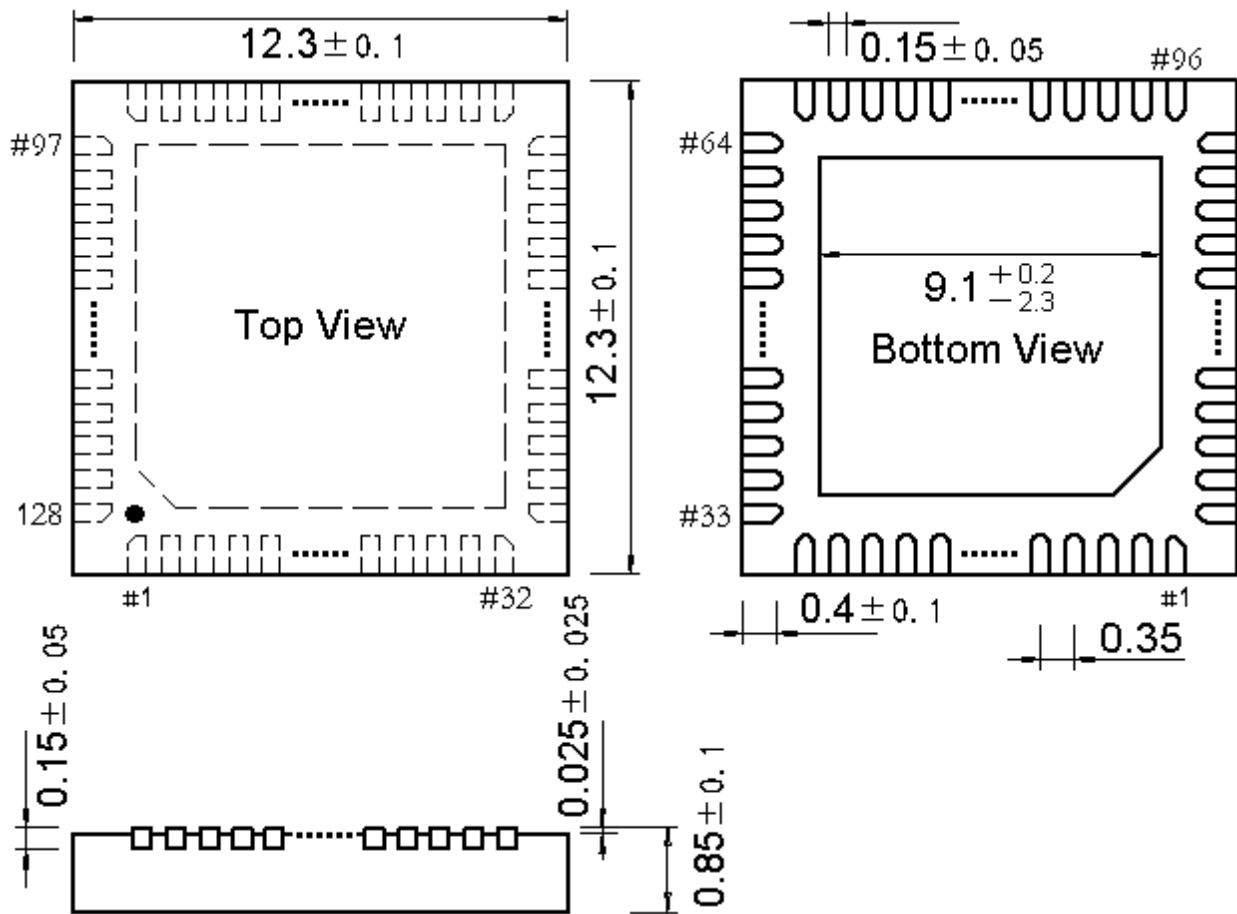
• QFN88X9 (QFN88-9\*9-0.35)



• QFN96 (QFN96-10\*10-0.35)

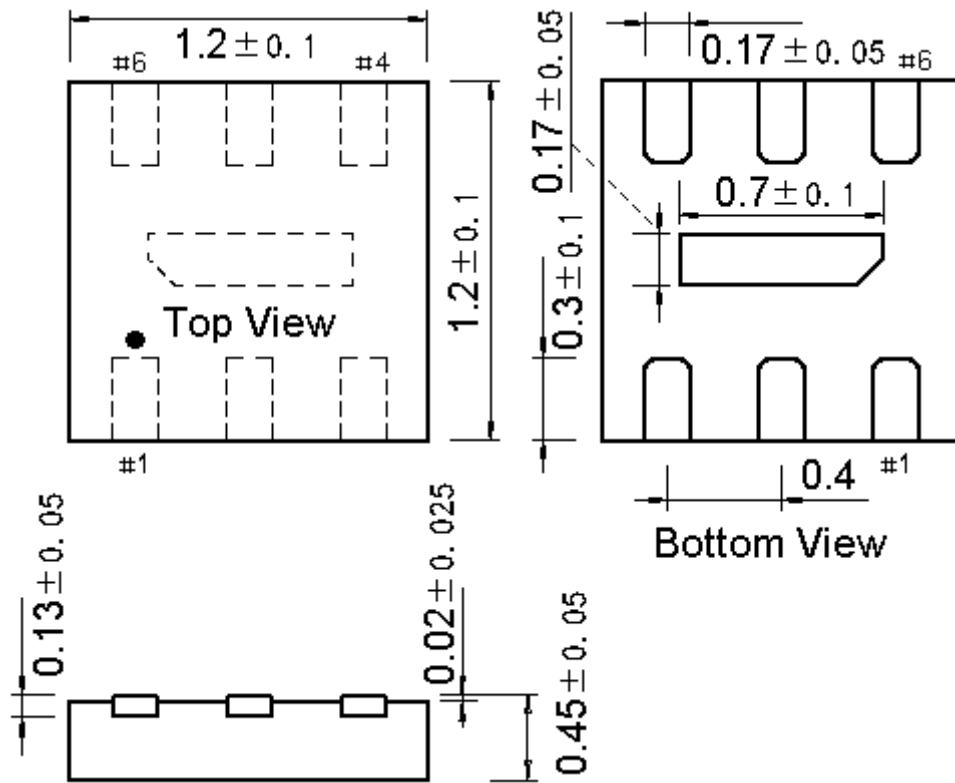


• QFN128 (QFN128-12.3\*12.3-0.35)

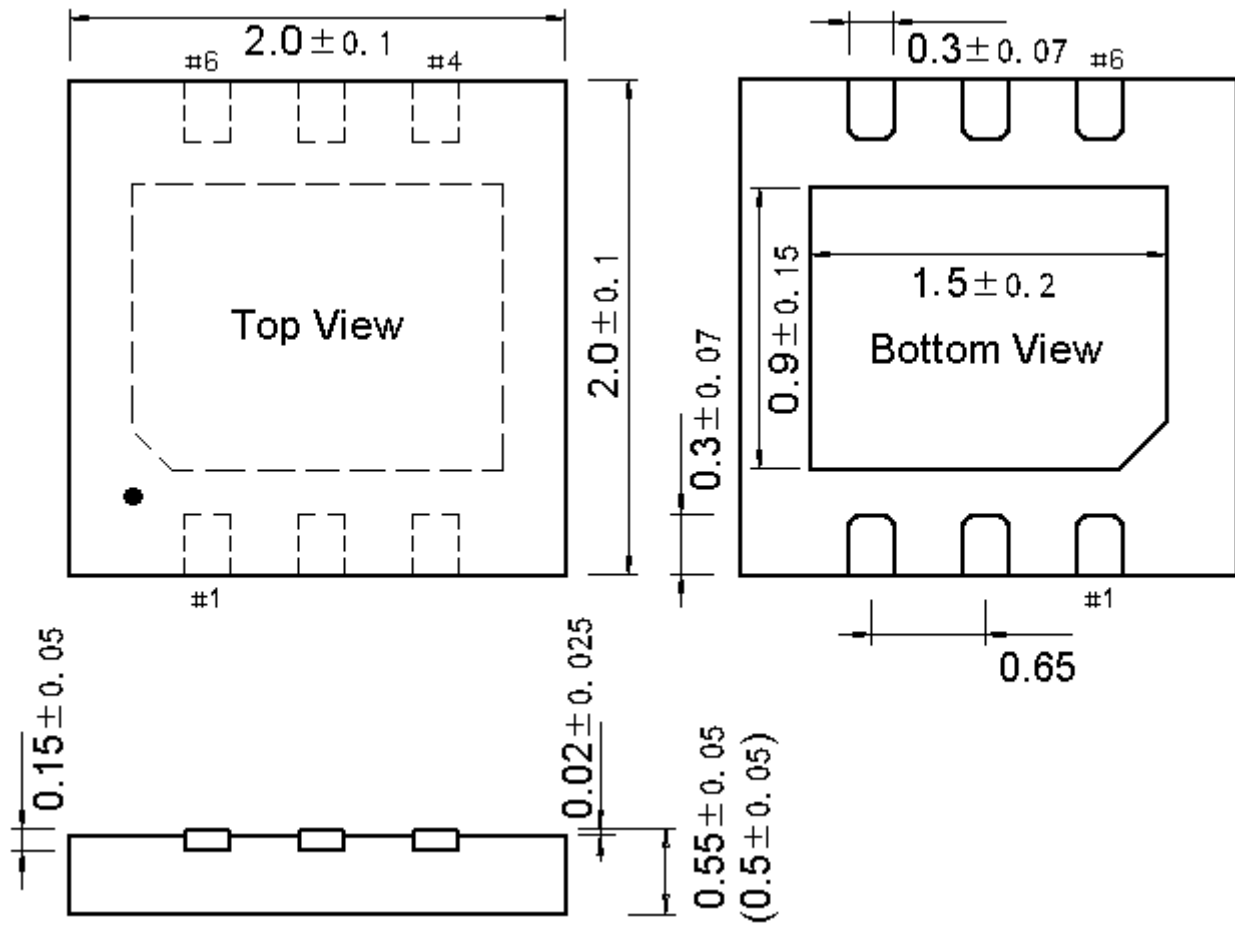


## DFN Type

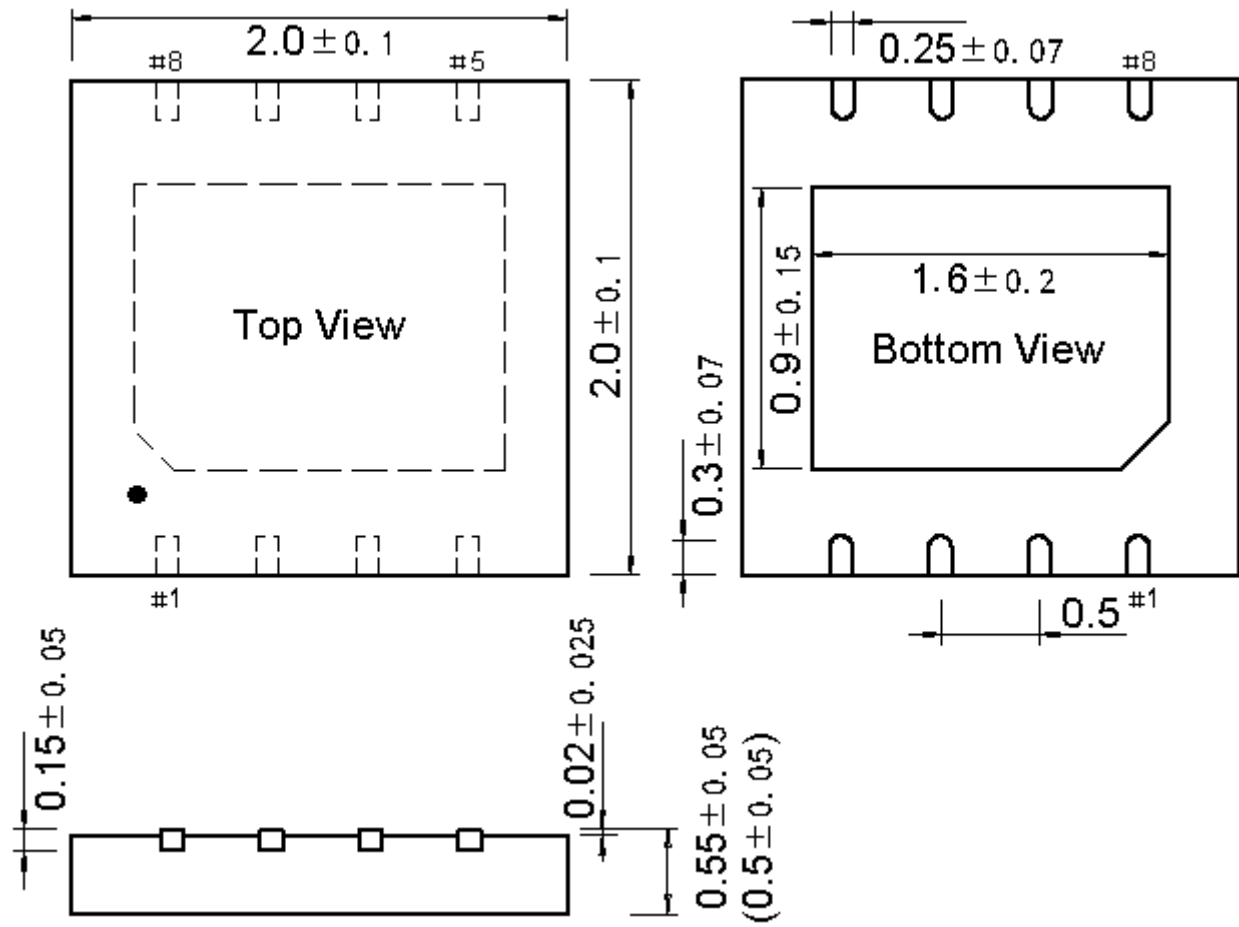
### DFN6X1.2 (DFN6-1.2\*1.2\*0.45-0.4)



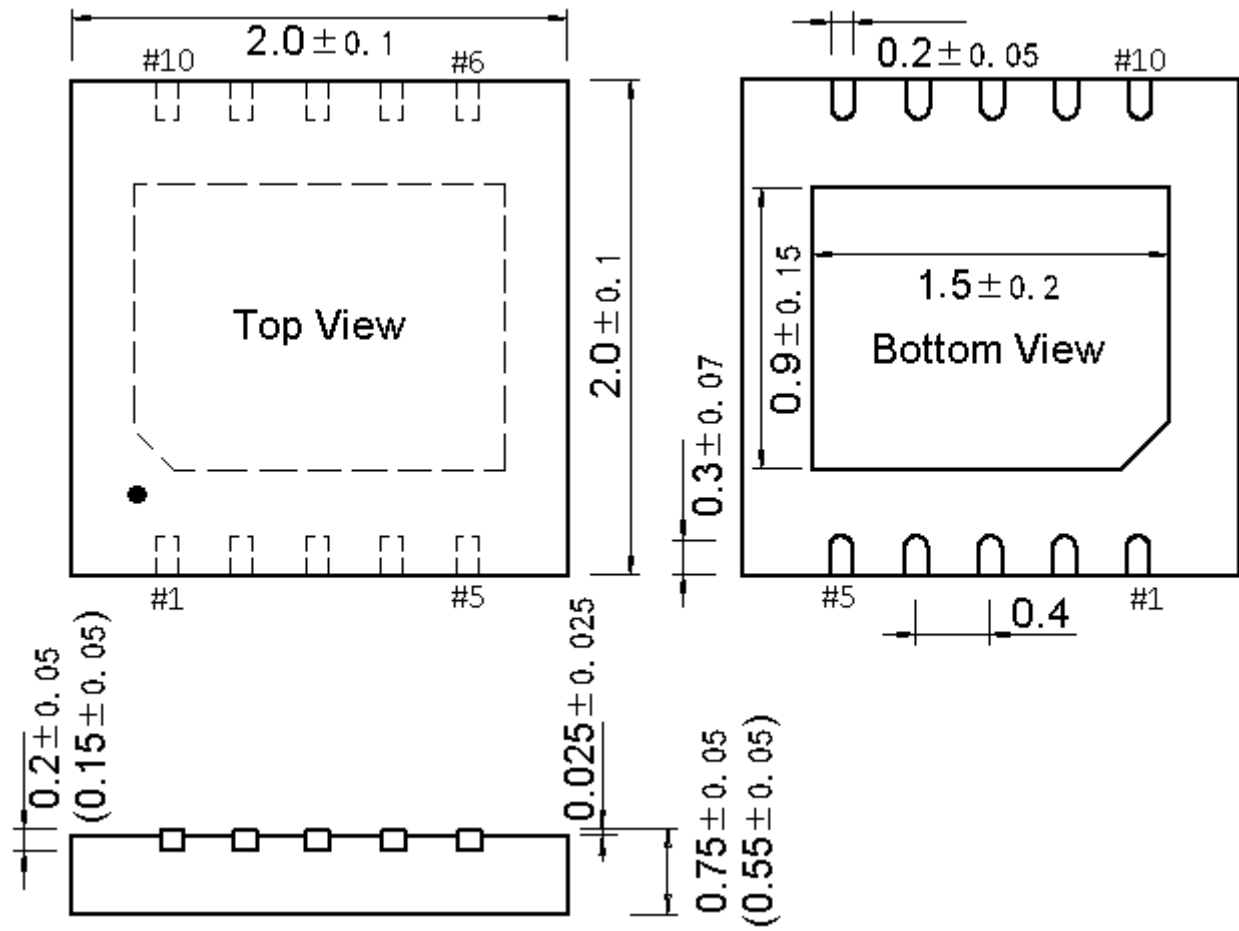
## . DFN6 (DFN6-2\*2-0.65)



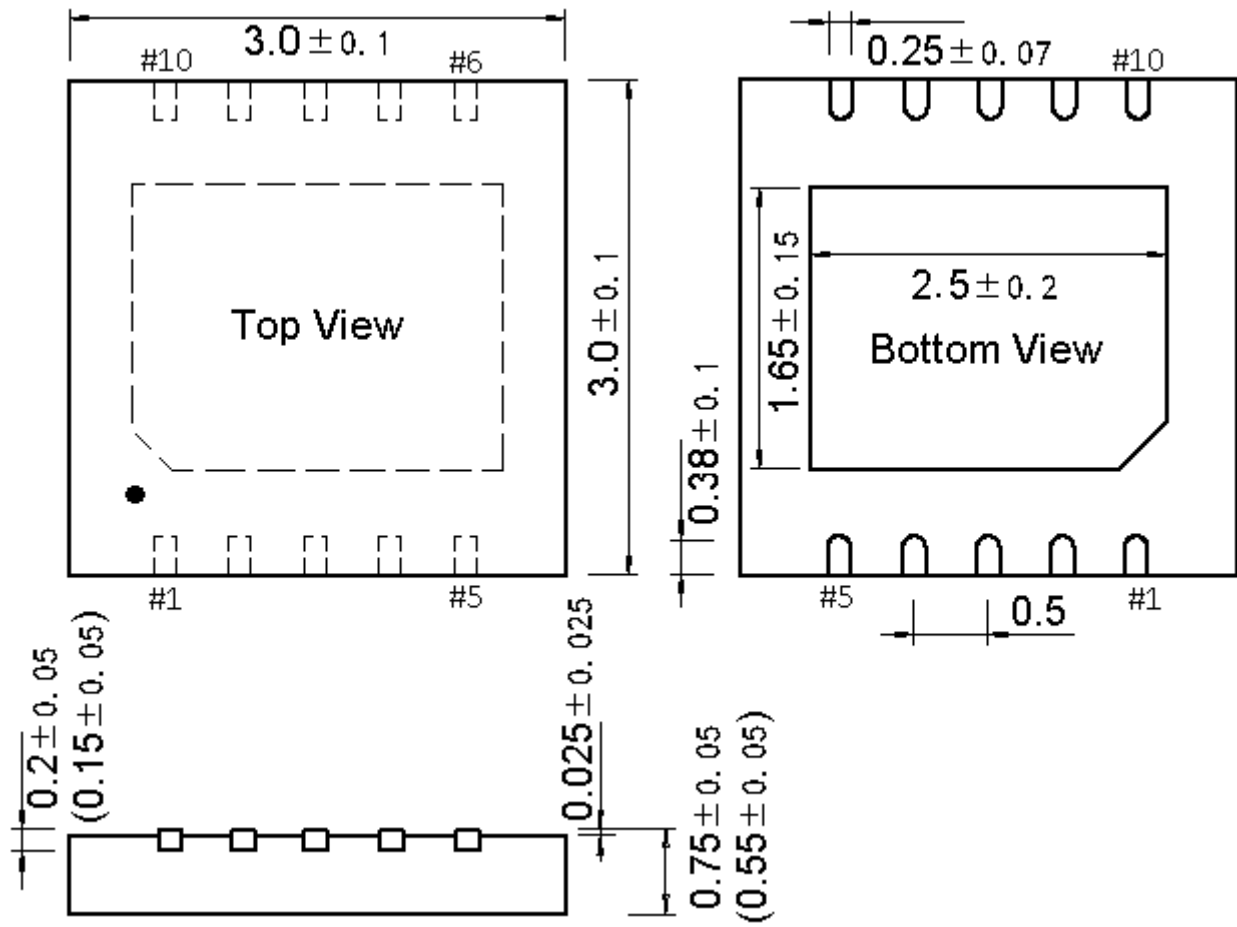
## . DFN8 (DFN8-2\*2-0.5)



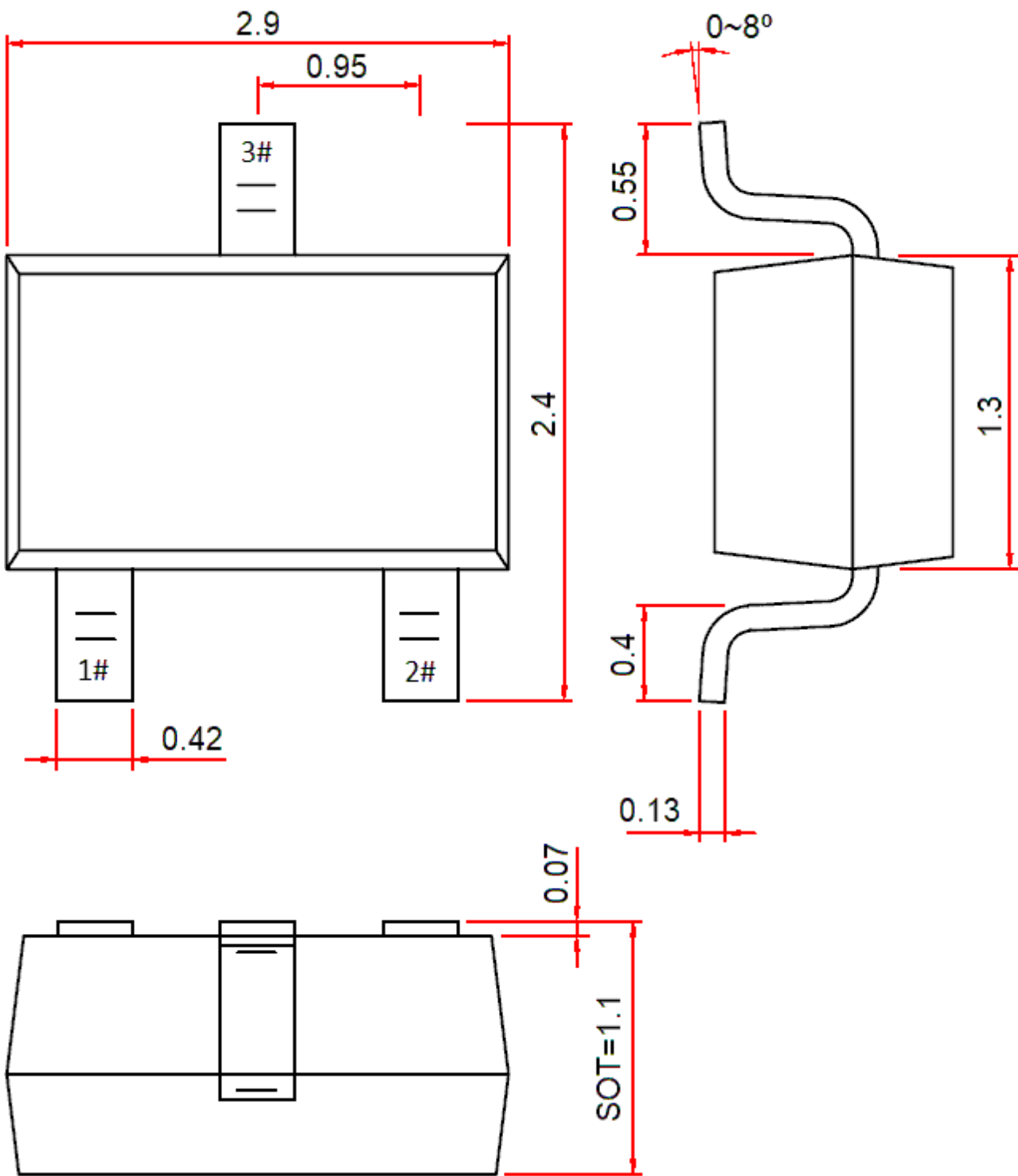
• **DFN10 (DFN10-2\*2-0.4)**



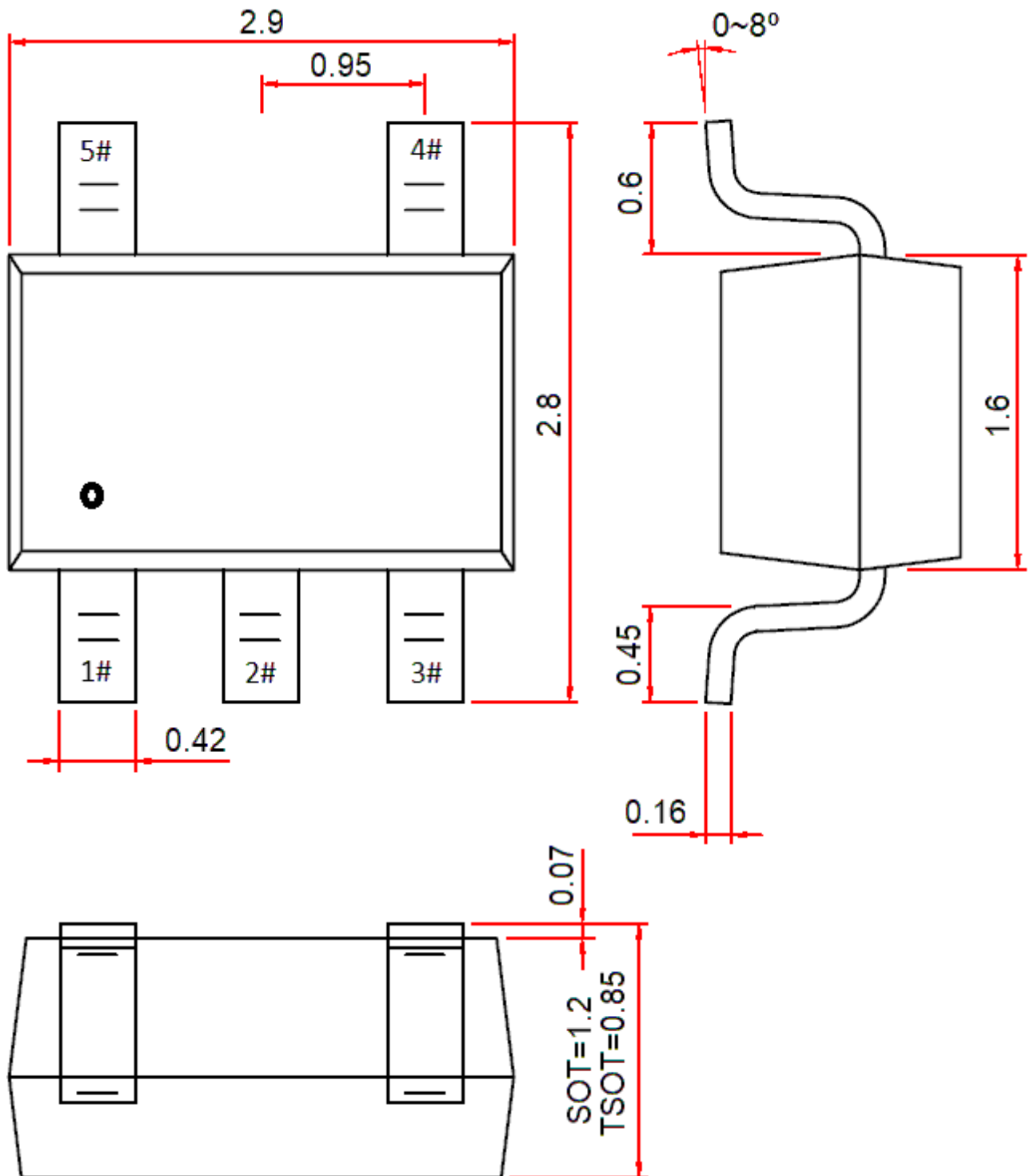
• **DFN10X3 (DFN10-3\*3-0.5)**



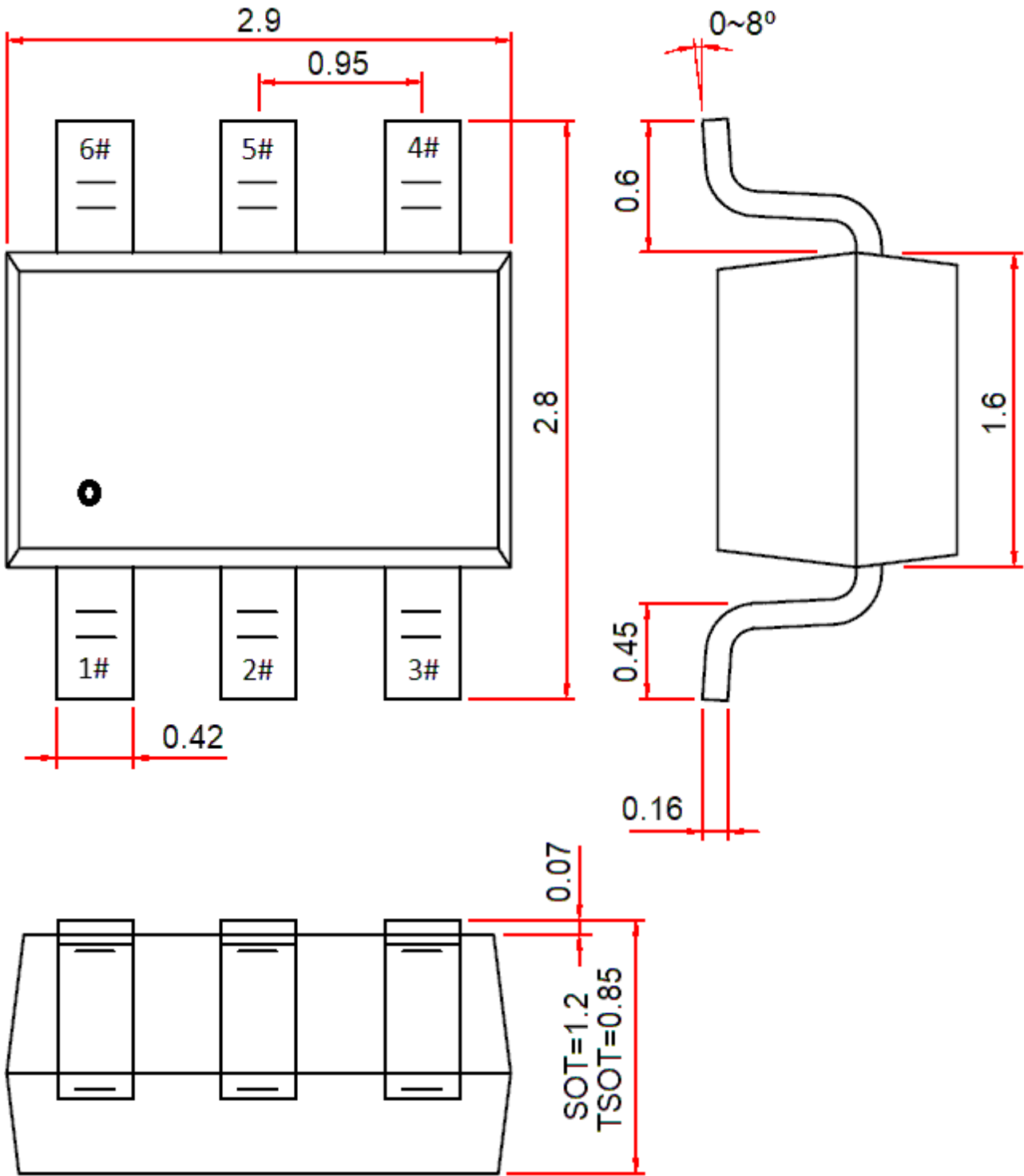
SOT Type  
· SOT23-3



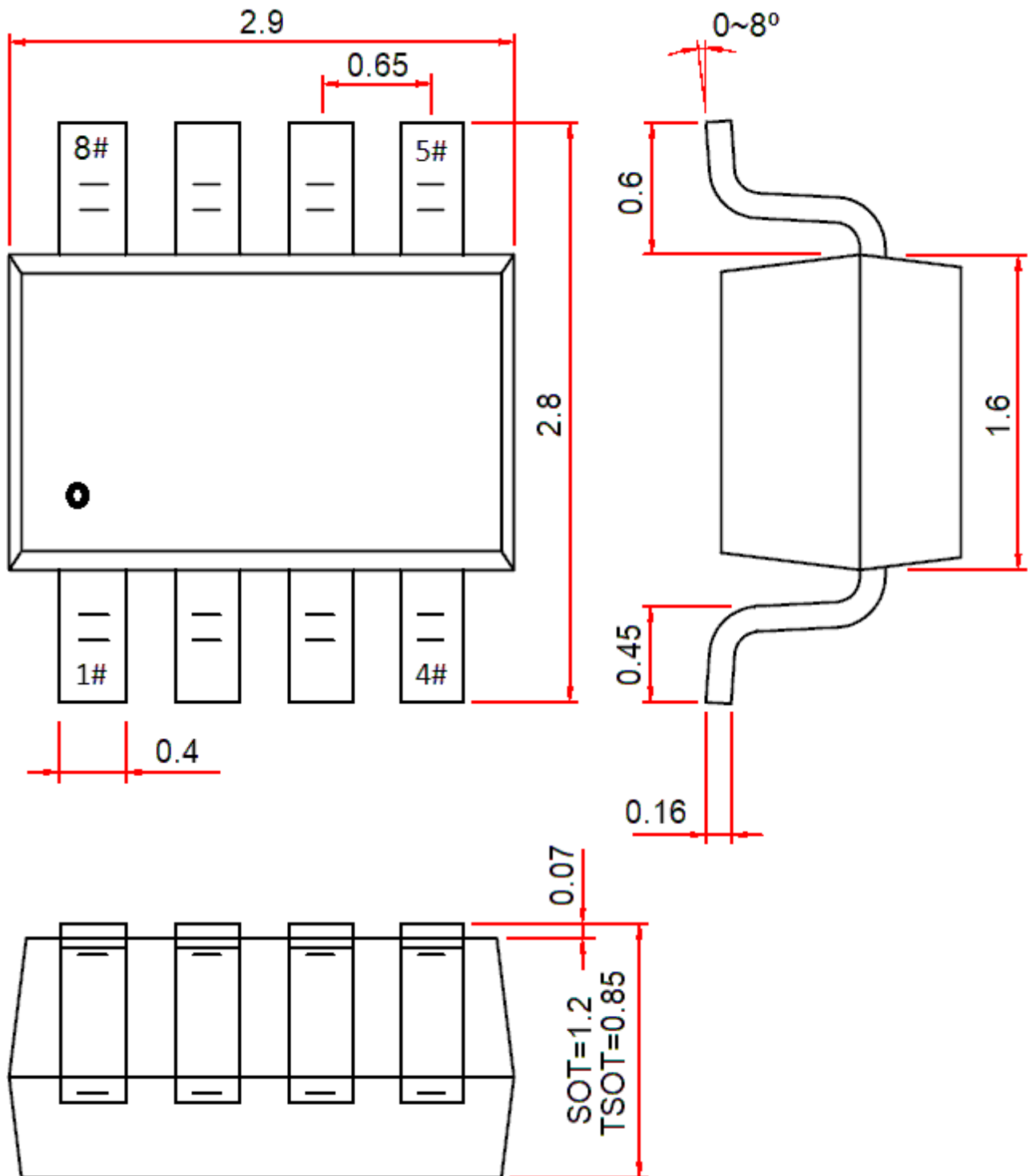
## . SOT23-5, TSOT23-5L, SOT25



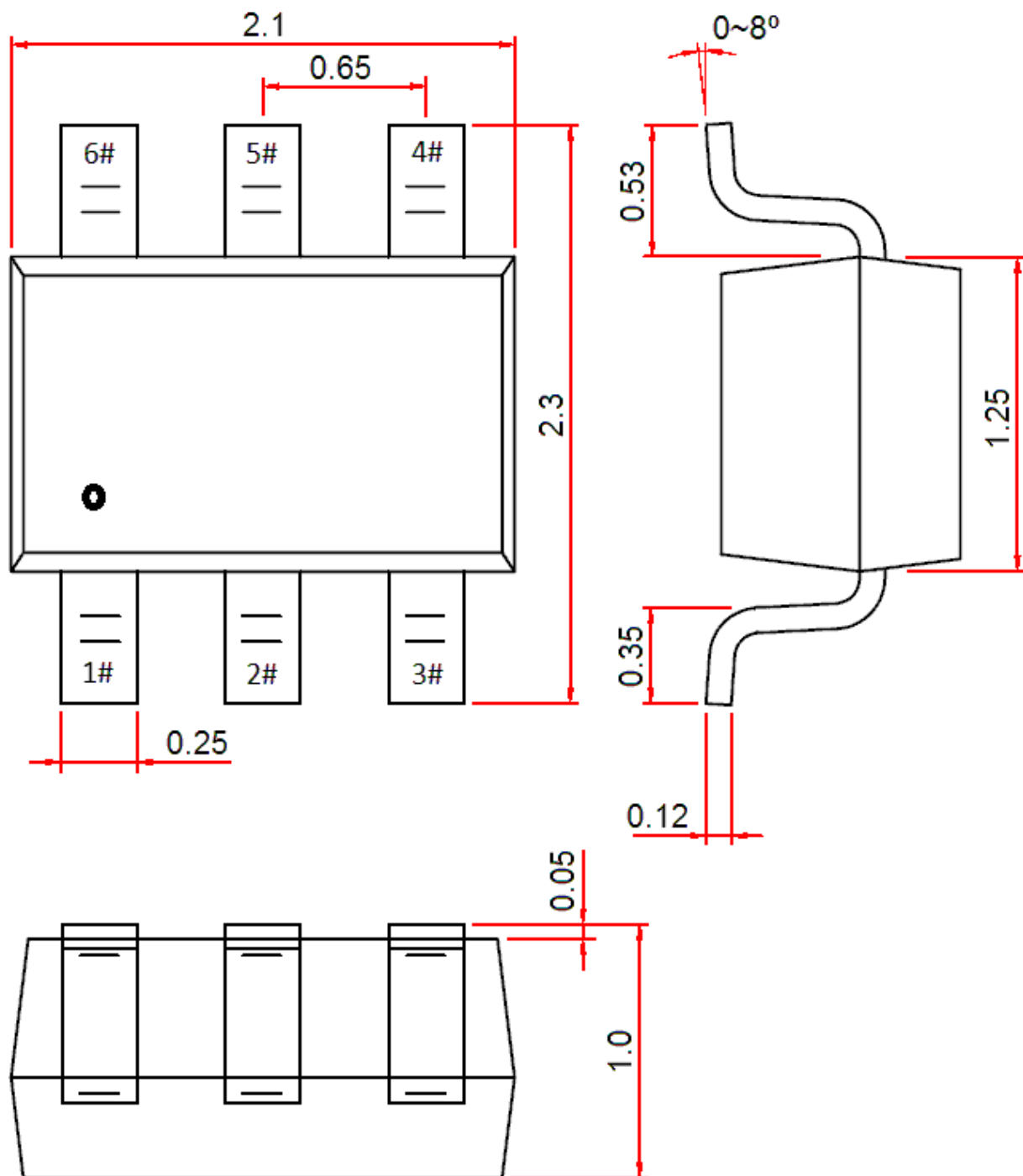
• SOT23-6, TSOT23-6L, SOT26



## . SOT23-8, TSOT23-8L, SOT28



## . SOT363 (SC70-6L)



## . SOT353 (SC70-5L)

